STATE UNIVERSITIES AND COLLEGES – ASSOCIATION OF COLLEGES OF AGRICULTURE IN THE PHILIPPINES (SUC-ACAP)

REVIEW MATERIALS FOR
THE LICENSURE EXAMINATION FOR AGRICULTURISTS

Animal Science

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SUC-ACAP-LEA

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PREFACE

Republic Act 8435 or the Agriculture and Fisheries Modernization Act of 1997, simply known as AFMA, was enacted on December 22, 1997. It is a policy instrument defining measures to modernize Philippine agriculture for the country to compete in the global market. A basic human resource development framework in modernizing agriculture under the Act is to enhance the professional competency of agriculturists by upgrading the standards of the agriculture and fisheries education that will ensure to similarly upgrade the quality and sustainability of, and promote global competitiveness, at all levels.

One strategy for professionalizing the agriculture sector is the Licensure Examination for Agriculturists (LEA). First implemented in 2003, LEA serves as a standardized evaluation tool to identify graduates of Agriculture and Agriculture-related degree programs who are academically and technically fit to be admitted in the agricultural profession. The three-day examination consists of multiple-choice type questions in the following component subjects: Crop Science, Soil Science, Crop Protection, Animal Science, Agricultural Economics and Marketing, and Agricultural Extension and Communication.

The COVID-19 pandemic has been a major concern of all higher education institutions. When review sessions for licensure board exams are already supposed to be grinding, SUCs are confronted by the reality that face-to-face review is undeniably not possible. Under the circumstance, universities face new pressures to provide an engaging, motivating, and effective teaching and learning experiences for both reviewees and reviewers.

Following this, the SUC-ACAP, with its goal of advancing the agriculture professional competency among their member-institutions, and the College of Agriculture and Food Science, UP Los Baños embarked on a joint initiative to enhance the capability of the SUC faculty particularly those engaged in the conduct of review classes in preparation for the Licensure Exam for Agriculture (LEA). To achieve this, the joint initiative conducted three online lectures, namely, 1) Innovative Teaching Methods and Tools Designed for Online Review Classes by no less than the Chancellor of the UP Open University, Dr. Melinda dP. Bandelaria. This online course aimed at introducing the faculty participants to the particularities of conducting review classes online with the use of innovative methods and tools; 2) Test Construction: the Multiple-Choice Type Test from a seasoned UP faculty, Dr. Blanda R. Sumayao, which gave the faculty-participants the opportunity to review the principles and rules for constructing multiple-choice type tests; and 3) Design and Development of Review Materials for LEA, again with Dr. Bandelaria with the objective of providing the faculty-participants the necessary knowledge and skills in designing and developing review materials and planning for the delivery of online review sessions.

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The Project initiative also gave the faculty-participants the opportunity to review their course syllabus viz-a-viz CMO #14 S. 2008 or the CHED-prescribed outline which enabled them to design and develop this harmonized syllabus for each of the six core subject matter area. The review materials are thus a product of many minds and talents, that is, of faculty-participants from the different SUCs. I would like to acknowledge all the writers/contributors who willingly presented themselves to the initiative and for strictly following deadlines in submitting drafts; to all University Presidents and the Deans of the Colleges of Agriculture who have been highly committed to the objective of this Project initiative and for the support provided to the faculty-participants; to all Cluster and Sub-Cluster Coordinators for the constant reminders and follow-ups; and to the SUC-ACAP Secretariat for the untiring administrative support.

And lastly, we are grateful to the College of Agriculture and Food Science, UP Los Baños for being true to its commitment of being a public service University, providing scholarly and technical assistance to its sister and brother SUCs. Specifically, we thank the faculty of CAFS who served as Subject Area Coordinators/Specialists who unselfishly guided our faculty participants throughout their journey in the project, as well as the Technical Assistants and Administrative Staff who have lend full support to attaining the objectives of the joint initiative.

As I have always reiterated in many occasions, let us be reminded that student preparation for the licensure is just one of the factors that would determine performance. It is still incumbent upon us SUCs or higher educational institutions to ensure that we will be able to provide the necessary prerequisite services and institutional support in order for students to ultimately pass successfully and help advance the students' professional careers post licensure exam.

The preparation for publication of these review materials was supported by the SUC-ACAP.

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December 2021

FOREWORD

Unlike in *Lord of the Rings* and *Harry Potter* films, there are no wizards, Balrogs, Voldemort, precious rings, or elderly wands, on the journey to becoming a professional agriculturist. But there are narrow bridges and mazes that you must travel and cross over. One of these challenges is the Professional Licensure Exam.

In the current climate where globalization, changing natural conditions, production, and social composition; migratory movements, among others, have significantly impacted most agricultural landscapes in the developing world, more than ever, the Licensure Exam for Agriculture has become all the more relevant as it prepares graduates of Agriculture to embrace the tasks ahead of them with utmost professionalism by way of recognizing their expert knowledge and practice. The Licensure Examination for Agriculture was established to recognize the expert knowledge and practice of aspiring professional agriculturists.

This Review Material produced by the faculty participants of the joint initiative of the UPLB College of Agriculture and Food Science (UPLB-CAFS)- State Universities and Colleges-Association of Colleges of Agriculture in the Philippines (SUC-ACAP) is a comprehensive resource that covers all relevant subject areas in the field of agriculture which hopes to serve as a valuable review tool to students of SUC-ACAP member institutions.

I wish you the best in your journey and beginning your career as a professional agriculturist.



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December 2021

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Unit I
ANATOMY AND PHYSIOLOGY OF FARM ANIMALS

ANIMAL PHYSIOLOGY

- Study of function of the parts or organ systems of the body
- Explains the physical and chemical factors that are responsible for the origin, development and progression of life
- It provides knowledge on the structure and function of the body and constantly, the care of the body

I. NERVOUS SYSTEM

- Network of specialized cells that communicate information about animals' surroundings and itself
- Composed of neurons and other specialized cells called glia, that aid in the function of t the neurons
- Controls the rapid activities of the body such as muscular contraction, secretions of some endocrine glands, heart rate, respiration rate, gastrointestinal motility and rapid reflex action
    - Basic units of NS: Brain, Spinal Cord and Nerves
    - Nerve Cells or Neurons - specialize in impulse conduction or the relay messages from effector organs to the nervous system and vice versa. Consist of a cell body (soma) and nerve fibers, namely axon (a single process that extends from the cell body and ends up on a synapse or an effector organ) and one or more dendrites (has distal end called receptors that receive stimuli that initiate conduction of impulses to the cell body of neurons).
    - Neurons may be classified according to the direction of impulse conduction as follows:
        - Afferent (sensory) neuron - transmits nerve impulses from effector organs to the spinal cord or brain.
        - Efferent (motor) neuron - transmits nerve impulses from the spinal cord or brain to the effector organs
        - Interneuron - conducts impulses from an afferent neuron within the Central Nervous System (CNS)

Two separate divisions of the Nervous System

1) **Central Nervous System (CNS)** – command center of the body; contains the majority of the nervous system and consists of the brain and the spinal cord. It interprets incoming sensory information
2) **Peripheral Nervous System (PNS)** – part of the nervous system outside of the CNS. It resides or extends outside the limbs and organs. It is not protected by bone, leaving it exposed to toxins and mechanical injuries.

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Divided into:

- *Somatic nervous system* - innervates striated or skeletal muscles, controls the **voluntary movements** of skeletal muscles.
- *Autonomic nervous system* - innervates glands and visceral musculature, control **involuntary or automatic activities** in the body

Two subdivisions of Autonomic nervous system

1) *Sympathetic nervous system* – regulates involuntary activities that help the body respond to stressful conditions
2) *Parasympathetic nervous system* – regulates involuntary activities that help the body run smoothly under normal conditions

Brain = Important Central Regulation

- Feed intake regulation
  - Ventromedial = Satiety center (voracious eating)
  - Lateral = Feeding center (loss of appetite)
  - Preoptic Area = Thermoregulatory Center
  - Suprachiasmic Nuclei = Center for sleep and wakefulness

Neuroendocrine – involvement of nerve cells that releases chemical messengers, especially neurohormone, directly into the bloodstream

Forms of energy converted by receptors

- Mechanical (touch-pressure)
- Thermal (degrees of warmth or cold)
- Electromagnetic (light)
- Chemical energy (odor, taste, O_2 content of blood)

Sensory modalities

- Smell
- Vision
- Hearing
- Rotational and linear acceleration
- Taste
- Cutaneous senses with receptors in the skin to monitor touch-pressure
- Cold
- Pain

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II. ENDOCRINE SYSTEM

- Endocrinology - branch of physiology which deals with the coordination of various body tissues by chemical mediators (**hormones**) produced by restricted area of the body (**endocrine glands**) and transported through the circulatory system to the target organ or tissue
- Hormone – substance or chemical messenger produced by an endocrine glands and carried in the blood to distant parts of the body to exert its effect
- Functions:
    - Regulating growth and development
    - Mood
    - Tissue function
    - Metabolism (catabolism and anabolism)
    - Sexual function and reproductive processes
- Endocrine glands are sometimes called ductless glands because they have no ducts connecting them to specific body parts
- Hormones may be classified as simple protein, glycoprotein and steroids but they all have common characteristics and functions:

    (1) Hormones appear to regulate rather than initiate reactions;
    (2) Hormones are effective in biocatalytic amounts;
    (3) Hormones are not secreted in uniform rates;
    (4) Hormones are inactivated rapidly either at the site where it exerts its effect or at some other glands or organs; and
    (5) Hormones are transported through the circulatory system or blood stream.

- Not all hormones have specific target organs, like growth hormone or somatotropin (STH); but for those with specific target organs, the cells in the target organ contain receptors that specifically recognize the hormone.
- Hormone receptors bind specific hormone and directly or indirectly trigger a metabolic effect.

Pituitary gland or Hypophysis

- Located at the base of the brain in a concavity of the sphenoid bone called *sella turcica* which protects it from outside pressure. It has three lobes:
    - Anterior Pituitary Lobe or Adenohypophysis;
    - Intermediate Lobe or Pars Intermedia; and
    - Posterior Pituitary Lobe or Neurohypophysis

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The Adenohypophysis secretes the following hormones:

(1) Growth hormone or Somatotropic hormone (STH) - promotes growth of the long bones before the epiphyseal – diaphyseal plate is fused together in adulthood. Over secretion of STH results in Gigantism when this happens before adulthood and Acromegaly when this happens after adulthood in human. Dwarfism occurs when there is a deficiency of STH during growth development
(2) Adrenocorticotropic Hormone or ACTH – stimulates the adrenal cortex to produce glucocorticoids such as cortisol, cortisone and corticosterone
(3) Thyroid Stimulating Hormone or TSH – stimulates the thyroid gland to produce thyroid hormones (T_4 and T_3)
(4) Prolactin or Luteotropic Hormone (LTH) – stimulates milk secretion in lactating mammary gland
(5) Follicle Stimulating Hormone (FSH) – stimulates the ovary to produce grafian follicle; in the male, it maintains the integrity of the seminiferous tubules of the testis
(6) Luteinizing Hormone (LH) – stimulates ovulation in maturing follicle and the formation of corpus luteum in ovulated follicle as well as the production of progesterone by the corpus luteum. Also called Interstitial Cell Stimulating Hormone (ICSH) in males, it stimulates the cells of Leydig or insterstitial cells to produce testosterone, a male sex hormone

The neurohypophysis secretes two hormones:

(1) Oxytocin – stimulates milk-ejection in lactating females; and
(2) Vasopressin or Antidiuritic Hormone (ADH), which is important in conserving body water by reducing urine formation. It promotes water reabsorption in the kidney tubules

Thyroid gland

- Located in the neck area just below the larynx. Has two lobes connected by a bridge called isthmus
- It is not essential to life but in its absence, there is poor resistance to cold, mental and physical slowing and in children, mental retardation and dwarfism
- Maintains the level of metabolism in the tissues that is optimal for their normal function. It secretes the hormone:
    - Thyroxine (T4) and Triiodothyronine (T3) which increase the metabolic rate of the cells of the body
    - Functions:
        - Stimulates O_2 consumption of most of the cells

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- Helps regulate lipid and carbohydrate metabolism
- Necessary for normal growth and maturation
- Increase the basal metabolic rate of an individual

- Disorders:
  - Thyrotoxicosis – over activity of the gland (high T4 secretion)
  - Hypothyroidism – lack of the T4 Secretion, due to lack of iodine (TSH is stimulated resulting to enlargement of the gland)

Parathyroid gland

- Located behind the thyroid gland
- Two distinct types of cells in parathyroid: 1) chief cells, and 2) oxyphil cells
- Secretes hormone:
  - Parathyroid hormone (PTH)- secreted by chief cells; mobilizes calcium from bone reserves and increases urinary phosphates, thus increasing blood calcium level
  - Oxyphil cells – less abundant cells with large amounts of mitochondria; function unknown
  - Calcitonin or Thyrocalcitonin = the hormone that lowers calcium level in the blood (opposite of PTH) by increasing urinary calcium excretion and preventing bone resorption through activation of osteoblasts which stimulate bone formation, secreted by thyroid gland upon stimulation by a secretion coming from the parathyroid gland in response to high blood calcium level
  - Disorders:
    - Rickets (in adult)/Osteomalacia (in young) = characterized by a failure of normal mineralization of bone in young or adults; under-mineralization of cartilage and bone results in retardation of growth and development of skeletal deformities
    - Hyperparathyroidism = hypersecretion of PTH, resulting to hypercalcemia, hypophosphatemia, hypercalciuria, and hypophosphaturia
    - Hypoparathyroidism osteoclerosis = may occur due to increased amount of calcified bone, symptoms include hypocalcemia, hyperphosphatemia, hypercaliurea and hypophosphaturia

Adrenal gland

- There are two endocrine organs 1) Adrenal cortex and 2) Adrenal medulla
- Outer adrenal cortex secretes steroid hormones
- Three types of cell making up three zones:
  1) Zona glomerulosa – secrete Aldosterone

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2) Zona fasciculata – secrete Glucocorticoids
3) Zona reticularis – secrete Glucocorticoids

- Aldosterone – regulates sodium metabolism by reabsorbing sodium from the kidney tubules; secretion is stimulated by extra cellular fluid volume (ECF) and/or blood pressure
- Glucocorticoids- include Cortisol, Cortisone and Corticosterone; important for carbohydrate metabolism, mobilizes glucose via glycogenolysis and gluconeogenesis; secretion is controlled primarily by ACTH from anterior pituitary
- Inner adrenal medulla secrete 1) Epinephrine = Adrenaline and 2) Norepinephrine = Noradrenaline. These secretions are not essential for life but help cope for emergencies

Pancreas

- Both an exocrine and endocrine gland. Located at the duodenal loop of the small intestine
- It functions as exocrine gland when its acinar cells secrete pancreatic juice containing digestive enzymes.
- The endocrine function is limited to the cells of the islets of Langerhans which are found throughout the pancreas.
- The *alpha cells* of the islets of Langerhans secrete *glucagon* which is responsible for increasing blood sugar level; and the *beta cells* secrete *insulin* which is responsible for lowering blood glucose level by facilitating entry into target cells, thereby increasing its utilization and to the adipose cells (stored fats)
- Types of diabetes
    - Type I – Insufficient insulin secretion
    - Type II – Tissues resistance to insulin action
- Somatostatin – third hormone, secreted by *delta cells*. Regulate islet cell secretion.

Pineal Gland

- Also called epiphysis cerebri
- Located at the epithalamus near the center of the brain
- Source of melatonin – hormone that plays a major role in the regulation of daily and seasonal rhythms for many vertebrates; associated with different behavioral and physiological responses in animals
- Examples: seasonal breeding for livestock, effect of daylength in stimulating egg production

III. CARDIOVASCULAR SYSTEM

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- Includes the heart, blood and blood vessels through which blood flows in circulation
- Functions:
    - Conveys nutrients absorbed from the digestive tract to tissues
    - Carries O_2 from lungs to tissues and CO_2 from tissues to lungs
    - Removes waste product of metabolism and take them to excretory organs for disposal
    - Transports hormones from one part of the body to another
    - Helps in maintaining water equilibrium in the body
    - Regulates hydrogen ion concentration in the body
    - Increases the flow of blood to meet increased energy demands during exercise and regulates body temperature
    - Conveys disease-fighting elements of immune system, such as white blood cells and antibodies
    - Sends clotting cells and proteins to the affected site, which quickly stop bleeding and promote healing

- Heart - located in the middle mediastinal space
    - A muscular organ located at the chest (thoracic) cavity and enclosed in a fibrous sac, the *pericardium*. The walls of the heart are composed primarily of cardiac muscle, the *myocardium*
    - Divided into four chambers: upper 2 chambers are the left and right atrium, and the lower 2 chambers are the left and right ventricle
    - Atrio-ventricular valve (A-V valve) – prevents backflow of blood from the ventricle to the atrium during ventricular systole; valve on the right side is the tricuspid valve and the one on the left side is the bicuspid (or mitral) valve
    - Aortic valve (valve at the aortic orifice) and Pulmonary valve (valve at the pulmonary orifice) – prevent backflow of blood from blood vessels (aorta and pulmonary artery) into the ventricles during diastole
- Blood
    - Thick suspension of cellular elements in an aqueous solution of electrolytes and some non-electrolytes
    - Separated into two categories of plasma and cells by centrifugation
    - Plasma
        - the fluid portion of the blood containing a number of ions, inorganic molecules and organic molecules which are in transit to various parts of the body or which aid in the transport of other substances
        - Contains water, gases, proteins, non-protein nitrogen substances, inorganic salts and minerals, enzymes, hormones, vitamins, immune substances, etc.
        - Normal plasma volume is about 3-5% of BW

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- Blood cells – suspended in the plasma
  - White Blood Cells (Leukocytes) - act to defend body against infection; phagocytic, produce antibodies and counteract toxins
    - Three types
      1) Granulocytes
         - Neutrophils – affinity to neutral dyes
         - Eosinophils – affinity to acidic dyes
         - Basophils – affinity to basic dyes
      2) Lymphocytes
      3) Monocytes
  - Red Blood Cells (Erythrocytes) - are biconcave disks manufactured in the bone marrow; these non-nucleated cells are soft and contain *hemoglobin*, a complex iron-containing conjugated protein
    - Hemoglobin binds O_2 to form oxyhemoglobin by attaching to Fe^{2+} in the heme; reacts with 4 molecules of O_2 to form Hb_4O8 (reaction is called oxygenation)
    - In muscles, O_2 is taken up by myohemoglobin
  - Platelets (Thrombocytes) – small oval disk-like granulated bodies; when blood vessels walls are injured, platelets collect at the site, sticking to the vessel wall and liberating serotonin, which leads to local vasoconstriction. They also liberate *thromboplastin* which aids in blood clotting
- Blood coagulation – conversion of soluble plasma protein fibrinogen to insoluble protein fibrin, a reaction catalyzed by enzyme thrombin.
  - Thrombin is formed from its inactive form called prothrombin in the presence of calcium ions by the action of activated thromboplastin
  - Prothrombin is synthesized in the liver with vitamin K involved in the hepatic synthesis

- Blood Vessel
  - Artery - blood vessel that carries blood away from the heart
  - Vein - blood vessel that carries blood back to the heart
  - Artery - blood vessel that carries oxygenated blood (exception: pulmonary artery)
  - Vein - blood vessel that carries unoxgenated blood (exception: pulmonary vein)
- Pulse rate (rate of heartbeat) may be taken by feeling the artery
  - Horse -external maxillary artery or about the middle of the lower jaw
  - Cattle and Carabao – similar location as the horse but slightly in the outer surface; coccygeal artery at the base of the underneath of the tail
  - Sheep and Goat – femoral artery
  - Pigs and others – auscultation method using stethoscope at the cardiac or chest region

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- Average pulse rate per minute
  - Horses – 38
  - Cattle and Carabao – 54
  - Goat – 78
  - Chicken – 200-400
- Systemic Circulation – carries oxygenated blood from the heart to all the tissues in the body except the lungs and returns deoxygenated blood carrying waste products, such as carbon dioxide, back to the heart. Includes the following systems of blood circulation:
    - Coronary circulation – supplies blood to the heart itself
    - Hepatic circulation – supplies arterial blood to the liver
    - Cerebral circulation – supplies arterial blood to the brain
    - Renal circulation – supplies arterial blood to the kidney
    - Splanchnic circulation – supplies arterial blood to the digestive tract
- Pulmonary circulation – carries this spent blood from the heart to the lungs. In the lung, the blood releases its carbon dioxide and absorbs oxygen. The oxygenated blood then returns to the heart before transferring to the systemic circulation

- Lymphatic System - is composed of lymph nodes, lymph vessel and the lymph.
  - Lymph nodes - function for the production of lymphocytes. Another function is to stop foreign materials that come to them in the lymph. They become swollen or inflamed during severe bacterial infections
  - Lymph capillaries - more permeable walls than the blood capillaries, all metabolites of big molecular size which cannot be absorbed by the lymph capillaries. Eventually, the lymph fluid will enter the circulatory system through the right lymphatic duct and the thoracic duct
  - Lymph vessels - contain thin-walled valves which prevent the back flow of its content

IV. RESPIRATORY SYSTEM

- Main function is to provide oxygen to the cells of the body and to remove excess carbon dioxide from them
- Interruption for more than a few minutes can lead to serious, irreversible damage to tissues followed by the failure of all body systems, and ultimately death
- Important functions:
  - Intake of oxygen and removal of carbon dioxide
  - Helps regulate the balance of acid and base in tissues, a process crucial for the normal functioning of cells
  - Protects the body against disease-causing organisms and toxic substances inhaled with air

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- Houses the cells that detect smell
- Makes use of two systems that cooperate to supply the needs of the tissues
    1) a blood circulatory system – carry to and from tissue cells large quantities of O_2 and CO_2 with the help of hemoglobin
    2) a respiratory system (gas exchanger) – load blood with O2 and remove excess CO_2

Respiratory Apparatus

- Air passages leading to the lungs
    - Nasal Cavity – passage of air; line with epithelial cells with microscopic cilia; between epithelial cells are goblets that produce mucus; the role is to filter the air to remove potentially disease-causing organisms
    - Pharynx – a short, funnel-shaped tube about 13cm (5 in) long that transports air to the larynx; lined with a protective mucous membrane and ciliated cells that remove impurities from the air
    - Larynx – a structure about 5cm (2in) long located approximately in the middle of the neck; transport air to the trachea; its mucous membrane and cilia-bearing cells help filter air
    - Trachea, Bronchi, and Bronchioles
        - A tube about 12 to 15cm (about 5 to 6 in) long located just below the larynx
        - Formed of 15 to 20 C-shaped rings of cartilage that keeps it open
        - Trachea branches into two tubes, the left and right bronchi, which deliver air to the left and right lungs, respectively
        - Within the lungs, the bronchi branch into smaller tubes called bronchioles
    - Alveoli – Each bronchiole branches into several subdivisions producing about million terminal tubes, at the end are the alveoli or alveolar sacs; gas exchange occurs at the alveoli
- Lungs
    - Tidal Volume - the amount of air that moves into the lungs with each inspiration or that moves out with each expiration
- Thorax
- Pleura
- Respiratory Muscles
    - Inspiratory muscles
        - Diaphragm
        - External intercostal muscles
    - Expiratory muscles
        - Internal intercostal muscles
        - Muscles of the anterior abdominal wall

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- Respiratory Centers
  - The flow of air in and out of the lungs is controlled by the nervous systems
  - The brain respiratory centers include:
    - Medullary center
    - Pneumotaxic center
    - Apneustic center
- Regulation of respiratory center activity – controlled by sensory receptors particularly chemoreceptors sensitive to chemical changes in the environment such as 1) medullary chemoreceptors, 2) carotid bodies, and 3) aortic bodies

Mechanism of inspiration and expiration

- Inspiration – active process; contraction of diaphragm that increases longitudinal diameter of the chest and contraction of external intercostal muscles elevating the ribs resulting in increased diameter of the thorax
- Intrapleural pressure is about -2.5 mmHg at start of inspiration which then decreases to -6 mmHg when chest volume is increased. Air flows into the lungs as pressure in the airway becomes slightly negative.
- Expiration – enlarged thorax returns to resting position by purely passive forces following inspiration. Pressure in airways become slightly positive and air flows out of lungs.
- Labored breathing is accompanied by active expiration

Regulation of respiration

- Spontaneous respiration is completely dependent upon discharge of respiratory center in the *medulla oblongata*

V. SKELETAL SYSTEM

BONES

- A highly specialized supportive tissue which is characterized by its rigidity and hardness
- Four main functions are to provide mechanical support (e.g. ribs) to permit locomotion (e.g. long bones) to provide protection (e.g. skull) to act as metabolic reservoir of minerals

Composition

1. Cells: the bone is composed of four cell types

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a. Osteogenic Cells
   - Mesenchymal cells
   - Located in the inner layer of the periosteum (fibrous membrane that covers the surface of the bone) and by differentiation, they give rise to osteoblast

b. Osteoblasts
   - These are bone secreting cells
   - During bone formation, they first secrete the amorphous and fibrous (collagen) components of unmineralized bone, and later they deposit their stored calcium to mineralize or harden the bone
   - They line the trabecular (bone material) surfaces of the bone

c. Osteocytes
   - Mature bone cells that stop secreting the bone substance
   - They are actually osteoblasts which have been trapped in spaces called lacunae in the bone substance during mineralization
   - Their main function is to maintain the integrity of the bone by supplying nutrient derived from the blood vessels in the canaliculi

d. Osteoclasts
   - Believed to rise from the fusing of osteocytes
   - Remodeling cells which are multinucleated in the structure and located in a concavity called the Howship's lacuna at the bone surface

2. Osteoid: a non-mineral matrix of collagen and glycosaminoglycan
3. Inorganic mineral salts deposited within the matrix

Types of Bones

- Bone is classified into two types based on the frequency of appearance of interosseous spaces
  - Interosseous space - is the space between adjacent trabeculae
  - Trabeculae - a portion of the bone substance which is also called the bone spicule

1. Cancellous Bone (Spongy Bone) - consisting of few trabeculae and many of the interosseous spaces
2. Compact Bone (Lamellar Bone) - consists of extensive lamellae and fewer interosseous spaces. This bone develops into existing trabeculae to form layers called lamellae

Osteon or Haversian System

- Basic unit structure of a compact bone
- Circular structure composed of a central canal called the Haversian canal surrounded by concentric layers (lamellae) of bone substance

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- Embedded in the lamellae are the bone cuniculi containing the cytoplasmic process of the osteocytes
- The Haversian canal connects a transversely positioned Volkmann's canal

Osteogenesis

- The process by which bone is formed through the secretory activity of the osteoblasts

- In embryo, it occurs in two ways:
  4. Intramembranous Ossification: bone develops in pre-existing mesenchymal connective tissue (bones of this type are called membrane bones)
  5. Endochondral Ossification: bones that developed from pre-existing cartilage (e.g. long bones like femur and humerus). This type of bone growth is responsible for the longitudinal growth of bones

Epiphyseal Plate (Epiphyseal Line)

- Site of bone growth as seen in long bones
- Composed of different zones
  a. Zone of growth
  b. Zone of cartilage transformation
  c. Zone of ossification

CLASSIFICATION OF BONE BASED ON GROSS APPEARANCE

1. Long bones - great dimension
   - e.g., Forelimb = humerus, radius, ulna, metacarpals, phalanges
     Hind limb = femur, tibia, fibula, metatarsals, phalanges

   Functions:
   a. act as levers
   b. for support, locomotion

2. Short Bones - cuboids or equal shape
   - e.g. carpals, tarsals

   Function: absorbs concussion

3. Flat Bones - thin and expanded
   - e.g. skull, ribs, scapula, pelvic bones

   Functions:
   a. protect vital organs e.g. brain, lungs
   b. provide areas for muscle attachment

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4. Sesamoid Bones - sesame-seed shape develops along the course of tendons
   - e.g. patella
   Functions:
   a. reduce friction
   b. change the course of tendons

5. Irregular Bones - unpaired bones in median plane
   - e.g. vertebral column, some bones of the skull
   Functions:
   a. for protection and support
   b. for muscle attachment

6. Pneumatic Bones - presence of air spaces or sinuses that communicate with the exterior
   - e.g. flying birds, maxillary and frontal bones

Axial Skeleton - includes all bones except those of the limbs or appendages composed of the skull, vertebrae and ribs

Appendicular Skeleton - made up of bones of the limbs bones of the front (pectoral) limbs and bones of the hind (pelvic) limbs

VI. DIGESTIVE SYSTEM

Simple Stomach (Monogastric animals)

- Mouth
- Esophagus
- Stomach
- Duodenum
- Jejunum
- Ileum
- Cecum
- Colon
- Rectum
- Vent

Modified simple stomach (Poultry Species)

- Beak
- Esophagus
- Crop
- Proventriculus
- Gizzard

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- Duodenum
- Jejunum
- Ileum
- Ceca
- Colon
- Rectum
- Anus/Vent

Herbivorous monogastric (horse/rabbits)

- Mouth
- Esophagus
- Stomach
- Duodenum
- Jejunum
- Ileum
- Cecum (large and functional)

Ruminant animals

- Mouth
- Reticulum
- Rumen
- Omasum
- Abomasum
- Small intestine
- Large intestine

Accessory glands - not part of the digestive system but plays important roles in digestion process

- Salivary glands
- Pancreas
- Liver
- Gall bladder

Digestion in Monogastric Animals

- Ingestion of food comes salivation
- Chewing reduces the particle size and increases surface area for enzyme action
- Swallowing
- Stomach secretes HCl, mucus, digestive enzyme, pepsin and gastrin (pepsin= protein to polypeptide; gastrin=hormone that stimulates secretion of HCl)
- Chyme passes through the duodenum (pancreas secretes, lipases, trypsin and chymotrypsin and other enzymes; duodenal cells secrete secretin, pancreozymin, CCK)
- Fibrous materials pass and fermented at cecum

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Digestion in Ruminant Animals

- Predigestive fermentation occurs in the rumen and reticulum
- Bacteria and protozoa use the roughages consumed by animals as nutrient sources for their growth and multiplication
- The rumen environment is ideal for microorganism growth because it is anaerobic, pH 6.9-7 and warm (39-41°C)
- Excess microbes are continuously removed from rumen and reticulum along with the small feed particles through the omasum and abomasum
- Abomasum is acidic and functions similar to monogastric stomach, the small intestines then digest the microbes as source of nutrients. Pancreas secretes enzymes into the duodenum similar to what is happening in monogastric animals
- The rumen fermentation process produces VFA (acetic, propionic and butyric acids). The ruminants use this as major source of energy
- In the process of fermentation, the animal releases methane gas, ruminants release this gas by belching. When the gas releasing mechanism fails, the gas accumulates in the rumen and causes bloating
- A young, suckling calf consumes little or no roughages. Largest compartment of a young ruminant stomach is abomasum
- Milk is directed immediately into the abomasum in the young ruminants by esophageal groove, allowing milk to bypass fermentation in the rumen
- Complete development of the rumen, reticulum and abomasum requires about 2 months in sheep and goat and 3 to 4 months in cattle. The development is influenced by the type of feed the animal consumes

VII. EXCRETORY SYSTEM

- Kidney - the main organ of excretion; the kidneys are commonly described as excretory organs, but actually, they are organs which primarily regulate volume and composition of the internal fluid environment. Their excretory function is incidental to their regulatory function. each kidney is composed of over a million units of nephrons
- Kidney aids in keeping blood plasma composition constant by:
    - Excretion of urea and other nitrogenous waste products of metabolism
    - Excretion of excess inorganic salts
    - Excretion of excess water
    - Excretion of non-volatile, soluble foreign substances that are in blood
- Structure of the kidney
    - Paired bean-shaped organ against the rear wall of the abdomen, on other side of the spine

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    - Supplied with blood by renal artery, branches on the pelvis, pass between *calyxes* and penetrate *parenchyma*
    - Outermost layer of the kidney is the *cortex*
    - Beneath the cortex lies the *medulla*, an area that contains between 8 and 18 cone-shaped sections known as pyramids, which are formed almost entirely of bundles of microscopic tubules
    - The center of the kidney is a cavity called the *renal pelvis*
    - Approximately one million *nephrons* compose each bean-shaped kidney
- A nephron consists of several parts:
    - Glomerulus – the filtration unit of the nephron that regulates the concentration within the body of important substances, and removes substances not needed by the body such as drugs and food additives
    - Bowman's capsule and glomerulus are collectively called as renal corpuscle or malphigian
    - Proximal tubule – is joined to Bowman's capsule by a short connecting segment.
    - Loop of Henle (descending limb and ascending limb) – becomes extremely narrow, extending down away from Bowman's capsule and then back up again in a U shape
    - Distal convoluted tubule
    - Collecting ducts
- The renal pelvis drains into the ureter and the latter enters the urinary bladder.

Urine Production

- Begins with the process of filtration that goes on at renal corpuscles or nephron
- Glomerular filtration = as blood courses through the glomeruli, much of its fluid, containing both useful chemicals and dissolved waste materials, soaks out of the blood through the membranes (by osmosis and diffusion) where it is filtered and then flows into the Bowman's capsule
- Urea is formed in the body to eliminate the very toxic ammonia formed in the liver
- The total glomerular filtration rate of both kidneys is 125ml/min
    - 125 ml/min x 60 min/hr = 7.5L
    - 7.5L/hr x 24/hrs/day = 180L.day
- Re-absorption – the movement out of the renal tubules back into the blood capillaries (peritubular capillaries)
    - Reabsorption starts at the proximal convoluted tubules and continues in the loop of Henle, distal convoluted tubules, and collecting tubules
    - Descending loop of Henle = permeable to water

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- Ascending loop of Henle = salts are extruded
- In water diuresis, blood ADH is low, the epithelium of the distal tubules and collecting ducts are impermeable to water

- Secretion – is the process by which substances move into the distal and collecting tubules from blood in the capillaries around these tubules. In short, secretion is the reverse of absorption
  - Re-absorption moves substances out of the tubules and into the blood
  - Secretion moves substances out of the blood and into the tubules where they mix with the water and other wastes and are converted into urine

Vasopressin or ADH

- Conserves body water by:
  - ADH is a vasoconstrictor which reduces medullary and papillary blood flow in the renal interstitium, thus increasing hypertonicity of the interstitium; reabsorption of water by osmosis would increase
  - ADH stimulate sodium pump in ascending loop of Henle - ↑ rate of sodium transport from the lumen of the tubule to the interstitium and concentration gradient of sodium between interstitium and fluid in the lumen of the tubules
  - ADH may dilate the "pres" of the collecting duct – facilitates water reabsorption

VIII. REPRODUCTIVE SYSTEM

Forms of Reproduction

- Sexual – male and female sex organs are involved in the process of propagation; union of sex cells (ovum and sperm) needed to form individual
- Asexual – does not require the sex organ

  Cloning
  - Dolly from mammary cells (Roslin Institute, Edinburg)
  - Mouse from a mouse tail (University of Hawaii)
  - Cow from reproductive tract (cumulus cells) (Kinki University, Nara, Japan)

Female reproductive system

- Ovaries-primary sex organs
  - *Functions*: (a) the production of sex cell or ovum and (b) production of female sex hormones – estrogen and progesterone
- Accessory reproductive tract includes

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- Infundibulum – funnel-shaped structure which picks up the egg when released by the ovary
- Oviduct – tubular structure connecting the infundibulum to the horn of the uterus; serves as the passageway of the egg on its way to the uterus
- Uterus - site of fertilization
- Horn of uterus - site of implantation for the fertilized egg
- Cervix – neck of the uterus; serves as the sperm receptacle in certain animals. Its opening called os uteri closes when the animal gets pregnant to protect the uterine contents
- Vagina - primary organ for copulation; serves as sperm receptacle in many species
- Vulva - common passageway for the products of reproduction and for urine; comparable to the ventral portion of the cloaca of birds and homologous with the scrotum of the male

Male reproductive system

- Testis - main sex organ
- Scrotum - serves as the external covering of the testes
- Birds – two testes are located within the body cavity in the dorsolumbar region
- Farm animals – testes are located outside the body cavity within the scrotum
- Testes - have 2 main functions: 1) the production of sperm cells and 2) production of male sex hormone - testosterone
- Testosterone – responsible for development of secondary sex characters (e.g. muscular development, aggressiveness, libido)
- Seminiferous tubules – site of spermatozoa production in the testis
- Penis is the male organ of copulation and serves to introduce spermatozoa into the female reproductive tract
- Three accessory glands that contribute to bulk of the semen
    1) Seminal vesicles or vesicular glands
    2) Prostate gland
    3) Cowper's glands or bulbo-urethral gland
- Semen – consists of sperm cells plus secretions of the three accessory glands
- Secretions serve as vehicle for transport of spermatozoa from vagina to the oviduct, stimulates spermatozoa activity, serve as lubricating substance during copulation
- Life span of ejaculated spermatozoa in the female reproductive tract
    - Most mammals – about 24 hrs (20-30 hrs)
    - Chickens – about 14 days

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- Both Follicle stimulating hormone (FSH) and Luteinizing hormone (LH) are required for normal spermatogenesis
    - FSH – stimulates germinal epithelium lining the seminiferous tubules to initiate spermatogenesis
    - LH) – stimulates interstitial cells or Leydig cells to secrete testosterone

Puberty and Estrous cycle

- Puberty is defined as the first estrus accompanied by ovulation in animals
- Indicates that the female has reached sexual maturity-capable of producing offsprings.
- The age of puberty varies between breeds of animals and among female animal with same breed

Species	Male	Female
Bovine	11 mo. (7-19)	11 mo. (9-24)
Ovine	7 mo. (6-9)	7 mo. (4-14)
Porcine	7 mo. (5-8)	6 mo. (5-7)
Equine	14 mo. (10-24)	18 mo. (12-19)

- Estrous Cycle
    - The period from estrus to the next estrus
    - In many farm animals such as carabao, cattle, pigs, and horses, estrus comes every 21 days if the female animal is cycling regularly, although it could vary from 12 to 30 days
    - Animals may be classified based on the occurrence of their estrus cycle as:
      (1) Monoestrus - the animal comes in heat only once a year, such as dogs/bitch
      (2) Seasonally Polyestrus - if it comes in heat at certain seasons only, such as sheep
      (3) Polyestrus - if it comes in heat all throughout the year, like cattle, swine and carabao
    - An estrous cycle may be divided into four portions:
      (1) Proestrus - which is characterized by follicular growth
      (2) Estrus - which is under the influence of estrogen
      (3) Metestrus - characterized by the formation of CL
      (4) Diestrus - which is under the influence of progesterone secreted by the CL
    - Estrogen
        - Hormone which causes estrus in females

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- Production is greatest when developing Graafian follicle (GF) in the reaches largest size; growth and development of GF in the ovary is caused by FSH secreted by the anterior pituitary glands
- Presence of estrogen stimulates LH production
- At peak of estrogen production, LH increases which coincides with the production of inhibin in the ovary (hormone that inhibits FSH production)
- Causes behavioral changes (Signs of Heat)
    - Stands for mounting other animals
    - The female stands still when mounted by the male
    - Reddening and swelling of the vulva
    - Mucus discharged from vagina
    - Nervousness and restlessness
    - Frequent urination
    - Increase in body temperature
    - Loss of appetite
    - Bellowing

- LH is the hormone that causes ovulation of GF; initiates formation of corpus luteum (CL)
- Summary of sequence of events in an estrous cycle
    1) Anterior pituitary secretes FSH which causes development og GF in the ovary
    2) GF secretes estrogen which stimulates LH production and at the same time causes heat or estrus in the female animal (best time to breed)
    3) Surge of LH released by anterior pituitary causes ovulation
    4) LH will also initiate development of CL and causes production of progesterone by CL
    5) Progesterone will prepare uterus for implantation of fertilized egg, if fertilization has taken place
    6) Progesterone maintains pregnancy if animal is pregnant
    7) The uterus will produce prostaglandin that causes destruction of the CL resulting in the cessation of the production of progesterone
    8) When this happens, FSH production would again continue and a new cycle begins

When to best breed or inseminate

Cattle and carabao

- Ovulation takes place 15-18 hrs from the end of estrus; best time is toward the end of estrus
- Duration of estrus is variable – 5-36 hrs, average is 18 hrs

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- It is recommended to breed (or inseminate) immediately once the animal is in standing heat

Swine

- Duration of estrus – 2-3 days
- Best time to breed (or inseminate) is on the second or third day of estrus

Mares

- Duration of estrus – 6-7 days
- Best time to breed (or inseminate) is on the fourth or fifth day of estrus

Fertilization and Pregnancy

- Fertilization means the sperm joins the egg, creating the zygote which is the first cell of the new individual
- Require three critical events
    a) sperm migration from site of deposition to oviduct and convergence of spermatozoa towards the ovum
    b) sperm attachment and penetration through the zona pellucida
    c) Fusion of sperm and ovum plasma membranes
- In polytocous species such as pigs, embryos undergo uterine migration and equidistant spacing prior to implantation. Pig blastocysts begin to attach to the uterine surface on day 13 and completed between days 18 to 24
- In ruminants, initial placentation occurs at day 22 in cattle and day 15 in sheep. Placental attachment is through caruncles or cotyledons, which are finger-like villi or papillae that allow nutrient intake by the developing conceptus.
- Gestation period (days)
    - Cattle – 278-290
    - Horse – 335-345
    - Sheep – 145-155
    - Swine – 112-115

Placental development

- Two general types of placentation

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- Sows and mare – have diffuse placenta, which consists of a simple apposition of fetal and maternal epithelia
- Sheep, goat, cows and carabao – have cotyledonary placenta, cotyledons from fetal placenta are attached to the caruncles of the maternal placenta through which uterine blood flows

Parturition and lactation

- Parturition (or labor) is defined as the physiological process by which the pregnant uterus delivers the fetus and placenta from the maternal organism
- Signs of approaching parturition
    - Obvious enlargement of mammary glands
    - Teats become swollen
    - Presence of milk in the mammary gland
- Triggered by the fetus and is completed by complex interaction of endocrine, neural and mechanical factors
- Parturition is divided into three stages
    1) Dilation of the cervix (1st stage)
    2) Expulsion of the fetus (2nd stage)
    3) Expulsion of the placenta (3rd stage)
- Onset of milk secretion follows parturition
- Prolactin initiates milk secretion in the mammary gland

IX. BODY TEMPERATURE REGULATION

Classification of animals based in their ability to regulate body temperature

1. Homeotherms (warm-blooded animals) – body temperature is largely independent of environment
2. Poikilotherms (cold-blooded animals) – body temperature directly varies with environment

Homeotherms attain body temperature constancy by

1. Chemical regulation
    - Altering metabolic rate by adjusting production of hormones such as thyroxin and/or epinephrine
    - Shivering and changing muscle tension
2. Physical devices and activities
    - Sweating with evaporation of water from the skin
    - Changing rate of breathing to alter vaporization of lungs
    - Altering volume of blood flowing to the body surface by vasodilation or vasoconstriction

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- Changing amount of internal or external thermal insulation such as subcutaneous fat, feathers, fur or hair
- Changing "free" body surface area by lying down and huddling when it is cold or by standing in stretched position when it is hot
- Locating more favorable environment such as shade, dry or wet areas, shelter from wind, etc.

Normal rectal temperature range (Body temperature)

- Horse, ruminants – 36-39°C
- Pigs, rabbit – 38-40°C
- Chicken, turkey, goose, duck – 40-41°C
- Quail – 41-43°C

Different processes of heat dissipation

1) Conduction
2) Convection
3) Radiation
4) Vaporization

Thermal Neutrality

- Thermoneutral zone, environmental temperature at which the heat loss is equal to minimal heat production (Basal Metabolic Rate/metabolism at rest)
- At this point, body temperature is normal without much regulation

Heat Stress

- As environmental temperature rises, body temperature is maintained until the upper critical temperature (UCT)
- UCT depends on species, ranges from +3 to +6°C of thermoneutral zone
- Beyond UCT, animal is unable to maintain its body temperature even with thermoregulating devices
- "Critical" body temperature refers to body temperature at which 50% of species dies

Cold Stress

- As environmental temperature decreases, thermogenesis begins to increase until it reaches lower critical temperature (LCT) to balance thermolysis
- Beyond LCT, animal is unable to maintain its normal body temperature, hence body temperature goes down correspondingly as ambient temperature declines
- LCT is -20°C of the normal body temperature of the animal

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Unit II
ANIMAL NUTRITION

Definition of Terms

Nutrition - is a science that deals with the series of processes by which an organism takes in and assimilates food for promoting growth and replacing damaged or injured tissues; it encompasses several fields of discipline

Nutrients - any feed constituent or a group of feed constituents that aids in the support of life

Digestion - process of breaking down of feed particles into suitable products for absorption

Absorption - transfer of substance from gastro-intestinal tract (GIT) to the circulatory system

Metabolism - combination of anabolic and catabolic reactions occurring in the body with the liberation of energy

- Catabolism - also known as *destructive metabolism;* and is the process that produces the energy required for all activity in the cells; i.e. cells break down large molecules (mostly carbohydrates and fats) to release energy

- Anabolism - also called *constructive metabolism*; is all about building and storing; small molecules are changed into larger, more complex molecules of carbohydrate, protein and lipids, etc.

Enzyme - a complex protein produced in living cells that causes changes in other substances within the body without being changed itself (*organic catalyst*)

Classes of Nutrients

Water

1. Cheapest and most abundant nutrient.
2. Makes up 65 to 85% of animal body weight at birth and 45 to 60% of body weight at maturity.

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3. Percentage of body water decreases with animal age and has an inverse relationship with body fat.
4. Found in the animal body as:
    a. Intracellular water – mainly muscle and skin
    b. Extracellular water – mainly interstitial fluids, blood plasma, lymph, synovial and cerebrospinal fluid
    c. Water present in urinary and gastrointestinal tract

Functions and Deficiencies of Water in the Animal

1. Functions
    a. Transportation of nutrients and excretions
    b. Chemical reactions and solvent properties
    c. Body temperature regulation
    d. Maintains shape of body cells
    e. Lubricates and cushion joints and organs in the body cavity
2. Deficiencies or restrictions
    a. Reduced feed intake and reduced productivity
    b. Weight loss due to dehydration
    c. Increased excretion of N and electrolytes such as Na and K

Sources of Water to the Animal

1. Drinking water
    a. Factors that affect drinking water consumption
        1) Environmental temperature and humidity (increased water consumption at heat stress)
        2) Dry matter consumption (water consumption directly correlated at moderate temperatures)
        3) Dietary factors
            a) High water content of feed reduces drinking
            b) High fiber, salt or protein content of the diet increases drinking.
    b. Function of the animal (lactating sows vs. dry sows)
    c. Type of urinary system (mammal vs. avian)
    d. Water quality
        1) Good water should have less than 2500 mg/L of total dissolved solids.
        2) Water containing over 1 g/L of sulfates may cause diarrhea.
        3) Levels above 100-200 ppm of nitrates are potentially toxic.

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2. Water contained in or on the feed
3. Metabolic water
    a. Results from oxidation of organic nutrients in the tissues.
    b. May account for 5 to 10% of total water intake.

Water Losses from the Animal Body

1. Urine
2. Feces
3. Vaporization from the lungs and dissipation through the skin (insensible losses).
4. Sweat from the sweat glands.

Carbohydrates

General

1. Includes sugar, starch, cellulose, and gums.
2. Very little occurs as such in animal body.
3. CHO serve as structural components in the plant.
4. CHO make up approximately ¾ of plant dry weight and thus form the largest part of the animal's food supply.

Classification (by Number of Sugar Molecules)

1. Monosaccharides (1 sugar molecule)
    a. Hexoses (6 C)
        1) Glucose
        2) Fructose
        3) Galactose
        4) Mannose
    b. Pentoses (5 C)
        1) Arabinose
        2) Xylose
        3) Ribose
2. Disaccharides (2 sugar molecules)
    a. Sucrose (glucose + fructose)
    b. Maltose (glucose + glucose)
    c. Lactose (galactose + glucose)

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d. Cellobiose (glucose + glucose)

3. Polysaccharides (many sugar molecules)
    1. Starch – α-linkage of glucose
        1) Amylose – unbranched-chain plant starch
        2) Amylopectin – branched-chain plant starch.
        3) Glycogen – branched-chain animal starch.
    b. Cellulose – β-linkage of glucose
    c. Mixed polysaccharides
        1) Hemicellulose – mixture of hexoses and pentoses
        2) Pectins – hexoses and pentoses mixed with salts of complex acids.
        3) Gums - mixture of hexoses and pentoses
    d. Lignin
        1) Not considered a true CHO as the proportions of C, H. and O are different from the other polysaccharides.
        2) Essentially indigestible by all animals and may reduce the digestibility of other nutrients.

Function

1. Source of energy
2. Source of heat
3. Building blocks for other nutrients
4. Stored in animal body by converting to fats.

Fats / Lipids

General

1. Insoluble in water but soluble in organic solvents.
2. One gram of typical fat yields about 9.45 kcal gross energy (heat) when completely combusted, compared with about 4.2 kcal for a typical CHO; thus, fats produce 2.25 times more energy than CHO.

Structure of Fat

1. Composed of fatty acids attached to glycerol by an ester linkage. It may have a mono-, di-, or triglyceride, which is an ester linkage of a fatty acid with 1, 2, or 3 hydroxyl groups of glycerol, respectively.

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2. Fatty acids consist of chains of C atoms ranging from 2 to 24 or more Cs in length and characterized by a carboxyl group on the end.
    a. Saturated fatty acids are those in which each C atom in the chain (except the carboxyl group) has 2 H atoms attached to it (3 Hs at the terminal C).
    b. Unsaturated fatty acids have one or more pairs of Ca atoms in their chain attached by a double bond, and H has been removed.

Functions

1. Dietary energy supply
2. Source of heat, insulation and protection for the animal body
3. Source of essential fatty acids
    a. Linoleic acid (C18:2) and linolenic acid (C18:3) apparently cannot be synthesized by animal tissues, or at least not in sufficient amounts to prevent pathological changes, and so must be supplied in the diet.
    b. Arachidonic acid (C20:4) can be synthesized from linoleic acid, and therefore is required in the diet only if linoleic is absent.
4. Serve as a carrier for absorption of fat-soluble vitamins.

Natural Sources of Fat

a. Most feeds contain low levels of fat, generally less than 10% (cereal grains, animal products, etc.). Unprocessed oilseeds may contain up to 20% fat (soybean, cottonseed, sunflower, etc.).

Proteins

General

1. Protein is the principal constituent of the organs and soft structures of the animal body.
2. All proteins have one common property: their basic structure is made up of simple units, amino acids. The essential component is an α-amino group (NH2) on the C atom adjacent to the carboxyl group (-COOH). The R represents the remainder of the molecule attached to the C atom associated with the α-amino group of the amino acid.
3. Twenty-two amino acids are commonly found in proteins: they are linked together by peptide bonds that couple the α-carboxyl group of one amino acid residue to the α-amino group of another residue.

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Functions

1. Basic structural unit of the animal body
    a. Collagen
    b. Elastin
    c. Contractile proteins
    d. Keratin proteins
    e. Blood proteins
2. Body metabolism
    a. Enzymes
        1) Digestion process
        2) Degradative process
        3) Synthesis process
    b. Hormones
    c. Immune antibodies
    d. Hereditary transmission
3. Source of energy after deamination

Deficiencies

1. Symptoms of protein deficiency include reduced growth rate and feed efficiency, anorexia, reduced N balance and serum protein concentration, anemia, fatty liver, infertility, reduced birth weight of newborn, reduced milk production, and reduced synthesis of certain hormones and enzymes.
2. Amino acid deficiency – a deficiency of an individual amino acid results in the animal body deaminating the other amino acids and using the C chains for energy. Symptoms of a deficiency may resemble those described for protein deficiency with certain amino acid deficiencies producing specific lesions.

Protein Terminology

1. True protein – protein composed only of amino acids.
2. Nonprotein nitrogen (NPN) – compounds that are not protein in nature but contain N and can be converted to protein by bacterial action (example: urea).
3. Crude protein (or total protein) – protein composed of true protein and any other nitrogenous product.

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Crude protein = %N x 6.25

4. Digestible protein – the portion of CP which the animal can digest
5. Indispensable (or Essential) amino acids – those amino acids that are essential to the animal and must be supplied in the diet because the animal body cannot synthesize them fast enough to meet its requirement. The following are considered indispensable AA for normal growth.

Phenyalanine*	Tryptophan	Histidine	(for poultry)
Valine	Isoleucine	Arginine	Glycine
Threonine	Methionine**	Leucine	Proline
		Lysine	

*About ⅓ to ½ of the requirement can be met (or spared) by tyrosine.

**About ½ of the requirement can be met (or spared) by cystine.

6. Dispensable (or Non-essential) amino acids – those amino acids that are essential to the animal but are normally synthesized or sufficient in the diet and need not be supplemented

Alanine	Cystine	Hydroxyproline
Asparagine	Glutamic acid	Proline
Aspartic acid	Glutamine	Serine
Cysteine	Glycine	Tyrosine

Vitamins

General

1. Organic components of natural food but distinct from CHO, fat, protein, and water.
2. Present in food in minute amounts and effective in the animal body in small amounts.
3. Essential for development of normal tissue; necessary for metabolic activity but do not enter into structural portion of the body.
4. When absent from the diet or not properly absorbed or utilized, results in a specific deficiency disease or syndrome.
5. Cannot be synthesized by the animal and therefore must be obtained from the diet (or microbial synthesis in the digestive tract).
6. General comparison of fat- vs. water-soluble vitamins:

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a. Chemical composition – fat soluble vitamins contain only C, H, and O, whereas the water-soluble vitamins contain these 3 elements plus either N, S, or Co.
b. Occurrence – fat-soluble vitamins differ from the water-soluble B vitamins in that they may occur in plant tissues in the form of provitamin (a precursor of vitamin), which can be converted into a vitamin in the animal's body. No provitamins are known for any water-soluble vitamins.
c. Physiological action – water-soluble B vitamins almost collectively are concerned with the transfer of energy. Fat-soluble vitamins are required for the regulation of the metabolism of the structural units.
d. Absorption - fat-soluble vitamins are absorbed from the intestinal tract in the presence of fat. The absorption of water-soluble vitamins is a simpler process because water from the intestine is constantly resorbed into the bloodstream.
e. Storage – any of the fat-soluble vitamins may be stored whenever fat is deposited; the storage increases with intake. The water-soluble B vitamins are not stored in the same way or to the same extent.
f. Excretion – fat-soluble vitamins are excreted wholly in the feces. Water-soluble B vitamins may be present in the feces, but their chief pathway of excretion following metabolic use is through the urine.

Classification

1. Fat-soluble vitamins
   a. Vitamin A
   b. Vitamin D
   c. Vitamin E
   d. Vitamin K
2. Water soluble vitamins
   a. Vitamin B_1 (Thiamin)
   b. Vitamin B_2 (Riboflavin)
   c. Vitamin B_6 (Pyridoxine)
   d. Vitamin B_{12} (Cobalamin)
   e. Niacin
   f. Folic Acid
   g. Pantothenic acid
   h. Biotin

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i. Choline
j. Vitamin C

Functions and Deficiencies

1. Since vitamins play various roles as regulators of metabolism, they are necessary for growth and maintenance of life. Therefore, proper vitamin levels in the diet are not only important from the standpoint of preventing specific deficiency symptoms, but also to promote general health, vigor and the ability to combat stress and disease. For example, most vitamins are apparently involved in antibody synthesis whereby animals acquire immunity to specific infections.
2. Vitamin requirements may also increase in old age due to difficulties in absorption and utilization. Today, suboptimal vitamin levels in the rations fed to livestock that cause undetected reductions in performance are probably of greater importance than gross deficiencies with the resulting typical deficiency symptom. On the other hand, vitamin supplementation should not be used as a "cure-all" treatment.

Natural Sources

1. A – feeds rich in carotene, yellow corn, whole milk, fish oils
2. D – Sunlight
3. E – seed germ or germ oils from plants
4. K – fish meal, menadione (synthetic Vitamin K3)
5. Riboflavin – milk or milk products, meat or fish meal, distillers or brewer's by-products
6. Pantothenic acid – brewer's yeast, fish solubles, most feedstuffs fairly good sources
7. Niacin – animal by-products, brewer's yeast, some present in ingredients but is largely unavailable in grains in non-ruminants
8. Pyridoxine – most feedstuffs fairly good sources, cereal grains and by-products, rice and rice bran, yeast
9. Biotin – widely distributed, esp. rich in egg yolk, liver, kidney, milk and yeast
10. Folic acid – some animal proteins
11. Choline – most commonly used feedstuffs are fair to good sources
12. B12 – animal proteins, fermentation products

Minerals

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General

1. Inorganic, solid, crystalline chemical elements.
2. The total mineral content of plants or animals is often called ash.
3. Make up 3 to 5% of animal body dry weight, depending on species; calcium accounts for nearly ½ of the total body mineral; P, approximately ¼; and all other minerals approximately ¼.

Classification

1. Major minerals (macrominerals)
   a. Minerals normally present at greater levels in the animal body or needed in relatively large amounts in the diet.
   b. Include Ca, P, Na, Cl, Mg, K, S.
2. Trace minerals (microminerals)
   a. Minerals normally present at low levels in animal body or needed in very small amounts in the diet.
   b. Include Co, Cu, F, I, Fe, Mn, Zn, Mo, Se.
   c. Small amounts of F and SE are considered beneficial in certain geographical areas, but toxic if fed in excess.

Functions

1. General functions of minerals
   a. Skeletal formation and maintenance – Ca, P, Mg, Cu, Zn
   b. Protein synthesis – P, S, Zn
   c. Oxygen transport – Fe, Cu
   d. Osmotic balance – Na, Cl, K
   e. Activators of enzyme systems – Ca, P, K, Mg, Fe, Cu, Mn, Zn
   f. Mineral-vitamin relationships – Ca, P, Co, Se

Utilization of Nutrients

- Digestion - involves the processes used to prepare food for absorption
- Absorption - includes the processes that move small molecules through membranes of the gastro-intestinal tract (GI tract) into blood so the molecules may be used for their specific function
- Types of Digestion
  - Mechanical Digestion - includes chewing (mastication) and muscular contractions of the GI tract
  - Chemical Digestion

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- Includes action of acids produced in the GI tract
- Activity from enzymes produced in the GI tract
- Enzymes produced by microorganisms located in various parts of the GI tract

Types of animals according to structure of their stomach

- Monogastric
  - Also called "simple-stomach" animals
  - Animals with one-compartment stomach, e.g. pigs
- Modified Simple-stomach
  - Has crop (for storage of feed), proventriculus (secretion of gastric enzymes for digestion) and gizzard (for mixing and grinding feed)
- Herbivorous monogastric
  - Have large and functional cecum that serves as site for microbial fermentation
  - The well-developed cecum enables them to utilize roughages as energy sources
- Ruminants
  - Also called "compound stomach animals"
  - Have a four-compartment stomach- rumen, reticulum, omasum, abomasum

Digestion and Absorption of Different Nutrients

1. Carbohydrates

CHO
- Crude fiber (cellulose, hemicellulose, lignin) - poorly digested
- Nitrogen-free extract (soluble sugars and starches) - readily digested

Carbohydrates must be broken down to monosaccharides for absorption from the digestive system. This requires digestive enzymes elaborated by the host or by microflora inhabiting the digestive system of the host. A high proportion of these monosaccharides will be converted to glucose in the wall of the small intestine. The remainder is modified by the liver or metabolized in other ways. The carbohydrate-splitting enzymes are effective in hydrolyzing most complex carbohydrates to monosaccharides except for those with the β (beta) linkage such as cellulose. Microflora of the rumen of ruminants and the cecum of some nonruminants, such as the horse and rabbit, produce cellulase, so that these

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species can utilize large quantities of cellulose. Anaerobic fermentation of carbohydrates by these microflora results in the production of large quantities of volatile fatty acids (VFA), mainly acetic, propionic and butyric acids. These acids provide the host animal a large proportion of the total energy supply.

2. Lipids

1. The digestibility of fat is quite high (usually exceeding 80%). The triglyceride molecule undergoes hydrolysis in the small intestine to monoglyceride and free fatty acids. These small units rarely pass into the blood circulatory system directly but normally are absorbed into the lymph system following hydrolysis and resynthesized during the absorption process.

2. Breakdown of long-chain fatty acids proceeds by stepwise removal of two carbons at a time beginning at the carboxyl end. Acetyl-CoA is the form in which the C2 fragments are removed. The acetyl-CoA released in oxidation is available for synthesis of fatty acids, for synthesis of steroids or ketones or for entry into the Krebs cycle.

3. Protein and Amino Acids

1. Dietary proteins must be broken down to amino acid fragments for absorption from the digestive system. An exception to this is in early postnatal life of mammals (in most species during the first 24 to 48 hours), when intact protein is absorbed across the intestinal epithelium.

2. Hydrolysis of dietary proteins is accomplished by proteolytic enzymes elaborated by epithelial cells lining the lumen of the intestinal tract and by the pancreas. The efficiency with which hydrolysis occurs determines the degree of absorption of individual amino acids and thus contributes to the nutritional value of the dietary protein. Another important factor contributing tp nutritional value is the balance of essential amino acids. Even proteins easily hydrolyzed in the intestinal tract do not have a high nutritional value if they have a deficiency or an imbalance of one or more amino acids.

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3. The fate of amino acids after absorption can be divided broadly into three categories:
    a. Tissue protein synthesis
    b. Synthesis of enzymes, hormones, and other metabolites
    c. Deamination and use of the carbon skeleton and entrance of the carbon skeleton into the Krebs.

5. Minerals

1. Mineral elements are absorbed primarily from the small intestine in the ionic form.

2. Absorption occurs as the result of active absorption (Ca, P, and Na) or diffusion (all other minerals).

3. Factors affecting mineral absorption

    a. Age of the animal (young more efficient than old in absorbing minerals)
    b. Form of the element (organic vs. inorganic) – inorganic generally more available.
    c. pH of the intestinal tract – lower pH enhances absorption.
    d. Binding or chelating components (oxalates, phytates, fats, etc.) may reduce absorption.

4. Excesses or interactions within minerals.

6. Vitamins

1. Little is known about the digestion of vitamins, but they probably can be used as such within the body without conversion to simpler compounds.

2. Most of the vitamins are absorbed in the upper portion of the small intestine, with the exception of vitamin B12, which is absorbed in the ileum. Water-soluble vitamins are rapidly absorbed, but the absorption of fat-soluble vitamins relies heavily on the fat absorption mechanisms, which generally are slow.

Nutrient Digestibility

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A careful assessment of the digestibility of energy and nutrients is critical for evaluating the nutritional value of feed ingredients and for estimating their energy and amino acid requirements. Methods for estimating nutrient digestibility should be accurate and easy to use in practice and yield values that are additive in mixtures of feed ingredients. Therefore, assessing the value of feed ingredients should be based on their digestible fractions, not their total concentrations.

Factors affecting digestibility

1. Age or body weight
2. Feed source and composition - feeds may vary in the type of fiber and/or ether extract material present and thus influence the digestibility.
3. Level of feed intake - as the level of feed intake increases above a certain value, the digestibility of all nutrients tends to decrease.
4. Rate of passage through intestinal tract
   a. Rapid - lack of time for complete digestive action and complete absorption.
   b. Slow - may be excessively subject to wasteful fermentations.
   c. Highly influenced by particle size and degree of feedstuff preparation.
6. Nutrient excess or deficiency - may cause poorer absorption and utilization of other nutrients.
7. Digestibility of a mixture is not necessarily the average of values for its constituents determines separately or indirectly.
8. Disease and/or parasites (worms)

Method to measure nutrient digestibility

A. Chemical analysis is the starting point for determining the nutritional value of feeds but the actual value of ingested nutrients depends on the use the body can make of them. The first consideration here is digestibility, since undigested nutrients do not get into the body proper.

B. A digestibility trial consists chiefly of:
   1. Running a proximate analysis of feed
   2. Feeding an animal a given amount by use of a marker or collecting feces at a given time on a constant rate feeding.
   3. Running a proximate analysis of feces
   4. The difference is the apparent digestible portion of the feed
   5. Computed as follows

$$\text{Apparent digestibility (\%)} = \frac{\text{Nutrient intake} - \text{Nutrient in feces}}{\text{Nutrient intake}} \times 100$$

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C. Methods Employed for Fecal Collection

b. Marker-fed with the diet at beginning and end of the collection period

a. Desirable properties of markers

(1) Physiologically inert

(2) Contain no element under investigation

(3) Will not diffuse

b. Types of markers

(1) Carmine

(2) Ferric oxide

(3) Chromic oxide

c. Use of markers is not desirable in animals with larger and more complicated digestive tracts (ruminants)

d. Using the marker method requires accurate measurement of the total amount of feed.

c. Indicator method – involves the use of an "inert reference substance" as an indicator

a. Ideal specifications of indicators

(1) Totally indigestible and unabsorbable

(2) Have no pharmacological action on the digestive tract

(3) Pass through the tract at a uniform rate

(4) Are readily determined chemically

(5) Preferably a natural constituent of the feed under test

b. By determining the ratio of the concentration of the reference substance to that of a given nutrient in the feed and the same ratio in the feces resulting from the feed, the apparent digestibility of the nutrient can be obtained without measuring either the feed intake or feces output.

c. The calculation is made as follows

$$\text{Apparent digestibility} = 100 - \left[100 \times \frac{\%\text{ indicator in feed}}{\%\text{ indicator in feces}} \times \frac{\%\text{ nutrient in feces}}{\%\text{ nutrient in feed}} \right]$$

d. Indicator materials

(1) Chromic oxide

(2) Titanium dioxide

d. Metabolism cages – used to confine the animal for quantitative collection of the feces uncontaminated by the urine. An essential feature of these stalls is that the animals must have freedom of movement, particularly as regards lying down or getting up. Can also be designed to collect urine separately.

e. Digestion trials generally consist of two periods:

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a. Preliminary or adjustment period- to free digestive tract of any prior undigested feed and accustom the animal to the facility.
b. Collection period- time period in which measurement of feed and collection of feces occur.
c. For pigs, preliminary and collection periods of 3 to 5 days each are commonly used.

5. Errors in a digestibility trial
    a. The nutrient digestibility is only apparent because the feces may contain portions of nutrients from sources other than from the consumed feed such as
        (1) Enzymes excreted into the digestive tract.
        (2) Nutrients of bacterial origin.
        (3) Nutrients (primarily N) from abraded intestinal mucosa.
        (4) Some mineral excretion into the digestive tract.
    b. Feed or fecal spillage or wastage.
    c. Errors in proximate analysis.

Nutrient Requirements

Livestock and poultry have daily requirements for absolute quantities of specific nutrients. Various factors affect voluntary feed intake and performance, and thus, these will affect the concentration of nutrients that should be provided in the diet. Underfeeding nutrients can result in suboptimal performance and overfeeding often increases feed cost and nutrient excretion. Thus, understanding the factors involved and adjusting diet formulations accordingly is essential to ensure optimal production at an economical cost. Nutrient needs are influenced by phase of production, genetics, gender, environment, health status, and overall management of pigs, equipment, and facilities. Taking these factors into account and providing allowances for variation in individual animal performance and feed nutrient composition results in nutrient recommendations for diets that should allow for success in the livestock production feeding system.

Fate of Nutrients

I. Maintenance

Maintenance means maintaining an animal in a state of well-being or good health from day to day. It is the amount of nutrients (or feed) required to adequately support an animal doing no non-vital work, making no growth, developing no fetus, storing no fat, or yielding no product. The nutritive requirements for maintenance are the first to be met. The nutritive needs of an animal for other purposes are for

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the most part over and above those for maintenance. As much as 100% of an animal's rations may go for maintenance. On the other hand, with full-fed animals as little as ⅓ or even less of an animal's nutrient intake may be required for maintenance purposes. On the average, about ½ of all feed fed to livestock goes for maintenance. While it is sometimes profitable to hold animals on a maintenance ration from a period of low prices to a period of high prices, generally speaking, only those nutrients fed over and above the maintenance requirements are available for economic production. The requirements for maintenance are as follows:

II. Growth

A. Growth is largely an increase in muscle, bone organs, and connective tissue. Since meat is basically muscle, then growth is basic for meat production. Also, it is only through growth that an animal is able to attain a mature status. It should be recognized in this connection that the nutritive requirements for growth as outlined below are in addition to those listed above for maintenance. The primary nutritive requirements for growth are as follows:

1. Protein
   a. The dry matter of muscle and connective tissue, and to a considerable degree also that of bone, is primarily protein. Hence, protein is one of the major nutritive requirements of growth.
   b. Protein for growth must be of good quality – that is, it must contain the proper proportions and amounts of essential amino acids at the tissue level.
2. Energy
   a. Animal tissue produced as the result of normal growth ordinarily contains a limited amount of ether extract. Energy in the form of net energy must be provided to meet this need in addition to that in the protein of tissue. Also a certain amount of additional energy is used by the body for growth.
3. Minerals
   a. Since bone formation is a primary activity of growth and since bone is high in calcium and phosphorus content, these two minerals are especially essential for growth.
   b. Other minerals are involved in the digestion and utilization of other nutrients needed for growth.
4. Vitamins
   a. Vitamin D is essential for bone formation.

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b. Certain other vitamins function in various metabolic processes related to nutrient utilization for growth.

5. Water

Fat-free muscle tissue is about 75-80% water. Hence, water is a major requirement for growth. Supplying adequate water for livestock growth, however, is not a major consideration sine the amount needed for growth as compared to maintenance is rather inconsequential. However, it behooves the livestock producer to take advantage insofar as possible of the fact that growth, especially muscle growth, is largely water, and water is cheap.

B. Rate of growth

1. Varies in amount per head daily among the different species – more or less in accordance with the mature size of the species.
2. Varies among different breeds largely in accordance with the size of mature animals of the respective breeds.

III. Fat Deposition

A. Fattening in an animal is simply the deposition of unused energy in the form of fat within the body tissues. Fattening is of two general types.

1. Abdominal, intramuscular, and subcutaneous deposition. This is for the most part undesirable but unavoidable if marbling is to be realized.
2. Intramuscular deposition. Commonly referred to as marbling. This is what is wanted. Difficult to obtain without excessive abdominal, intramuscular and subcutaneous deposition.

B. The object of fattening is to make meat tender, juicy and of good flavor. While most people do not like fat meat, they like the lean meat only if it contains a certain amount of fat as marbling. Fat represents the most costly form of gain in livestock. Consequently, livestock are ordinarily fattened only to a point that they will be sufficiently marbled to make their meat acceptable to the consumer.

C. Requirements for fattening

1. The primary requirement for fattening is energy. This energy must be in the form of net energy. Net energy for fattening meat come from any of several forms of feed energy such as:

Starch	Protein
Sugars	Fat
Cellulose	

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2. Fattening increases an animal's need for protein over and above that required for maintenance and growth only to the extent that additional protein may be necessary to promote good digestion. Fattening increases an animal's metabolic requirements for protein little, if at all.
3. Fattening may increase the need for certain of those vitamins related to energy metabolism.
4. Fattening animals are usually full fed since only that net energy over and above that required for maintenance and growth is available for fattening. There may be some loss in feed digestibility with full feeding, but this is more that offset by an overall increase in the efficiency of feed utilization for growth and fattening.

D. Fattening vs. growth in producing weight gains

1. Weight gains in animals are derived from the following:
   a. Growth
   b. Fattening
   c. Fill or increase in content of feed and water
2. Growth is a much cheaper form of gain than is fattening
   a. Gain from growth is primarily in the form of protein tissue and bone. Gain from fattening is largely in the form of fat.
   b. Protein tissue is about 25% protein and 75% water. While protein is one of the more costly nutrients, water is essentially free. Hence, protein tissue is a less expensive form of gain.
   c. Bone formation in growth requires considerable calcium and phosphorus. Hence, growth utilizes the calcium and phosphorus of feed more completely for weight gain than does fattening. Even if the requirements for growth make it necessary to add calcium and or phosphorus supplements, these are usually an economical source of nutrients for gain.
   d. Gain from fat is generally relatively inexpensive. Little, if any, water and minerals are laid down in fattening. In fact, during fattening, fat nay actually replace water in the tissues, which makes for a very uneconomical exchange. Also, about 2.25 times as much energy is required to form a kg of body fat as is required to form a kg of body protein. While protein is usually a costly source of net energy, it is seldom 2.25 times as expensive as other sources.
   e. Since gain from growth is usually more efficient and cheaper than gain from fattening and since animals do most of their growing while young,

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young animals make more efficient and less expensive gains than do older animals.

f. Older animals are more easily fattened than are younger animals because a larger percentage of their energy consumption is available for fattening.

IV. Milk Production

A. On a dry matter basis, the milk of all animals is fairly similar in composition, being high in protein, fat, NFE, and other minerals.

B. Milk is produced and secreted by the mammary glands. Nutrients for milk production are carried by the blood to the mammary glands. The nutrients are removed from the blood by the mammary glands, converted into milk and secreted into the udder. Milk is secreted into the udder more or less throughout the day. Nutrients for milk production must come from the feed, either directly or indirectly via body reserves of nutrients, which came originally from the animals feed.

C. Nutritive requirements for milk production

1. Are in proportion to the amount of milk produced.
2. Are over and above those for other physiological phases of production such as maintenance, growth, fattening, fetal development, etc.
3. The major nutritive requirements follow:

a. Protein

- Must be of good quality at the glandular level.
- Animals will not produce milk low in protein.
- If diet is deficient in protein, tissue reserves of protein may be used for milk production.
- Prolonged shortage of protein will limit milk production.

b. Energy. Energy over and above that for milk protein is required for the formation of milk fat and milk sugar.

- Must be in the form net energy.
- May come from ration carbohydrates, ration fat, or even excess ration protein.
- An animal will not produce milk extremely low in energy.
- If diet is low in energy, body reserves of energy may be used for milk production.
- While dietary fat is not essential for milk fat, a small amount of ration fat helps milk production.

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- Prolonged shortage of protein will limit milk production.

c. Calcium and phosphorus
   - May come from the feed or from supplemental sources.
   - Animals will not produce milk that is low in these minerals.
   - If diet is deficient in calcium and/or phosphorus, the animal will draw upon its bones for calcium and phosphorus with which to produce milk. This will weaken bones and may result in broken bones.

d. Vitamin A and/or carotene
   - May not be essential specifically for milk production.
   - Animal may produce milk low in these factors.
   - Milk rich in these factors is to be preferred.
   - Carotene and/or vitamin A for milk production may come from the ration or body reserves.
   - Vitamin A and carotene content of milk can be increased by increasing ration vitamin A and carotene.

e. Vitamin D
   - Some evidence that vitamin D is essential for assimilation of calcium and phosphorus for milk production.
   - Vitamin D content of milk can be stepped up through vitamin D feeding.

f. Sodium and chlorine (salt)
   - Essential for digestion of nutrients for milk production.
   - Milk normally contains some salt.

g. Other minerals and vitamins
   - Not too well defined.
   - Several involved in nutrient utilization.
   - Most essential minerals and vitamins normally in milk; must come from feed, from water, or from rumen or body synthesis.

V. Fetal development

Two major considerations in feeding during gestation are:

A. To provide nutrients for development of fetus and fetal membranes.
   1. Fetal development is basically just prenatal growth. Nutritive requirements for fetal development are qualitatively similar to those for growth- that is, protein, calcium, phosphorus, and vitamin D in particular and others indirectly.
   2. Quantitatively the nutrient requirement for fetal development is not great.

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3. If a diet is deficient in calcium, phosphorus, protein, and/or energy, an animal may draw upon its own body for nutrients to develop the fetus.
4. If an unbred animal is severely deficient in nutrients, she probably will not come into heat.
5. If an animal is severely deficient in nutrients during early gestation, she may cease fetal development through resorption or abortion.

B. A second consideration of feeding during gestation is to build up the nutrient reserve of the animal's body to meet the requirements for milk production following parturition.
1. Few animals can consume, digest, and metabolize nutrients sufficient to meet their needs during heavy lactation. Most must call on body reserves of fat, protein, calcium, phosphorus, vitamin A, and possibly other nutrients for this purpose. Hence, an animal should be fed during late gestation to encourage some buildup of reserves of these nutrients.
2. Termination of lactation during late gestation will promote this nutrient buildup in the body.
3. Too liberal feeding, however, during late gestation may result in an oversized fetus and difficult parturition.

Energy

Energy fuels maintenance, growth, reproduction and lactation processes and physical activity. Total energy content of a feed ingredient (gross energy) is determined by its chemical composition and is the sum of the gross energies of carbohydrate, fat, and protein fractions. The proportion of gross energy utilized by the animal is dependent on energy digestibility and how the energy is utilized by the animal. Energy concentration represents the largest cost component of swine diets. Thus, information regarding energy values of feed ingredients and their contribution to the diet is of particular importance to reduce diet costs and maximize performance.

Energy is not a nutrient per se, but oxidation of carbohydrates (starch, sugar, and fiber), protein (amino acids), and lipids (fats and oils) by living cells produce energy. This energy can then be either released as heat from the animal or retained for use in powering the animal's metabolic processes. Because pigs fed ad libitum will generally consume feed to meet their requirement for energy, it is important to have accurate estimations of the energy concentration of feed ingredients. Accurate estimation of energy concentration then allows for an estimation of daily feed intake and

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formulation of diets needed to meet the animal's requirements for amino acids, minerals, and vitamins based on dietary intake.

Expressions of energy, and their advantages

Energy concentration in feed ingredients is often expressed as kilocalories (kcal) or megacalories (Mcal) of gross energy (GE), digestible energy (DE), metabolizable energy (ME), or net energy (NE). The energy concentration of the diet is often expressed as kcal per unit of diet (kcal/kg) and energy requirements are often listed as kcal or Mcal per day. Gross energy (GE) of a substance is the energy (heat) liberated when that substance is combusted. The proportions of carbohydrate, fat, and protein in a feedstuff determine the GE concentration. Digestible energy (DE) is dietary GE intake minus the GE in the excreted feces. Metabolizable energy (ME) is defined as DE minus the GE of urinary and gaseous losses. The energy released through gaseous losses is very small, and, thus, is often ignored. Net energy (NE) is the difference between ME and heat increment. Heat increment is the heat produced from the digestion and metabolism of nutrients in an animal's diet and by fermentation in the intestinal tract. Normally, the heat increment is wasted energy, but it can be used to maintain body temperature in environments colder than the thermoneutral temperature of the animal. Net energy is the energy that is retained by the animal for maintenance and productive purposes. Therefore, NE is considered to be the most accurate estimate of an animal's need for energy.

Amino acids

The amino acid composition of a feedstuff or a diet can be determined by chemical procedures, commonly acid hydrolysis followed by ion-exchange chromatography with colorimetric or fluorimetric detection of the amino acids. Chemical procedures do not however, determine the amounts of amino acids that are available to the animal. To be "bioavailable," an amino acid must be absorbed and presented to the tissues in a form that can be used for normal metabolic functions.

Functions of amino acids

Animals of all ages and stages of the life cycle require amino acids to enable them to grow and reproduce. Amino acids are the structural units of protein. During digestion, proteins are broken down into amino acids and peptides. The amino acids and peptides are absorbed into the body and are used to build new proteins, such as muscle. Thus, pigs require amino acids, not protein. Diets that are "balanced" with respect to amino acids contain a desirable level and ratio of the 10 essential amino

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acids (of the 20 amino acids) required for maintenance, growth, reproduction, and lactation. Those 10 essential amino acids for swine are: arginine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine. These amino acids must be provided in the diet at a minimum level in order to meet the pig's requirement for the development of body protein.

Limiting amino acids

In most practical diets, the amino acid of greatest concern is simply a deficiency of one or more amino acids. When the dietary protein content is inadequate to meet the requirements for all essential amino acids, swine performance will be restricted. The amino acid that is present in the least amount relative to its requirements is said to be the **first-limiting amino acid**, and the extent to which it is adequate will determine animal performance. If the deficiency of this amino acid is corrected, then the amino acid next lowest in relation to its requirement (second-limiting) will dictate animal performance. Information about which amino acids are most limiting in natural feedstuffs is important in formulating swine and poultry diets.

Expressions of amino acid needs and their advantages

Not all of the amino acids in feedstuffs are biologically available. Many attributes can limit digestion and absorption of the amino acid from the feedstuff. Therefore, the digestibility values of feedstuffs are being determined and utilized for diet formulation.

Apparent total tract digestibility is the amount of nutrient in the diet minus the amount in the feces divided by the amount of nutrient in the diet. However, protein that disappears from the large intestine is not used effectively by monogastrics; thus digestibility at the end of the small intestine (terminal ileum) before the digesta enters the large intestine have been determined to be more accurate. This type of digestibility is referred to as ileal digestibility.

Ileal digestibility values may be expressed as **apparent (AID)**, **standardized (SID**), or **true (TID)**. These various terms reflect how ileal endogenous amino acid losses are accounted for and separated. Ileal endogenous amino acid losses are separated into basal losses, which are not influenced by feed ingredient composition and specific losses, which are influenced by feed characteristics such as fiber and antinutritional factors. Apparent ileal digestibility (AID) values are calculated by deducting the total ileal outflow of amino acids (sum of endogenous losses and non-digested dietary amino acids) from dietary amino acid intake. However, when apparent ileal

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digestibility (AID) is corrected for the basal endogenous losses of amino acids then standardized ileal digestibility (SID) values are established. Furthermore, when AID is corrected for both basal and specific endogenous amino acid losses then true ileal digestibility (TID) values are calculated. Currently, SID values should be used for feed formulation until more information on TID values becomes available.

Different Classification of Feedstuffs

- Roughages - feed materials containing more than 18% crude fiber and are generally low in energy, e.g. grasses and legumes
- Concentrates - feeds that are high in energy and low in crude fiber
  Two types of concentrates
  1. Basal or Energy Feeds
  2. Protein Feeds
- Supplements - feedstuff that is mixed with a primary grain and/or roughage to provide all the nutrients required to support the form of the production for which it is intended

Ration Formulation

- Important considerations in feed formulation
  - Acceptability
  - Digestibility
  - Cost
  - Presence of anti-nutritional factors and toxins

- Methods of Ration Formulation
  - Pearson square Method
  - Simultaneous Equation Method
  - Two-by-two matrix method
  - Trial-and-Run method
  - Linear Programming

- Ration Formulation for Ruminants
  - Nutrients: dry matter basis
    - Total digestible nutrients (TDN, kg/day)
    - Crude protein (CP g/day)
    - Calcium (Ca g/day)
    - Phosphorus (P g/day)
  - Pointers
    - Listing of nutrients requirements of animals
    - Nutrient analyses of feed ingredients and cost to be used

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- Usage and limitation of feed ingredients
- Check ration for each nutrient content cost/kg of mixed feed

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Unit III

GENETICS AND LIVESTOCK IMPROVEMENT

Additive Effect - The effect of each allele in the locus where no interaction exists

Additive Gene Action – Type of gene action whereby each of two alleles contributes equally to the production of qualitative phenotypes; neither allele is dominant

Allelic Genes - Genes that occupy the same locus in the chromosome

Animal Biotechnology - Any biology-based technology for the manipulation of animals, its parts or products to provide knowledge, goods and services. Animal reproductive biotechniques include the following: Artificial insemination, cryopreservation of semen, estrus synchronization, superovulation and embryo transfer, cryopreservation of embryos, in vitro fertilization, intracytoplasmic sperm injection, cloning, and genetic engineering

Animal Breeding - The art and science of genetic improvement of farm animals. Its objective is on the accumulation of and proper combination among the genes that are favorable to the expression of superior performance

Artificial Insemination - Technique of introducing semen (spermatozoa) into the oviduct without the benefit of the male effective if the superior animal is the male

Assortive Mating - Individuals that are more phenotypically similar tend to mate more often that would be expected by chance

Autosomal Linkage - State when the genes are located in the same autosome; during the prophase stage of meiosis, when the chromosomes double, segments of the homologous chromosomes usually cross-over and exchange segments

Autosome - The type of chromosome other than the sex chromosome

Bateson, William - English biologist who found out in 1906 that Mendelian inheritance applies to animals. Utilized chickens to observe inheritance of traits

Bases - The chemical component of the gene consisting of adenine, guanine, thymine and cytosine

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Best Linear Unbiased Prediction (BLUP) – a statistical technique for estimating genetic merits; used for the prediction of breeding values and genetic trends in a given population

Breed – Group of animals that only breed among themselves and show persistent differences from other members of similar breeds or species

Carnivores - Species of animals that live on meat of other animals

Chromosomes - The colored bodies in the nucleus that contain the DNA

Chromosome number - Diploid (2n) complement; haploid (n) complement; composed of autosomes and sex chromosomes

Cloning - Process of replicating an individual out of its own body

Culling – Opposite of selection; process of removing breeding females from the herd that may exhibit characteristics that reduce production performance or do not lead to breeding goals. Examples of reason for culling are poor reproductive performance, injury, and poor genetic qualities.

Culling Rate – Number of females that are removed from the herd.

Crick, Francis - Cambridge University Scholar, proposed the biochemical nature of the gene together with James Watson in 1956

Crossbreeding - The breeding of individuals from one population to individuals from another distinctly different population; increases the frequency of heterozygotes at the expense of the homozygotes

Dam Line – Line, breed or crossbreds selected for a combination of good female reproductive performance and growth performance and used as dams in a terminal crossbreeding program

Darwin, Charles - Proposed the theory of evolution in his book "The Origin of Species" in 1859

Disassortive Mating - Individuals which are less phenotypically similar tend to mate more often that would be expected by chance

DNA - Abbreviation of deoxyribonucleic acid, a biochemical substance that bear the chain of nucleotides called polynucleotide. Each nucleotide consists of Phosphate (P), a sugar (S), and a base (B)

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Dominance Gene Action – Type of gene action whereby one allele is expressed more strongly than the other. There are two types: complete dominant gene action and incomplete dominant gene action.

Domestication - Process of enabling animals to survive and reproduce normally under man's care

Environment - The, external factor, other than the gene that affects the character of an individual; non-genetic; modifies phenotypic expression; nutrition, climate, management, season, housing, etc.

Epistasis - A type of gene action where one gene suppresses or alters the qualitative phenotypes produced by a second gene. This interaction usually results in the production of new qualitative phenotypes.

Equilibrium Population - A population in which the gene and genotypic frequencies do not change generation after generation

Estimated Breeding Values (EBV) – estimate of the genetic value of an animal as a potential parent of the next generation; calculated using a statistical technique called BLUP (Best Linear Unbiased Prediction)

Estrus Synchronization - A management technique that makes use of hormones to bring animals into estrus at the same time

Evolution - The long process of change in the characteristics of organisms that result into new species; change in frequency of alleles in the gene pool of a population

Expected Progeny Differences (EPD) – equal to half of an EBV of an animal

Fertilization - The process of introduction of the sperm cell into the egg to form a zygote which develops as a new individual

Fisher, Ronald - British scientist responsible for the integration of the concept of Darwinian evolution and Mendelian inheritance into the discipline of population genetics

Fitness - Is the capability of a phenotype and a corresponding genotype to survive and reproduce in a given environment

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Gamete - The collective term for the reproductive cells from either the male or female

Gametogenesis - The process of production of gametes

Gene - The carrier of hereditary characteristics:

- Segment of the DNA that determines the base sequence of the nucleotide in the messenger ribonucleic acid (mRNA) that makes up the code for a certain biological function
- Occurs in pairs
- Coined by Johannsen
- Helical structure by Crick and Watson
- Located in the chromosomes
- Located in the sex chromosomes
- Functions:
    - Store and transmit genetic information from cell to cell and from parent to offspring
    - Copy or replicate itself with great consistency and precision
    - Undergo mutation or error in copying which would subsequently copied and replicated

Gene Action - the action of genes as they influence genotypic values; may be any or a combination of the following types: Additive, Dominance, and Epistasis

Gene Frequency - the proportion of a particular gene at a given locus

Gene Function - Copy and replicate itself, store, transmit genetic information and undergo mutation

Generation Interval - Average age of parents when offspring are born

Genetics - The study of the nature and function of genes

Genetic Engineering - Also called genetic modification, is the direct manipulation of an organism's genome using biotechnology; manipulation of DNA to produce new types of organisms, usually by inserting or deleting genes

Genetic Improvement - is effected by the purposeful manipulation of the genetic constitution of animals which determines the expression of their inherent characteristics

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Genetic Parameters - Heritability (narrow sense; broad sense), repeatability, covariance, correlation

Genetic Progress – change in the average of EBVs for a herd or whole population over time; influenced by selection intensity, heritability and generation interval

Genetic Variation – Amount of variation that is controlled by genes

Gene Pool - The totality of the genes that could potentially be transmitted by individuals in a population to the next generation

Genotype - The specific combination of genes in the diploid phase representing a phenotype; Assembling of gene in an individual; determined at the time of fertilization; genetic make-up; genetic constitution

Genotypic Frequency - The proportion of a particular phenotype present in the population or a breeding

Genotype x Environment Interaction - When certain genotypes perform well under certain environments than other genotype

Grading-Up (or Upgrading) – Mating procedure whereby sires of a high-performing exotic breed are mated to a local breed of female and their female offspring generation after generation

Hardy-Weinberg Law - Gene and genotypic frequency remain constant from generation to generation if population is large, there is random mating and there is absence or if there is balance of selection, migration and mutation

- Violations of HWE: small population leads to genetic drift; assortative or disassortative mating; natural or artificial selection; movement of animals through importation of live animals, semen or embryo: change in the biochemical structure of gene

Herbivores - Species of animals that live mainly on plants

Heritability - The degree by which a character is genetically transmitted from a parent to offspring

Heterogametic sex - The sex that carries the different sex chromosomes; e.g. X and Y in mammals

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Homogametic sex - The sex that carries identical sex chromosomes; e.g. X and X in mammals

Hybrid Vigor or Heterosis - A condition where the heterozygote is better than the average of the homozygote. F1 progeny is better than the average of the parents; better than the average of the better parent; occurring on lowly heritable traits; caused by non-additive gene action (dominance, epistasis) higher in the F1, than in the F2

Inbreeding - The process of mating animals that are closely related by descent; non-random mating or special case of assortive mating; increases genetic homozygosity; inbreeding depression; lowly heritable traits; closest possible inbreeding in animals is full-sib mating and parent-offspring mating

Inbreeding Coefficient – Measure of the level of inbreeding equal to the probability that alleles carried by that individual will be identical

In Vitro Fertilization IVF) – process of fertilization where an egg is combined with sperm outside the body

Law of Independence - In Mendelian inheritance, members of one pair of genes segregate independently with other pairs of genes

Johannsen, William Ludvig - Danish geneticist who coined the term gene in 1909

Line Breeding – Mating of animals in the same genetic line or selected groups which are not highly related

Linecrossing – Mating of animals from different inbred lines

Local Breed – Breed that only occur in only one country

Locus - The specific location of a pair of genes in the chromosome (pl., loci)

Lush, Jay - American scientist who introduced the application of the various theories of the Genetics of populations to the practice of animal breeding

Maternal inheritance - Inheritance of maternal characteristics through the distinctive maternal phenotype; e.g., lack of teats in the udder of the sow could retard the growth of some piglets

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Mating Systems – Alternative strategies for the mating of animals selected to be parents as they influence gene combinations received by progeny and the degree of homozygocity or heterozygocity of the population

Meiosis - The process of cell multiplication in which the number of chromosomes in the reproductive cells are reduced from diploid to haploid as in gametogenesis

Mendel, Gregor - An Austrian monk who founded genetics through his studies on crossbreeding of garden peas in 1865. He formulated and published his hypothesis on the mechanics of inheritance of characters

Mendelian Traits - Governed by few pairs of genes; qualitative; discreet classes; independent segregation; independent assortment; coat/plumage color; present/absence of horns

Migration - Process of movement of individuals from one population to another population

Mitosis -Process of multiplication of identical cells

MOET - Acronym for multiple ovulation and embryo transfer - a process of hastening the reproductive rate of female animals; Effective if the superior animal is the female

mRNA - Acronym for messenger-RNA - the biochemical substance that transcribes the genetic code in the process of protein synthesis

Mutation - Spontaneous change in the biochemical or structural characteristic in the genetic material that results in a different biological action

Native Breed – Breed that inhabit or originated from a particular region or area; also called indigenous (endemic)

Non-Mendelian traits - Governed by many pairs of genes interacting with one another; quantitative traits; continuous distribution; sex-limited; sex-influenced; sex-linked; cytoplasmic inheritance

Non-nuclear inheritance - In plants, the chlorophyll-bearing plastids are carried in the cytoplasm. Since pollens carry very little, if any cytoplasm with it; the plastids are transmitted from generation to generation through the ova

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- In animals, non-nuclear or more specifically, cytoplasmic inheritance, is quite rare
- If, there are cytoplasmic genetic materials affecting economically important traits in animals, females would play a larger role than the males in influencing the characteristics of the progeny

Non-Random Mating - occurs when some individuals do not have the same chances of mating with individuals of the opposite sex; because of certain physical, physiological, or psychological factors, some individuals tend to mate more often together than with some other individuals; important forms of non-random mating are assortive mating, disassortive mating and inbreeding

Objective of Animal Breeding - increase/improve production, efficiency, product quality and aesthetic value

Omnivore - Species of animals that live on plants, meat of animals and other food materials

Oogenesis - Process of production of female gamete

Outbreeding - Mating of individuals closely related than the average of the population; *crossbreeding* is an example of crossbreeding; crossbreeding is mating of different breed group; breed-group can be breed variety, strain, line or genetic group. Other form of out breeding includes upgrading, top crossing, inter specific (between species) and intra-specific (within. species).

Outcrossing – Mating of unrelated animals within a particular breed

Pedigree - An individual's family tree

Phenotype - The observable effect of a gene action

- May be expressed mathematically as P=G+E + (G x E) where G x E may be defined as the interaction between the genotype of the individual and the environment under which it is raised
- A function of gene and environment; outward manifestation of a genotype; traits; characteristics; measurable; expressed at different stages in life; same as genotype if effect of environment is zero; qualitative or quantitative; production, reproduction and adaptation; economically important or not

Pleiotrophy - process whereby one gene may affect two or more traits

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Population - a group if individual sharing a common gene pool

Qualitative phenotypes - Phenotypes that are described: for example, color.

Quantitative phenotypes - Phenotypes that are measured: for example, weight.

Random Mating - Process where individual of one sex has an equal chance of mating with any other individual in the opposite sex

Recessive - The state where the expression of a gene is suppressed by it alternative allelic gene

Recombinant Type Gamete - A gamete that carries two or more genes resulting from the crossover of segment of the homologous chromosomes during meiosis

Regulator genes - serve to control or regulate the function of the other genes; may function in terms of quality, quantity, or timing of the activity of certain structural genes

Replacement Rate – Percentage of replacement females used in the operation for continuous genetic improvement

Reproductive Rate – Number of offspring by a sire or dam in a year

Reproductive Techniques - artificial insemination; multiple ovulation and embryo transfer; super ovulation; in-vitro fertilization; in-vitro maturation

Response to Selection - function of heritability and selection differential; selection differential is difference between the mean of the selected parents and the mean of the base population

Segregation - The separation of the pair of genes -in the process of gametogenesis

Selection - Process where certain individuals are able to contribute more progeny in the next generation than other individuals

Selection Criteria – Traits that can be improved corresponding to traits in the breeding objectives

Selection Intensity – Relative measure of how much the average performance of individuals selected as parents will exceed the average of all individuals in a population

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Selection Methods - Based on records of individuals and/or relatives

Selection Procedures for Simultaneous Trait Improvement - Tandem, independent culling level and index selection

Sex chromosome - The chromosome that determines the sex of the individual; Male is heterogametic (XY) in mammals; male is homogametic (ZZ) in poultry

Sex-linkage - The state where a gene is located in the sex chromosome

Species Hybridization – Mating of animals belonging to different species (Inter-specific cross) or different genus (Inter-generic cross)

Spermatogenesis - Process of producing sperm cells

Structural Genes - directly responsible for the synthesis of certain biochemical products during cell metabolism

Test-crossing - Crossing of an individual of unknown genotype

Terminal Crossbreeding – Crossbreeding program where progeny are not used as breeders or do not enter the breeding population

Terminal Sire Line – Line, breed or crossbreds that are selected for growth performance and used as sires in the terminal crossbreeding program

Transboundary Breed – Breed that occurs in more than one country or region

Transcription - The process whereby the sequence of the bases in the DNA is transcribed by the mRNA in the nucleus for subsequent translation into the corresponding protein material

Translation - The process of synthesis of the polypeptide chain based on the base sequence in the mRNA

Watson, James - Proposed the biochemical nature of the gene together with Francis Crick in 1956

Wright, Sewall - American biologist who contributed significantly to the integration of evolutionary and genetic theories into the study of population genetics

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Unit IV

SLAUGHTERING, PROCESSING, AND MARKETING OF FARM ANIMALS

DEFINITION OF TERMS

- Meat - the carcass or carcass parts coming from animals sufficiently mature and healthy; flesh of animals used for human food.
- Beef - meat from cattle more than one year old.
- Veal- meat from cattle less than one year old.
- Carabeef - meat from carabaos more than one year old.
- Chevon - meat from goats
- Pork - meat from swine
- Mutton - meat from sheep
- Lamb - meat form lambs
- Venison - meat from deer
- Hot meat - meat coming from an illegal source
- Fresh meat - meat that has not undergone any substantial physical, chemical and microbiological changes from the time of slaughter
- Meat by-products - the edible by-products from the slaughterhouse
- Abattoir - the premises approved and registered as premises for slaughtering animals for human food
- Cold slaughter - slaughtering an animal dead of some causes prior to slaughter
- Fabrication - process of cutting carcasses into standard wholesale and retail cuts
- Flaying - act of removing hide from animals

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- Corned meat - prepared meat cured by soaking with or without injecting into it a solution of common salt with or without one or more of nitrate, nitrite, sugar, dextrose, syrup, honey and with or without the use of spices

- Curing - the addition of salt with or without nitrate, nitrite and sugar for the purpose of lengthening shelf life and/or enhancing the development of odor, color and flavor

- Quick freezing - freezing products such that the deepest portion hardens in 30minutes

- Deep freezing - freezing products at a temperature of -20C or lower

- Green hams - unsmoked cured hams

- Hot boned meat - meat deboned before the development of rigor mortis

Milk terms

- Pasteurization - it is a process of heating a foodstuff, usually a liquid, for a definite time at a definite temperature and thereafter cooling it immediately. US Public Health Service recommends heating milk to at least 145°F for 30 minutes or 161°F for at least 15 seconds

- Homogenization - fat globules are subjected to mechanical treatment, which breaks them down into smaller globules, uniformly dispersed in the milk.

- Certified milk - fresh milk of higher purity to be delivered to the customer within 36 hours

- Pasteurized milk – milk that has been subjected to pasteurization

- Homogenized milk – milk that has been subjected to homogenization

- Evaporated milk or unsweetened condensed milk - fresh milk that has been concentrated to 74% water and 26% total solids. Sweetened condensed milk- concentrated milk to which sugar has been added

- Dried or powdered milk - milk obtained by evaporating the moisture from fresh milk from 87% to around 1.5 to 5%

- Filled milk - milk wherein the butterfat is substituted with vegetable oil such as coconut, corn and other plant oils

- Recombined milk - milk prepared by mixing together skimmilk powder/ buttermilk powder, butter oil and water

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- Reconstituted milk - liquid milk prepared by mixing together milk powder and water to approximate the original composition of fresh milk

Basic Principles of Selecting Animals for Slaughter

Selecting animals for slaughter depends mainly on the purpose for slaughtering and the use and cost of meat produced. The basic considerations are:

- Sex
  - No distinct differences in meat quality between barrows and gilts
  - Boar taint becomes apparent at 7 months of age or upon sexual maturity
  - Castrated males tend to deposit more fat than uncastrated counterparts and females of same age
  - Meat from entire male is generally less tender, darker, coarser and lower in overall acceptability than that of castrated males
  - Steers and heifers have similar meat characteristics. Steers tend to have more marbling, more subcutaneous fat and less longissimus dorsi area than heifers and bulls
  - Bull carcasses contain 8% more muscle and 38% less fat than steer carcasses. Boneless bull carcasses contain 58% less crude fat and 23% more crude protein than steer carcasses

- Age
  - The recommended ages for meat processing and fresh meat retailing are 4-8 months for swine, 3 years or younger for cattle and carabaos and about a year old for goats
  - Meat from older animals has the tendency to be darker, tougher, fatter and usually poorer in quality than meat from younger animals. Meat from old animals is also flavorful and has high water binding and emulsion capacity and is associated with high degree of marbling

- Size
  - 80-100 kg for hogs (90-115 for larger commercial slaughterhouses)
  - 300-450 kg for cattle and carabaos
  - No standard size recommended for goats

- Class
  - For swine, the following may be ranked from most desired to least desired: barrows and gilts, shotes, sows and boars. Meat from sows and stags have higher water binding capacity, therefore are more suitable for the production of comminuted products
  - Classes of beef and carabeef include steers, heifers, cows, bulls, stags and bullocks

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- Health
  - Only healthy animals must be slaughtered

Handling/Management of animals prior to slaughter

1. Fasting - feed is not given but water is provided
   a. 6 to 12 hours for chicken
   b. 12 to 24 hours for pigs and other simple stomachs
   c. 24 to 48 hours for ruminants

   Advantages: savings of feeds, ease of eviscerating the carcass, ease of cleaning the entrails, thoroughly bled and brightly colored carcass, long shelf-life and low shrinkage of resulting meat, reduce volume and contents of waste water

2. Rest and relax the animal- no form of stress should be given

   Disadvantages of stress: loss of muscle glycogen, high temperature of carcass, low water holding capacity of resulting meat, low aroma, flavor, texture and juiciness scores. With less severe stress, pork tends to be pale, soft and exudative (PSE), however, if stress is very severe, meat tends to be dark, firm and dry (DFD) particularly in beef

3. Handle animals gently - blood clots and red spots results when animal is whipped, kicked or boxed. Blood clots are due to breakage of blood vessels and are not desirable because they support microbial growth and cause meat spoilage

4. Clean/ bathe the animal - good sanitation practice is best started by keeping the animal clean, one of the worst contaminants of meat is the microflora abundant in feces

Basic considerations in slaughtering

- Cleanliness of meat produced and health of the animal
- Hygiene of production
- Efficiency of meat inspection
- Adequacy of meat preservation

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The Slaughterhouse

- Must meet the standards set by the National Meat Inspection Service

The following are the requirements for the accreditation of slaughterhouses:

1. Must have facilities for the proper care of animals prior to slaughter. Lairage areas must be wide enough to allow for relaxation of animals to be slaughtered. Adequate ventilation and water must be provided at all times
2. Must have separate section for slaughter of swine and ruminants
3. Must have separate area for handling meat and offal to avoid contaminating the meat
4. Must have separate equipment and tools for handling meat and by-products
5. Facilities for hanging carcasses, refrigeration units (optional)
6. Adequate water supply must be provided. Water requirement is at least:
    - 60 gals per head of cattle and carabao
    - 30 gals per hog
    - 15 gals per goat
    - 7 gals per head poultry
7. Must have good drainage
8. Adequate ventilation
9. At least 100 meters away from residential areas
10. Must have adequate waste treatment facility

Classification of Slaughterhouses in the Philippines

The slaughterhouses in the Philippines are presently classified by the National Meat Inspection Commission into four categories, namely:

- AAA - Those with facilities and operational procedures so adequate that the meat processed herein is eligible for sale in any market in and out of the country
- AA - Those with facilities and operational procedures sufficiently adequate that the meat processed herein is eligible for sale in any market in the Philippines
- A (Locally Registered Meat Establishment or LRME)- Those with facilities and operational procedures of minimum adequacy as defined by the National Meat Inspection Commission, the meat processes herein is eligible for sale only in the city or municipality in which the plant is located

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- Non-accredited - Those with facilities and operational procedures of less than minimum standards that must be closed until minimum standards are provided or achieved. This also includes slaughterhouses which do not get accreditation even if they are qualified for it

The step by step slaughtering of animals

1. Ante mortem inspection - to determine whether the animal is fit for slaughter. Sick, newly parturiated, vaccinated, administered with antibiotics and other similar treatment are rejected for slaughter

2. Stunning - rendering the animals unconscious without killing them. This is done to make restraining easy and slaughtering humane. May be done using captive bolt or stunning gun or cartridge pistol, electric stunner, sledgehammer, CO_2 chamber, or the matador technique which is commonly used for stunning cattle and carabaos

3. Sticking - must be done within 3minutes after stunning. If done later, the animal may regain consciousness or die. The animal is bled by cutting the jugular vein or carotid artery. Thorough bleeding is accomplished in 3 to 5 minutes in properly stuck animals

4. Cleaning the carcass- there are 3 methods:
   a. Flaying - synonymous to skinning and involves removal of the hide. Commonly done in carabaos, cattle, horses and goats

   b. Scalding - submerging the carcass in hot water to loosen the hairs and scurf. The temperature of scalding water must be 54.4 to 82.2C. Applied only to hogs and poultry. In hogs, scalding is followed by scraping or de-hairing and then shaving or singeing; while in poultry, scalding is followed by plucking or removal of feathers

   c. Singeing - involves burning the hairs and scraping the hairs as soon as there are burned. Accomplished using a blowtorch or by burning wood or even old newspaper and/or rice straw. Usually done in goats

5. Evisceration - involves cutting the carcass open and removing the visceral organs from the body cavity. Care must be exercised not to cut the internal organs. Spillage of contents of internal organs contaminates the carcass

6. Splitting - cutting the carcass at the backbone, with the tail going with the left side if not removed before splitting

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7. Washing - right after splitting, carcass is washed to remove remaining blood, bone dust and dirt

8. Post mortem inspection - to determine whether the carcass is fit for consumption or not. Judgment given as: condemned, passed, passed for refrigeration, passed for rendering, passed for sterilization and retained

9. Shrouding (*optional*) - is wrapping the carcass with cheesecloth previously soaked in lukewarm water. This is commonly done in carcasses in which the skin is removed. Cloth is first soaked in lukewarm water and wrapped around the carcass, smoothens the external fat covering, causes fat to appear white and dense, and prevents excessive shrinkage and oxidation

10. Chilling - storing at 0-4^{0}C. Beef, carabeef and horse carcasses should be chilled for 48 hours while goat, mutton and other small carcasses should be chilled for 24 hours. Chilling is done to firm up the carcass, allow *rigor mortis* to pass, and inhibit growth of microorganisms

11. Aging (*optional*) - is holding the meat for 7 to 14 days at 1.5-3^{0}C to improve tenderness and flavor. Pork and poultry is not commonly aged because it gets rancid fast and variation in tenderness of the different parts is not substantial

Fabrication of Carcasses

Fabrication is cutting carcasses into standard wholesale and retail cuts. Proper fabrication: lowers of cutting losses, makes buyers conscious of what they are buying, and gives confidence to both producers and consumers

There are four basic principles to follow in cutting carcasses:

- Thick portions must be separated from thin ones
- The tender parts must be separated from the less tender portions
- The high-priced cuts are separated from less expensive parts, and
- Cuts must be made across the muscle fibers.

COMPOSITION OF MEAT, MILK AND EGGS

General characteristics of meat

- High quality, concentrated, and easily digested source of nutrients; and is well balanced with regard to the relative amounts of specific nutrients

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- It is an excellent source of high quality protein, B-complex vitamins and certain minerals
- It provides satiety unequaled by other food items
- It can be prepared in several ways and it provides nutrients almost in the proportion and amount needed by the human body

Physical components of meat

The basic structures that can be seen in meat are lean, fat, bone, skin, connective tissues, and blood vessels. Some meat cuts have also noticeable bone marrow inside the bones.

- Lean or muscle is considered the most important among the components because of its palatability and nutritive value. Muscle also comprises the bulk of the carcass (about 50-65%)

- Fat tissues are the most variable constituents of meat. The amount of fat in the body is markedly influenced by the diet, age, exercise and other physiological factors

- Connective tissues are also important in meat because it determines the tenderness or toughness of the meat. The higher the amount of connective tissues, the tougher the meat is

Chemical components of meat

Chemical composition of meat from different animal species

Meat	Crude Protein (%)	Ether Extract (%)	Moisture (%)	Ash (%)
Pork	18.5	4.7	73.8	1.0
Beef	19.2	1.1	76.3	1.1
Carabeef	20.2	1.0	74.4	1.1
Chevon	18.6	2.7	75.5	1.0
Mutton	20.2	4.7	74.1	1.0
Rabbit	19.8	2.8	76.8	1.2
Venison	21.9	0.2	76.7	1.2

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Chicken	22.6	0.1	76.3	1.0
Turkey	20.3	2.1	76.6	1.0
Duck	19.8	6.0	73.1	1.1
Geese	23.0	3.9	72.1	1.0
Frog	20.9	0.6	77.5	1.0
Dog meat	19.8	2.4	76.1	1.0

Amino acid composition of fresh beef, pork, lamb, cured and processed meats and meat products

- Of the essential amino acids, leucine and lysine are the most abundant; ranging from 7.4-8.4% and the least is tryptophan, 1-1.4%
- Of the non-essential amino acid, the most abundant is glutamic acid, ranging from 12.9-14.5%

Nutritional Value of Meat

- Meat protein has high biological value
- Two (2) servings of meat daily provide the recommended intake of essential amino acids
- The essential and non-essential amino acids in meat is quite constant regardless of species or cut
- Connective tissue proteins have high amount of proline, hydroxyproline and glycine and low in tryptophan and tyrosine
- Meat is an excellent source of B vitamins, P and Fe but relatively low in Ca

MILK

General Characteristics of Milk

- Bluish white to almost golden yellow color
- Opaque due to suspended fat, protein, lactose and minerals
- Pleasantly sweet and salty
- Rich, creamy flavor
- pH between 6.5 to 6.7
- normal freezing point ranges from -0.512 to -0.550^{0}C
- specify gravity between 1.030 to 1.035

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Gross composition of milk of various species

Species	Water (%)	Fat (%)	Protein (%)	Lactose (%)	Ash (%)	Energy (Kcal/100g)
Cow	87.3	3.9	3.2	4.6	0.7	66
Water buffalo	82.8	7.4	3.8	4.8	0.8	101
Goat	86.7	4.5	3.2	4.3	0.8	70
Sheep	82.0	7.2	4.6	4.8	0.9	102

Constituents of Milk

- Water - principal constituents, 87%
- Fat
  - Present as fat globules
  - Very good absorber of flavor and odor
  - Comes in two forms:
    - True fat - glycerides
    - Associated substances - lecithin, cholesterol, Vit. A, D, E
- Non-fatty substances
  - Lactose - milk sugar, about 4.9%
    - Complex disaccharide made up of glucose and galactose
    - Easily fermented by microorganisms to lactic acid
  - Nitrogenous substances
    - Protein - composed of casein and milk serum proteins
    - Non-protein nitrogen
- Minerals- Ca, Mg, Phosphate and Citrate
- Other constituents
  - Enzymes -peroxidase, catalase, lipase, phosphatase

Nutritional Value of Milk

- Rich in high quality protein, calcium, phosphorus, riboflavin and other B vitamins

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- Poor source of Vitamins C and D, and Fe
- Milk protein is one of the highest quality protein, low in sulfur containing amino acids, methionine and cysteine but rich in lysine
- Milk lipids contain high saturated fatty acids and low in linolenic and linoleic acids
- The fat - soluble vitamins in milk are heat stable

EGG

General characteristics of eggs

- Egg is a compact package of well-balanced food nutrients
- Eggs have high protein content, low in carbohydrate, high in Vitamins A and D, riboflavin
- Egg is the richest source of iron among animal foods
- One chicken egg weighs on the average 50-60g
- The calcium in egg is in the form of $CaCO_3$

Chemical composition of various parts/components of an egg

Egg components	**% of part**	**% of Egg**
Yolk		31
Moisture	47-48	
Dry Matter	52-53	
Protein	16	
Ether extract	32-36	
Phospholipids	9.5	
Triglycerides	22.4	
Cholesterol	1.7	
NFE	1.0	
Albumen or White		59
Moisture	87-89	

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	Dry Matter		11-13	
		Protein	9.7-10.6	
		ash	0.5-0.6	
		Ether Extract	0.03	
	NFE		0.5-0.9	
Shell				10
	Calcium		98.2	
	Magnesium		0.9	
	Phosphorus		0.9	

Proximate composition of Eggs

	Moisture	**Protein**	**EE**	**Ash**
Chicken	72.8	12.14	11.0	1.0
Duck	70.6	11.7	12.6	1.0
Quail	73.9	11.8	10.6	1.0

Laboratory methods for chemical evaluation of Meat, Milk and Eggs

	Laboratory Analysis		
Composition	**Meat**	**Milk**	**Egg**
Moisture	Oven drying	Oven drying/lactometer	Oven drying
Total Solids	Oven drying	Oven drying/lactometer	Oven drying
Ash	Ignition	Ignition	Ignition
Crude Fat	Ether Extraction /Babcock test	Gerber/Babcock	Ether Extraction

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Crude Protein	Kjeldahl	Kjeldahl	Kjeldahl
Casein	N/A	Walker Method	N/A

Principles of Handling and Processing of Meat, Milk and Egg

Meat, milk and eggs are very nutritious and therefore favorable for the growth of spoilage microorganisms. Eggs have their own natural protection (shell) and thus can keep longer at room temperature than milk and meat

Basic principles of handling meat, milk and eggs:

- Causes of deterioration must not be allowed to enter or to proceed
- Forces of deterioration that were able to enter must not be allowed to grow and/or prosper

Causes of deterioration

- Biological - due to microorganisms, insects, rodents, human
- Chemical - oxidation of fats causing rancidity, and oxidation of pigment causing discoloration of meat
- Physical - dehydration and enzymatic changes or degradation

Processing of Meat

- Cold Storage - chilling (0-4^{0}C); freezing (-18^{0}C)
- Drying - lowering the available water in meat
- Salting - is accomplished by addition of salt, hydrochlorites of calcium and/or other salts
- Canning - hermetic sealing in cans then subjecting to thermal process
- Curing - use of salt, sugar, and nitrate/nitrite

Milk Processing

The following are the common dairy processes right after milk collection:

1. Straining - right after milking, the milk is strained using sterilized cheesecloth
2. Cooling milk - the milk should be cooled to 50^{0}F of lower as soon as it is milked out to minimized microbial growth

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3. Clarification and filtration - this process is done to further remove dirt, which were not removed during straining. Clarification is by the use of the principle of centrifugation while filtration is using the principle of filter paper
4. Separation - this is separating the cream and the skim milk through gravity method or using a mechanical cream separator
5. Pasteurization - treating milk with heat thereafter followed by rapid cooling to destroy bacteria. About 98.99% of bacterial cells die. Pasteurization can be done by either:
    a. Holding process - milk is heated to 145^{0}F for 30minutes then cooled rapidly
    b. Flash Process
        i. High Temp. Short Time (HTST) - milk is heated at 165 to 185^{0}F for 15min.
        ii. Ultra High Temp. (UHT) - milk is heated at 192-194^{0}F for 1 to 4 seconds
6. Homogenization - forcing milk through small opening under high pressure. Normal pressure is from 2,500 to 5,000 psi. fat particles are reduced to less than 2 microns in diameter

Forms of milk in the market

- Fresh whole milk
- Whole milk powder - water is removed
- Skim milk - fat or cream is removed
- Skim milk Powder (SKMP) - water is removed from the skim milk, also known as non-fat dry milk
- Evaporated milk - some water removed
- Condensed milk - more water removed + sugar
- Filled milk - SKMP = vegetable oil + vitamin A and D
- Homogenized - small fat globules (subjected to2500-5000 psi pressure)
- Flavored milk - SKMP + water + flavor
- Whey powder

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MARKETING OF LIVESTOCK AND LIVESTOCK PRODUCTS

Methods of Marketing Livestock in the Philippines

1. Livestock market/concentration yards- farmers or traders bring their animals to the concentration yards for sale. Markets usually have facilities for the care of animals

   The transactions could be:

   1) Trader to trader
   2) Farmer to trader
   3) Trader to farmer

2. Barrio agents - buy livestock from scattered backyard farms and sell these to viajeros, may act as a viajero or wholesaler and sell livestock directly to retailers. These agents are highly skilled especially in live weight estimation
3. Wholesalers or viajeros- buys livestock in wholesale quantity transport the livestock and resell them to wholesalers, retailers and/or processors
4. City dealers - are the intermediaries between the provincial viajeros and the wholesalers in the cities
5. Retailers -are those who buy animals and sell them to consumers. These are very common in goat marketing
6. Contract growing schemes - there are several approaches used in contract scheme. In all of the schemes, a mother company takes the animals produced and takes care of the processing before final sale to consumers

Problems commonly encountered in the marketing of livestock

1. Lack of integrated information
2. Difficulty of transporting animals
3. Cost of transportation is very high
4. Lack of suitable animal and carcass transport system

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Meat and Meat Products Marketing

Traditionally, the meat distribution system is multi-layered with many middlemen involved in the whole process. The most common is the five (5) level distribution system

Producer ---> Barrio agent ---> Wholesaler ---> Retailer ---> End consumer

However, recent trends indicate that more producers have resorted to direct marketing system in an effort to reduce or totally eliminate the middlemen involved and maximize profits

Producers →retailers

- wet market
- institutional markets
  - Supermarkets
  - Groceries
  - Hotels
  - Restaurants

Types of meat markets

1. Retailers
   a. Wet markets - are those meat stalls in public markets where meat are usually hung in bulk and sliced only when a customer comes. Wholesalers usually supply meats for these markets
   b. Supermarket style/meat shops - are components or grocery stores, supermarket or individual meat shops. Meats are properly fabricated and handled
2. Wholesalers
   a. Pack trading
   b. Packers to retailers
   c. retail outlets
   d. institutional outlets

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Unit V

POULTRY PRODUCTION

Definition of Terms

- Poultry - a collective term for all domestic birds rendering economic service to man. Domesticated animals are those that can live and multiply freely under the care of man
- Fowl - generally is a term applied to all poultry species specifically, applied to designate mature domestic cocks and hens, poultry can refer also to the dressed carcass of fowls
- Poultry species - kinds of birds that are included in the term poultry (chicken, ducks, quail, turkeys, pigeons, geese, etc.)
- Average live weight - average weight (in live form) of chicken during harvest/market age.
- Beak - the projecting mouthpart of the chicken and turkey, consisting of upper and lower mandibles; organs of prehension
- Bill - the projecting mouthpart of waterfowls, consisting of upper and lower mandibles
- Breeder - general term that designate the poultry raiser who produces fowls for any special purpose with the object of improving their value or in conformity with an agreed standard of excellence. The same term can be used to mean the animals used for breeding
- Breed - a race of domestic fowls which maintains distinctive characteristic shape, growth, temperament and shell colors of eggs produced. Breed is a broader term than variety. Breed includes varieties, examples: Barred, White and buff varieties of Plymouth Rock breed
- Broiler - meat type chicken commonly grown up to 35-42 days and weighing 1.5-2.0 kg live weight
- Brooder - a place where young chicks/ducklings are cared for after hatching until they have grown to a point where they no longer need additional heat
- Capon - caponized cockerels; usually grown up to 4 months and weigh up to 3 kg with more improved quality of meat

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- Chicken - the most popular poultry species; different from turkeys, ducks, quails, geese, etc.
- Chick - young domestic chicken while in the downy stage
- Clutches - group of eggs laid for successive days
- Cock/Rooster - a male fowl one year old or over
- Cockerel - a male fowl less than one year old
- Comb - made of vascularized tissue growing on top of fowl's head. This serves as an ornamental function, signs of status and condition of the male and for heat dissipation. The standard varieties of comb are single, rose, pea, V-shaped, strawberry, cushion and the buttercup: all other being modifications of these
- Crop - the receptacle in which the fowl's food is accumulated before it passes through the gizzard
- Culls - old hens that had passed their usefulness for commercial egg production
- Debeaking - cutting part of the upper and lower mandibles with the use of electrically controlled cauterizing blade having a temperature of 815°C to destroy the tissue responsible to generate beak growth. This is done to prevent cannibalism
- Down - the first covering of the chick whose major function is for insulation
- Drake - a male of the duck family
- Dubbing - cutting of the comb, wattles or earlobes so as to leave the head smooth
- Duckling - the young of the duck family in the downy stage of plumage
- Feed Conversion Ratio - volume of feed necessary to produce a kilogram gain in weight
- Hatching - the process where the fully developed embryo (chick) breaks out of the eggs
- Hatchery - a place where the facilities and process of incubation and hatching is done

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- Harvest Recovery -the number or volume of chicken sold/marked based on the number of day old chicks raised
- Incubation - period (in days) where embryonic development is takes place out of the body of the hen
- Layer - egg-type or dual-type 6 months old female that lays eggs
- Morbidity Weight - number of chicken afflicted by disease
- Mortality Rate - number of chicken that died based on the total number of chickens raised
- Oviposition - act of laying eggs
- Ovulation - mature ova released from the ovary which is affected by hormonal secretion (LH)
- Plumage - the feathers of a fowl
- Poult - the young of a domestic turkey; the term IS properly applied until sex can be distinguished and when they are called cockerels and pullets
- Pullet - female fowl 5-6 months of age intended for egg production
- Pullet duck - a female duck less than 6 months old
- Primaries - the longest feather of a wing, growing between the pinions and secondaries, hidden when wing is folded, otherwise known as flight feathers
- Quill - the hollow, horny, basal part of stem of a feather
- Secondaries - long, large quill feathers that grow between the first and second joints of the wing, nearest to the body, these are visible when the wing is folded
- Shank - the portion of a fowl's leg below the hock, exclusive of the foot and toes
- Single comb - a comb consisting of a single, fleshy, serrated formation extending from the beak backward over the crown of the head
- Strain - a family of any variety of poultry that possesses and reproduces with mark regularity, common individual characters which distinguish it from the families of the same variety

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- Variety - a subdivision of a breed, a term used to distinguish fowls having the standard shape and other characteristics of the breed to which they belong, but differing in colors of plumage, shape of combs, etc. from other groups of the same breed

- Wattle - the pendant growth at the sides and base of the beak

- Web of feet- the flat skin beside the toes

- Web - (web of feather) the flat portion of a feather, made up of a series of barbs on either side of the shaft

- Web of wings - the triangular skin attaching the three joints of the wings, visible when the wing is extended

Poultry Breeds

Chicken

- Common origin of domestic chickens: Red Jungle Fowl (*Gallus gallus*)
- Changes in characteristics of species, breeds and varieties are by:
    - Genetic mutation
    - Selection
        - Natural selection
        - Artificial selection
    - Crossbreeding

Classification of Breeds Based on Utility or Commercial Value

1. Egg-type
2. Meat-type
3. Dual Purpose
4. Fancy/Ornamental type
5. Game type

Genetic Groups of Philippine Native Chickens

1. Banaba – Batangas
2. Bolinao – Pangasinan
3. Boholano – Bohol
4. Camarines – Bicol
5. Darag – Panay Islands

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6. Paraoakan – Palawan
7. Zampen – Zamboanga Peninsula

Ducks

Classification

1. Based on body size
   a. Heavy – Pekin, Muscovy, Aylesbury, Rouen
   b. Medium – Cayuga, Crested, Swedish, Buff
   c. Light – Runner, Campbell, Magpie, Wesle Harlequin
   d. Bantam – Mallard, Cali, East Indies
2. Based on utility
   a. Egg-type – Mallard, Khaki Campbell, Indian Runner
   b. Meat-type – Muscovy, Pekin, Aylesbury, Rouen
   c. Ornamental – Crested White, Black East Indies

Quail

Breeds (Egg-type) available in the Philippines

- Japanese Seattle – good egg production
- Negro – black or grayish black plumage
- Tuxedo – white spot on the breast area over black plumage
- Silver – white plumage

Poultry Management

Incubation

- **Goal: To attain high fertility and hatchability rate of hatching eggs**
- Incubation period differs among species

Common Name	Scientific Name	Egg Weight (g)	Incubation Period (days)
Chicken	*Gallus gallus Linn*	58	21
Ducks Mallard	*Anas platyrhynchos* *Cairina moschata*	80-83 70	28 35-37

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Muscovy			
Turkey	*Meleagris gallopavo*	85	28
Quail	*Coturnix coturnix japonica*	12	16-18
Pigeon	*Columba livia*	17	18
Geese	*Cygnopsis sygnoides*	200	30-31
Swan	*Olor columbianus*	285	35
Ostrich	*Struthio australis*	1,400	42
Peafowl	*Pavo cristiatus*	95	28
Pheasant	*Phasianuis colchicus*	32	23-24
Guinea fowl	*Numida meleagis*	40	28

- Requires proper egg storage, incubation temperature, relative humidity, good air quality and frequent turning of eggs
- Eggs must be stored in a room with lower temperature (15-18°C) with at least 75% RH
- Physiological zero of an egg is 21°C (no embryonic development will take place inside the eggs)
- Hatching eggs are stored butt-end up to avoid pressure on the air cell
- Do not store eggs >7 days – will lower hatchability
- Optimum incubation temperature and relative humidity:
    - Commercial chicken = 37.5°C and 60% RH
    - Duck = 37.5°C and 80% RH
    - Ostrich = 37.5°C and 25% RH
- Two stages of incubation
- Setter stage – eggs inside incubator are turned at least 45°angle every hour to improve embryo development; candling are done twice (7th and 14th day of incubation)
- Hatcher stage – eggs are transferred to trays to allow development until hatching; no turning and wait for eggs to hatch
- Characteristics of good quality day old chicks:
    - Alert and active

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- Can stand firmly and walk well
- Down feathers are clean, dry and fluffy
- No genetic deformities
- Has healed and dry navel
- Fully retracted yolk sac

Brooding

- **Goal: To maximize the growth and survival of the chicks by enhancing the immune, digestive, skeletal and muscular development.**
- Brooding – stage or activity where chicks are placed in an area where heat is supplied until they are able to regulate their body temperature
- Natural brooding – by the mother hen
- Artificial brooding – usually last for 14 days depending on farm conditions
- Proper temperature = 30-32°C during first 3 days
- Sources of artificial heat: hoover lamps powered by electricity or LPG, incandescent bulbs, or burning charcoal in tin cans
- Provision of proper amount of light – 22-24 hrs of day length is recommended to allow chicks to easily locate feeders and waterers throughout the day
- Chick sexing – methods are auto sexing, vent sexing or feather sexing
- Beak treatment – trimming of beak via infrared or precision beak treatment to prevent cannibalism; done during first 10 days post-hatch

BREEDER CHICKEN MANAGEMENT

- Goal: Maximize production of good quality hatching eggs
- Phases
  - Brooding period – 0 – 6 weeks
  - Growing period – 6-21 weeks
  - Laying period > 21 weeks
- Male and female chicks are identified at day old and reared separately until 21-24 weeks old
- Healthy pullets are transferred to layer/breeder house at 18-22 wks of age
- Recommended stocking density (growing period)
  - Broiler breeder (10-105 days old)
    - Males – 2.7-3.6 birds/ft^2
    - Females – 1.4-2.7 birds/ft^2
  - Layer breeder (10-126 days old)

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        - Females – 1.1 birds/ft^2
- Laying period
    - Increasing amount of light
    - Provision of proper diet
    - Egg collection
    - Culling
- Mating system
    - Natural mating – cocks are allowed to mate with hens naturally.
    - Artificial insemination – deposition of semen into reproductive tract of female using means other than natural mating
- Measures of production efficiency
    - Hatching egg recovery rate = No. of good hatching eggs / No. of eggs collected x 100
    - Mortality rate = (No. of layers at start of production – No, of layers present) / No. of layers at start of production x 100

LAYER CHICKEN MANAGEMENT

- Goal: To maximize egg production throughout the laying cycle with eggs having excellent external and internal qualities.
- Strains of layer chicken
    - Babcock
    - Shaver
    - Lohmann LSL
    - Hy-line
    - Dekalb
    - Hi-sex
    - H and N
- Two starting points of layer production
    1) Purchase day old chicks, grow until sexual maturity, and keep layers for egg production
    2) Purchase ready-to-lay pullets and keep layers for egg production
- Phases
    - Brooding stage – 0 to 6 wks
    - Growing stage – 6 to 18 wks prior to lay
    - Laying stage > 18 wks

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- Laying management
- Increasing amount of light – laying birds must be rpvoded with light during the entire laying period. Recommended 14-16 light hrs per day
- Provision of proper diets – must be offered Layer 1 and Layer 2 diets to support egg production
- Egg collection – majority of hens lay between 10:00 am – 2:00 pm, starts at about 7:00 am and concentrates around noon/ Eggs can be collected thrice a day to avoid possible contamination from feces and breakage.
- Molting – periodic shedding and replacement of feathers in layers; involves temporary reproductive quiescence (hen tends to lay eggs at lower rate)
- Measures of production efficiency
    - Hen-day egg production = No. of eggs collected / No. of layers present x 100
    - Hen-housed egg production = No. of eggs collected/ No. of layers at the start of production x 100
    - Feed efficiency = total feed intake / total egg weight
    - Mortality rate

BROILER CHICKEN MANAGEMENT

- **Goal: To attain high harvest recovery of good quality finished broilers**
- Broilers are fast growing birds raised for meat production
- Raised up to 28 to 35 days depending on target weight
- Can have 6 to 7 production cycles per year
- Broiler strains
    - Arbor Acres
    - Cobb
    - Ross
    - Indian River
- Broilers can be raised free range, semi-confined and confined rearing system
- In confined rearing system, conventional open-sided and tunnel-ventilated houses are the most commonly used housing systems
- Broilers can be raised
    - All-In-All-Out System
    - Multiple-batch System
- Phases

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    - Booster – 0-2 wks
    - Starter – 2-4 wks
    - Finisher – 4-6 wks
- Brooding and growing management
    - Proper temperature
        - Heaters open 2 hrs before DOC arrival for temperature stabilization and avoid brooders' pneumonia
        - Heaters are open 24 hrs to maintain required temperature of 30-32°C and are regularly monitored; adjusted based on age and behavior of chicks
    - Ad libitum feeding
    - Clean drinking water
    - Proper amount of light – 24 hrs of light or 23 hrs light /1 hr darkness in first 7 days to encourage feed consumption
    - Proper floor space
    - Proper ventilation
    - Target: 7-day BW must at least be 4 times the day old chick weight
- Measures of production efficiency
    - FCR = total feed consumed / Body weight gain
    - Livability or Mortality Rate
    - Broiler Performance Efficiency Factor (BPEF) = Live weight (kg) x 100 / FCR (Note: A value of 100 or more is desirable)
    - Broiler Farm Economy Index (BFEI) = Ave live weight (kg) x Livability (%) / FCR x growing period (days) (Note: BFEI value > 2.0 indicates better management of the farm and optimal performance; < 1.3 indicates poor flock performance)

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Unit VI

SWINE PRODUCTION

Swine (*Sus scrofa)* - is a wild hog of continental Europe from which most domestic swine have been derived

Swine (*Sus vittatus)* - was the chief, if not the only species of the East Indian pig that contributed to domestic swine

Breeds of Swine

- Philippine Native Swine
  - They have the following general characteristics:
    - Small and late maturing
    - Mostly solid black or black and white in color
    - Have small ears
    - Sway back
    - With weak pasterns
    - Very strong motherly instinct
    - Most native pigs are named after the region and/or province where they are found
- Pure bred - Landrace, Yorkshire, Large White, Duroc, Tamworth, Pietrain, Limousine, spotted, Hampshire, Anglian Saddleback, Berkshire, Poland China and Meishan

❖ Landrace - originally developed in Denmark for bacon production

- General characteristics:
  - White (black freckles are common)
  - Drooping ears
  - Longest breed of swine
  - Backless arched
  - Prolific and good mothering abilities
  - Weakness: legs and pasterns

❖ Large White/Yorkshire

- Name interchangeable
- Originated in Yorkshire for bacon production
- General Characteristics
  - White
  - Erect ears

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- Farrow large litter
- Great milkers
- Weakness: some are short and with big belly; tends to develop carcasses with excess back fat

❖ Duroc
- Originally called Duroc-Jersey/Red Power
- General Characteristics:
  - Light golden to dark red
  - Drooping ears
  - Good growth rate and FE
  - Good muscle quality and resistant to stress
  - Weakness: well arched back and unsound legs

❖ Pietrain
- Muscle Pig
- Peitrain, Belgium
- General characteristics:
  - Spotted black and white
  - Ears pointed, some drooping
  - Extremely muscular
  - Thin backfat
  - Weakness: hind legs: highly susceptible to stress

❖ Hampshire
- The Belt
- Hampshire, England
- General Characteristics
- Black with white belt around the shoulder and body including the foreleg
- Feed efficiency excellent
- Weakness: small litter size at birth and weaning; poor mothering ability; black color being associated to being fat

❖ Meishan
- Purebred from China, considered Taihu pigs, deriving their name from the Taihu lake
- Slow growing and fat, but have a very good taste, resistant to some diseases
- Also known as the wrinkled face and skin
- One of the most prolific breeds of pig in the world

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- ❖ Poland China
  - Black with 6 white points

- ❖ Berkshire- long bodied
  - Hybrids-
    - Upgrades- native pigs bred with foreign breeds
      - Diani- upgrade of native pigs w/ Berkshire,
      - Berkjala- 5/8 Berkshire and 3/8 Jalajala pig,
      - Kaman- upgrade of native pig,
      - Miracle Pig- ½ Large White ¼ Landrace ¼ Native
    - Crossbred- Landrace x Yorkshire; Duroc x Pietrain; Triple Cross
    - Hybrids/Synthetic Breeds- Seghers, Hypor, Nieuw Dalland, Babcock; Camborough; PIC; Tyson; JSR; Minnesota #1, Cotswold

  - Other purebreds:
  - Spotted
  - Chester White
  - Large Black
  - Chinese Taihu
    - Funjung
    - Meishen
    - Jianxiang Black
    - Erbulian

Production Systems

- Growing-Finishing - weanling pigs →80kg-95kg slaughter pigs
  - Most small swine farmers in the Asian region are in this enterprise due to limited financial resource and technical know-how for the care and management of sows and piglets.
- Sow Herd
  - Farrow to feeder - pregnant sow → weanlings
  - Farrow to finish – pregnant sow→weanlings→growers→ finisher→ sold for slaughter
  - Farrow to breeder - starts with a pregnant sow/silt to produce breeder stocks, specifically junior boars, and replacement gilts
- Boar for Hire (Boar Stud) - boar is used to breed the gilts/sows in the community for a fee

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Management of Gilts/Sows

Prior to Pregnancy - Breeding female in a pig enterprise is intended to be pregnant, nursing a litter or getting bred

- Under good management, a sow has a tremendous capability to produce 2.3 litters per year or over 20 pigs annually

Selecting and raising gilts

- A steady supply of replacement gilts is required to replace sows, which die or are culled because of poor performance
- Plan to replace 40-55% of the herd annually, if the herd age is 5 litters or less
- Commercial female is preferred for commercial production. It has added advantage of hybrid vigor
- Raise them in all-female groups in dry, well-ventilated pens that provide 0.56 and 0.74 sq.m. of floor space per animal
- Base final selection on individual performance and their physical and structural soundness
- Gilts should be bred at 8 months old to farrow the first litter at about a year old; should be bred on the 2nd or 3rd heat cycle when the animal is in standing heat
- Flushing - the daily feed intake of gilts is increased by 0.5 to 1.0 kg for 10-14 days before expected date of breeding

Signs of heat - indicate that the gilt or sow is receptive to the boar

- Reddening and swelling of the external genitalia
- Mounting other animals or willingness to be mounted by others regardless of sex
- Restlessness, continuous moving around and grunting as if calling for its kind
- Frequent attempts to urinate with very little or no discharge at all
- Capricious appetite
- Assumes a characteristic mating position when touched on back either by a boar or caretaker
- Difficult to move or be herded in the presence of a boar

Estrous Cycle - 21days

Gestation period - 114 days

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Techniques which are generally used in verifying sexual receptivity:

- Haunch pressure test – the operator should approach the sow from behind and rub her side and thighs
- Riding the back test – applied by riding or merely pressing the back of the animal
- Semen on snout test – semen sample is squeezed on the snout of the female using a squirt bottle or atomizer. Synthetic boar odor preparation may be used
- Teaser boar method – allowing a vasectomized boar, to mount the sow
- Sound test – use of chomping sounds of the boar.

Housing

- The edges of the slats should be rounded to avoid damage to feet or teats
- In a penning situation, allow about 1.85 sq. m. per bred sow or gilt. Stall measurements of about 0.5 x 2.13 m is recommended

Feeding

- Normally, 1.8 to 2.3 kg of balanced 14% CP ration will meet the requirements of a gestating sow. Avoid underfeeding or over feeding a pregnant sow

Signs of farrowing

- She is restless, nervous and often bites the walls or stall partition
- She starts building a nest
- There is distinct swelling of the mammary glands
- There is slackening of the abdominal wall
- The presence of milk when the teats are stripped indicates the sow will farrow within 24 hours

Supervised farrowing

- The need for an attendant
  - An attendant should be present to assist the newly born pigs and the sow if necessary. No equipment can substitute human hands for the safety and welfare of the newly born pigs

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Management after farrowing

- Healthcare
  - Observe the sows regularly during the first few days after farrowing for signs of illness such as fever, loss of appetite, etc.
  - Inspect the mammary gland for congestion, inflammation, laceration, and other forms of injuries. If these are present, watch for MMA syndrome
    - Metritis – inflammation or infection of the uterus
    - Mastitis – inflammation of the udder
    - Agalactia – inadequate supply of milk
  - MMA can be avoided by injecting the sow (im) after farrowing as indicated by complete evacuation of the placenta
  - A small fraction of the placenta is usually expelled before the second half of the birth process, generally after passing a jet of urine
  - All of the placenta should have been evacuated 1-4 hours after the birth of the last piglet
  - Immediately after weaning, multiparous sows should be vaccinated against Hog Cholera at least twice a year

- Feeding during lactation
  - Do not feed the sow 12 hours from farrowing to prevent the production of too much milk in excess of the actual needs of the baby pigs
  - If the sow appears hungry, feed her lightly with a thin slop of the usual lactation diet
  - On the second day, start building up her daily feed allowance
    - This is done by gradually increasing the level of feeding by 0.5 kg to 1.0 kg daily until full feeding is attained on the 7th to the 10th day after farrowing.

Management of the Boar

Assessing the boar's breeding potential

3. Semen evaluation. This is the easiest and the fastest way of estimating the fertility of the boar
   a. collect semen samples
   b. Examine for volume, motility, concentration, and morphology

Volume, ml (GEL FREE)	150-200
Sperm concentration, M/ml	200-300

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Total Sperm per Ejaculate, B	30-60
Motile Sperm, %	70
Morphologically Normal Sperm, %	70
Color	Creamy White

4. Test mating. This provides an opportunity to observe the new boar's sexual behavior and his ability to serve the gilt normally
    - A training phase consists of teaching the young boar proper procedure for mounting a female and successfully breeding her

Feeding Boars

1. Feed boars with 2.3-3.0 kg ration with 13-14% CP. The exact level will depend on breeding load, conditions of the boar and climatic condition
2. Provide fresh water at all times. In the tropics, voluntary water consumption is 4-5 liters per kg of air-dried feed

Housing and environment

1. Keep boar in comfortable and individual pen or stall (0.6 x 2.1 x 1.1 m.) that is draft-free and dry. If the boar pen doubles as the service area, allow between 5-7 m^2 of floor area
2. Use individual pens or stalls to eliminate fighting, riding and competition

Reproductive Phenomena

- A boar should start serving at 8 months of age
- Sexual maturity is a gradual process in which the sperm production and sexual desire begins concurrently in increasing intensity

Definition of Terms

- Farrowing - refers to the act of giving birth in swine
- Litter size - refers to the number of pigs born in one farrowing
- Weanling - an animal that has just been weaned
- Weaning - refers to the process of separating the young from their mother

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    - Traditional weaning- pigs are weaned at 8 to 10 weeks of age
    - Conventional weaning- pigs are weaned at 5 to 7 weeks of age
    - Early weaning- pigs are weaned at 3 to 4 weeks age
    - Very early weaning- pigs are weaned from a few days from birth to 2 weeks
- Barrow - castrated male swine
- Sow - refers to an adult female swine
- Boar - male swine
- Shote - refers to a young pig of either sex, approximately weighing 60kg
- Primiparous sow - a female which has farrowed only once. It is sometimes referred to a young sow
- Open sow - a sow whose litter has been weaned nut which has not yet been bred or is not pregnant
- Monotocous - an animal which gives birth to only one individual at a time
- Polytocous - a litter bearing animal which gives birth to more than one individual at a time

Feeding System

- Ad libitum feeding - giving feeds without restriction and always available at any time
- Restricted feeding - controlled amount of feed given to the animals
- Combination of Ad libitum and restricted feeding - fed ad lib until they reach the weight of 50kg and fed restricted until they are marketed.

Care and management of the baby pig from birth to weaning

Temperature requirements

- The capacity of the pig to regulate body temperature starts to improve gradually only on the 7th day onwards up to the 20th day when full development of the system should have been attained
- Ambient temperature preference of swine ranges from:
    - 30-32 °C – birth to 1st week
    - 29-30°C – 2nd to 6th week
    - 26-30°C - >6th week (for 50kg pig)
    - 17-22°C - >6th week (for 100kg pig)

Cutting the Umbilical cord

- Tie the umbilical cord about 1-2 inches from the base with a sterile thread
- Cut the umbilical cord with a blade below the knot

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- Dip the stump of the cord in a strong solution of iodine or merthiolate to prevent infection
- Failure to disinfect may lead to "navel ill" which can cause lameness or death

Cutting the needle teeth

- Pigs are born with 4 pairs of sharp teeth (2 in each jaw) called black, needle or wolf teeth

Tail docking

- The best way of eliminating tail biting is cutting the tail (tail docking) to a short stump at birth or at a later age
- This is now a common routine management practice in commercial swine farms

Colostrum feeding

- Colostrum is rich in resistance - conferring immunoglobulins
- Newborn piglet should consume colostrum within hours after birth to receive adequate immunity against infective microorganisms.

Identification

- Ear notching
- Ear tattooing
- Ear tagging

Creep feeding – the term "creep feed" here means the pre-starter diet offered to piglets before and just after weaning until they can be changed to a cheaper starter diet

Castration

- The removal of the testes; done at 2 weeks old or earlier
- Male pigs are raised for pork production is castrated to prevent the formation of boar taint in the meat at market time

Weaning the piglets

- Early weaning (EW) – weaning at 3-4 weeks of age
  - During this period, the process of sexual recovery of the sow must have been completed and the reproductive tract is ready for a new pregnancy.
- Very Early Weaning (VEW) – included under this is artificial rearing (1-2 days weaning) and weaning from a few days (3-7) from birth to 2 weeks

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Growing-Finishing Pigs

- The period from weaning to a slaughter weight of about 80-100 kg is referred to as the growing-finishing stage in the life of a pig
- The care and management requirements of the pig during this period are not as demanding as in the earlier stage of growth
- However, the pig should be provided with ample protection and must be fed properly
- Scour Control
  - Scour or diarrhea is the most common cause of mortality and weight setbacks in weanling pigs
  - The occurrence of the disease especially in large holdings cannot be avoided completely, but can be minimized through proper feeding and management procedures
  - Some scour problems are difficult to control particularly those arising from transmissible gastroenteritis (TGE), *colibacillosis* and other similar diseases
- GIT parasite control
  - Deworming of growing-finishing pigs 1 or 2 weeks after weaning is recommended as part of a sound health program
  - A second treatment 1-1/2 to 2 months after the first treatment is necessary because it takes approximately about a month for the large roundworms to complete its entire life cycle so that the larvae not killed by the first treatment are killed as adults
- External parasite control
  - The major parasites are the mange mites such as the *Sarcoptes scabiei* and the *Demodex phylloides*
    - *S. scabiei* are the most common mite and burrow into the upper 2/3 of the dermis. The life cycle is about 15-25 days
- Vaccination
  - Growing-finishing pigs are best vaccinated against Hog Cholera 2 weeks after weaning or 1 week after deworming, if deworming precedes vaccination
  - If pasturing is practiced, keep the pigs indoors until about 2 weeks when full immunity must have been attained
- Feeding
  - The growing-finishing pigs may be limit-fed, full-fed, or ad libitum fed. Most economical gains are obtained when the animals are full-fed
  - However, they may me limit-fed from about 75-85% their full-fed capacity if sufficiently large premium is paid for leaner carcass

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Unit VII
RUMINANT PRODUCTION

LARGE RUMINANTS

CATTLE

- The goal in the controlled propagation of breeding cattle is to produce the type of animal that will:
  - Meet market demand
  - Be productive under adverse climatic condition
  - Convert feeds efficiently
- Number of cattle breeds worldwide: 277 identifiable breeds (OSU, 1995)
  - Beef breeds – 33
  - Dairy breeds – 51
  - Draft breeds – 18
  - Meat-dairy breeds – 54
  - Meat-draft breeds – 39
  - Dairy-draft breeds – 21
  - Meat-dairy-draft breeds – 61
  - The beef breeds are quite limited in terms of distribution and importance.

Beef Production Systems

- Ranching or Extensive System
  - Cow-calf operation - involves production of calves and raising them to weaning age; found in cheap, low carrying-capacity grazing lands in hilly areas (Nueva Ecija, Nueva Vizcaya, Masbate, Mindoro, and Bukidnon); uses grades or Phil cows and pure bred or high-grade bulls. May be integrated with orchard or coconut plantation
  - Purebred program or breeder farm operation – involves breeding of purebred or registered cattle, needs big capital, skills, and sound judgement; suited for those with experience, not for beginners; aim is to produce breeder stocks to be sold to other ranchers; popular in Batangas, Tarlac, and Pangasinan. Few operations are also found in Masbate, Bukidnon, Cotabato and Iloilo; uses high energy feeds to give a better finish (more fat covering the carcass) and appearance

- Feedlot fattening operation – it provides the farmer with extra income; gives him year-round work and allows the use of cheap and plentiful farm by-products; helps meet the urgent demand for high quality protein food in the diet

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- Objective: Add weight to the animal and increase its value
- Other features:
  - Returns from the difference between buying and selling prices, and the gain in weight
  - Turnover of investment is faster than in ranching. Feeder stocks require only 120-180 days fattening period from the date of purchase
  - Needs very small land area: a sheltered feeding area (1 ½ - 2 sq. m/head, and a lofting area (5 sq. m/ head)
  - Located in areas where large amounts of feeds are produced
  - May be integrated with pineapple and sugarcane enterprise.

- Backyard operation – Backyard sector constitutes 92% of total cattle population. The aggregate share of top 5 regions (Ilocos, Southern Tagalog, Central Visayas, Northern Mindanao, and Davao) accounts for 56.9%; aims to produce animals that can be sold to augment farm income
  - 1 or 2 heads of either fattening or breeding cattle is raised at farmer's spare time
  - Farmer tethers animals to graze and supplements with farm wastes like rice straw or corn stover when back in pen
  - Others give Ipil-ipil and 1-2 kg/ animal of rice bran, corn bran and/or copra meal
  - Batangas farmers do force feeding (supak) of feed mixture

Breed Classifications:

- Based on place of origin or adaptation

1. *Bos indicus* (Zebu breeds)
   a. Adapted to tropical conditions- has survived through centuries exposure to inadequate conditions such as food supplies, insect pests, parasites/diseases and weather extreme of tropical India

2. *Bos taurus* (European breeds)- adapted to temperate conditions

Characteristics of Zebu breeds as compared to temperate breeds

- Large hump over top of shoulder and neck
- Large pendulous ears
- Dewlap having large amounts of excess skin
- Highly developed sweat glands (perspire more freely)
- Oily secretion from the sebaceous gland (believed to repel insect)

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- Based on the type of products- basically, cattle are kept to provide meat, milk and labor (draft animal power); beast of burden
    - Beef Cattle - for meat
    - Dairy Cattle - for milk

Breed - is a group of animals having common origin and possessing certain distinguishing characteristics which set them apart from other groups

Zebu breeds	Country	Products
Brahman	U.S	Meat, hardiness
Indu-Brazil	Brazil	Meat
Kankref	India	Draft, meat, milk
Krishna	India	Draft, resistant to tick fever
Nellore	Brazil	Meat
Red sindhi	Pakistan	Milk
Sahiwal	India	Milk
Sta. Gertudis	U.S.	Beef quality
Tharparkar	India	Milk, draft
Temperate Breeds		
Angus	Scotland	Meat
Chianina	Italy	Draft (initially), meat
Simmental	Switzerland	Milk, butter, cheese, draft
Ayrshire	Scotland	Milk, butter, cheese
Brown Swiss	Switzerland	Milk cheese
Guernsey	France	Milk high butterfat

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Holstein-Friesian	Netherlands	High-producing dairy cow
Jersey	France	Milk, butterfat
Milking Shorthorn	Northeastern England	Milk, meat

SOME BEEF CATTLE BREEDS

Angus

- From Scotland
- Resistant to harsh weather, undemanding, adaptable, good natured, mature extremely early
- Black or red in color
- Have high carcass yield with nicely marbled meat

Brahman

- From India
- Strains used for breed development: Guzerat, Nelore, Gir and to a lesser extent, the Krishna Valley
- Color varieties: from very light grey to red to almost black
- Intermediate in size
- Usually thrifty, hardy, and adaptable to a wide range of feed and climate.

Charolais

- From France
- White in color with a pink muzzle and pale hooves
- Medium to large-framed
- Have demonstrated growth ability and efficient feedlot gains

Chianina

- From Italy
- Largest breed of cattle
- Color varies from white to steel grey
- Unsurpassed capacity for lean meat production

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Hereford

- From England
- Dark red to red-yellow with a white face, crest, dewlap and underline
- Known for vigor and foraging ability and for their longevity

Limousin

- From France
- Color is golden-red and a lighter color under the stomach, inside the thighs, around the eyes and muzzle and around the anus and the end of the tail
- Excellent carcass

Wagyu

- Refers to all Japanese beef cattle
- "Wa" – Japanese or Hapanese-style
- "Gyu" – cattle
- Coat color of black and red
- Where Kobe beef comes from

Braford

- Created in US and Australia (5/8 Hereford, 3/8 Brahman)
- Red like Hereford with white underbelly, head and feet
- Stockier than a Hereford
- Primarily used for beef
- Heat and insect resistant

Brangus

- 3/8 Brahman, 5/8 Angus
- Black or red
- Have a good temperament
- Very versatile being high performers on pasture and in the feed yard
- Resistant to heat and high humidity

Beefmaster

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- ½ Brahman, ¼ Hereford, ¼ Milking Shorthorn
- Dual-purpose breed
- Heat, drought and insect resistant
- Moderate in size
- Light red to dark red and some will have white mottle in faces

Santa Gertrudis

- 5/8 Shorthorn, 3/8 Brahman
- Red in color
- Broad, strong, and well-muscled
- Have outstanding foraging ability and can thrive under the harshest conditions
- Very disease resistant

Simbrah

- Combination of Simmental and Brahman
- Medium-sized
- Usually red with a bit of white on the face or black with a bit of white on the face.

BREEDS OF DAIRY CATTLE

- ❖ The conventional dairy breeds of cattle were developed mainly from the Taurine species. Among the *Bos indicus* breeds, Sahiwal and Red Sindhi are considered as milk cattle
- ❖ In choosing cattle for milk production, some factors used are the breed popular in the community and high salvage value of the animal. Holstein Friesian is preferred
- ❖ The crossbreed of Holstein and the *Bos indicus* breed Sahiwal, known as Australian Friesian Sahiwal (AFS) is the most common crossbreed used in the Philippines
- ❖ In the selection of the dairy cows, dairy operators look for properly attached udder and strong feet. These are the best indicators that the cow will remain a high producer for a long time

Holstein Friesian

- Originated from Holland

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- Weighs 1000kg for male and 681kg for female
- Normally black and white-colored
- Yields an average of 6850kg of milk annually
- Very low % fat: 3.70
- Very large udder
- *Bos primigenius* as ancestor

Ayshire

- The cattle from Holland (Holstein) and the Channel Islands (Guernsey and Jersey) were used to produce Ayshire with red varying from light to brownish shade to very dark mahogany
- Originated from Scotland
- Weighs 839kg for male and 545 kg for female
- Red and white-colored
- Intermediate milk yield/year: 5400kg
- 4.2% milk fat (intermediate)
- Large, strong udder
- *B. primigenius* and *B. longifrons* as ancestors

Brown Swiss

- Originated from Switzerland
- Weighs 908kg for male and 636kg for female
- Brown in color
- Yields 5650kg milk/year (intermediate)
- 4.2% milk fat
- Large, strong udder
- *B. longifrons* as ancestor

Guernsey

- Originated from Guernsey Island
- Weighs 773 kg for males and 500kg for females
- fawn-colored; with few white hairs
- low milk yield with only 4950 kg per year
- 4.2% milk fat
- Medium size, strong udder
- *B. longifrons* as their ancestor
- Dairy cattle breed known to produce milk high in beta-carotene.

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Jersey

- From Jersey Island
- Weighs 682kg for males and 500kg for females
- Gray, fawn, black hair color
    - The blackish hairs have white tips to give grey color, or red tips to give a fawn color; can also be solid black or white spotted.
- Yields only 4650kg annually
- Small, strong udder
- *B. longifrons* as their ancestor
- Smallest dairy cattle breed

Milking Shorthorn

- From England
- Weighs 908kg for males and 636kg for females
- Red, roan or white, and/or combination of hair colors
- Yields 4750kg of milk/year
- Low to intermediate milk fat: 3.90%
- With large, strong udder
- *B. longifrons* as their ancestor

Tharparkar

- Originated from West Pakistan
- White hair color
- Milk production: 2,500 kg/lactation

Sahiwal

- Originated from Pakistan
- Reddish with light brownish red hair color
- Produces 2,100kg milk/lactation

Red Sindhi

- Originated from West Pakistan
- Red with occasional white hair color
- Produces 1,700 kg milk/lactation

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Breeding - Art and science of genetic improvement

- Attained through proper mating of selected animals

Breeding Management

- Manipulative skills involved in producing the desired type of animals
  - The cow comes into estrus the whole year but chiefly in summer
  - These frequent periods of heat make it possible to arrange that every cow in the herd does not calve at the same time, and that the supply of milk is more or less distributed throughout the year
  - To achieve this, the cows should be about 45% of the dairy herd
  - An ideal dairy herd should have the following animal composition

	Optimum (%)
Cows-in-milk	45.0 (40-48%)
Dry Cows	9.5
Bulls	0.5
Total Mature Stock	**55.0**
Heifers (>10 mos.)	24.0
Young Stock (1 ½-10 mos.)	13.0
Calves (<1 ½ mos.)	8.0
Total Growing Stock	**45.0**
Total Livestock Inventory	100.0

Selection - Process in which some individuals are chosen over others as parents of the next generation

- Basis of selection: Individuals possessing desired traits and characteristics

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- Beef Cattle: Calf crop, rate of gain (ADG), carcass quality grade, FCE, etc.
- Dairy Cattle: Milk yield, milk fat yield, milk protein yield, FCE, etc.

Systems of Breeding

1. Inbreeding - is the mating of relatives (with common ancestor within the first 4 generations) results in increased homozygosity (the state in which an animal has more uniform offsprings because their chromosomes tend to be more alike)
    a. Line Breeding - is an inbreeding practice whereby several lines of pedigree are bred back to certain individuals; retains the favorable genetic traits of an ancestor to its offspring
2. Outbreeding - is the breeding/mating of unrelated individuals (heterozygosity or less uniform offsprings); offers opportunity to combine better genes through selection
    a. Crossbreeding - mating of individuals from two distinct breeds
    b. Upgrading - mating of an animal from an unrecognized breed with a distinct breed; grade or upgrade is the resulting offspring

Breeding Methods

1. Natural Method - conventional use of bull to impregnate a heifer or cow
    a. Handmating - a bull in good condition can serve 3-4 times/week or one service every other day:
        i. 18 months old - 1:12-15
        ii. 2 years old - 1:20-25
        iii. 3 years old - 1:40-50

    b. Pasture Mating
        i. 2-3 year old bull - 10-15 cows
        ii. 9 years old bull - 20-25 cows
2. Artificial Insemination
    - Process of inducing fertilization in female without the benefit of sexual contact between male and female animals
    - Involves collection of semen from bulls and artificially introducing it into the cows or heifers to achieve pregnancy
    - Requires detection of heat or estrus to determine when to inseminate
    - Requires detection of pregnancy to determine whether further insemination is needed

Outward physiological manifestation of in-heat animals (signs of estrus)

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- Reddening/swelling of vulva
- Mucus discharge from vagina
- Frequent urination
- Restlessness and lack of appetite
- Mounting other animals
- Female stands still when mounted

Some basic parameters related to breeding management

Parameters	**Goat**	**Sheep**	**Cattle**
Weaning age (months)	3	3	6-8
Age at 1st estrus(months)	7-9	7-8	8-10
Estrous Cycle (days)	18-24	15-19	18-24
Estrus(heat) Period (hrs)	24-72	18-24	8-30
Occurrence of estrus after parturition (postpartum)	2months	17days	3-8weeks
Ave. length of gestation	150	150	280-282
Suitable age for breeding (months) M F	 8 10	 12 15	 14-15
Male: Female (pasture mating)	1:20-25	1:20-25	

Measures of good breeding or reproductive management

- Calving interval = the average length of time (in days) between successive calving
- Calving period = time/period of giving birth to calf
- Conception rate (Pregnancy) = percent of breeding females that conceived versus the total number of females exposed to the bull or inseminated
- Percent non-returns = percent of breeding females confirmed pregnant at 90 or 120 days, by pregnancy diagnosis, versus the total number of females exposed to the bull or inseminated (similar to conception rate)

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- Calving rate = percent of breeding females that give birth versus the total number of females exposed to the bull or inseminated
- First heat after parturition = the occurrence of estrus after giving birth to a young
- Calf crop = (number of calves weaned/number of cows exposed to the bull) x 100
- Kg calf weaned per cow = average weaning weight of calves times percent calf crop
- Proven sire = bull which has sufficient data to show its good genetic potential and excellent breeding performance

HERD MANAGEMENT

A. Herd division - ensures the appropriate nutrition of various age groups of the herd, prevents premature breeding which seriously affect the growth of young bulls and heifers, and prevents high rate of abortion resulting from butting and fighting

Animals may be divided into the following herds:

1. Pregnant herd – composed of pregnant females. Cows are grouped with the breeding herd during the breeding season
2. Breeding herd – consists of dry cows and heifer ready for breeding. After breeding season, pregnant animals are transferred to the pregnant herd
3. Heifer herd – composed of heifers not yet ready for breeding. Heifer calves are included in this herd after weaning
4. Steers, feeders, or fattening herd – consists of growing cattle and those fattened for the market
5. Bull herd – consists of mature males kept mainly for servicing the breeder cows

B. Management of Breeding Females
   1. Lactating and pregnant cows – once the cows are determined pregnant, they should be separated from the herd to prevent injury and possible abortion due to riding, butting, and fighting with other animals. The signs are cessation of heat and enlargement of abdomen and udder. A more reliable pregnancy test is through rectal palpitation (manual examination of the reproductive tract by way of the rectum and colon to verify pregnancy in cattle) 60-90 days after breeding

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2. Open cows and replacement heifers – open or non-pregnant cows and replacement heifers should be given the right amount of feed daily to ensure that they are in the right physiological status
    - Animals with reproductive problems and those which fail to settle after the clean-up breeding period should be culled

C. Management of Calves, Growers, and Fatteners
1. Calves – calves should suckle colostrum from their mother within the first 1 to 12 hours after calving
2. Grower – Usually maintained in the pasture with very little attention. They are given salt and other mineral supplements
3. Fatteners – needs a shorter period to each slaughter weight, generally bigger, mature or nearing maturity. However, 1 ½ - 2 year-old animals weighing 200-300 kg are preferred
    - They may be fattened either in feedlot, on pasture, or in both areas

D. Management of breeding bulls
- Purebred breeding bull – should be selected because it contributes 50% of the genetic make-up of the offspring
- Bulls should be in good condition at the start of the breeding season
- Selected bull should be allowed to breed when they are at least 2 years old
- They should be given supplemental feeding of grains or concentrates 60-90 days before and after breeding period depending on their condition
- A breeding bull should be kept in the herd as long as he is aggressive enough to mate
- However, to avoid interbreeding, a bull should be kept out of the herd 2 ½ - 3 years when the offspring reaches the breeding age

Care of Calves

- Many dairy operators often castrate bull calves and grow them out for beef
- Heifer calves are grown to maturity and kept for one lactation to determine how much milk they can produce
- Dairy calves are allowed to nurse from their dams from 1-3 days old, or in some cases weaned from their mother's milk right at birth, and reared on the pail
- It is important to train or teach the calf to drink on the pail

HERD HEALTH MANAGEMENT

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- Hygiene and Sanitation Program - aimed at maintaining the contact of the animals with agents of infectious diseases or carriers/sources of these agents
- Hygienic measures:
    - Confine animals in suitable housing, whenever possible
        - Ample spaces for feeding, drinking, resting, clean house/disinfect; well-drained holding area
    - Regular removal/proper disposal of manure
    - Proper disposal of dead animals
    - Check the drinking water supply (potable and not contaminated with pathogenic microorganisms)
    - Quarantine of premises during outbreak of diseases
    - Regular grooming of animals
        - To remove dirt
        - Stimulate the skin's oil gland

Regular Vaccination Program

- Vaccination - involves exposing an animal to an antigen derived from an infectious agent so that immune response is stimulated and the animal acquire resistance to that infectious agent before natural infection occurs

Groups of classic vaccines according to the status of organism or bioactive agent included

1. Live antigen
2. Killed (inactivated) antigen
    a. Killed - is for bacterial vaccines
    b. Inactivated - is for vaccines used against viruses or toxins

Rendering immunity to animals against infectious disease

1. Passive immunization - provides temporary resistance by transferring antibodies
   e.g. colostral immunoglobulins
2. Active immunization- is the result of natural infection or vaccination which involves the administration of an antigen

Some Important Diseases in Cattle

1. Foot and Mouth (FMD) - Apthovirus types A, 0, C Phil
2. Hemorrhagic septicemia –*Pasteurella multocida*
3. Anthrax - *Bacillus anthracis*
4. Blackleg/malignant - *Clostridium chauveilsepticum*

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5. Tetanus - *Clostridium tetani*

Bloat

- Rapid fermentation of carbohydrates (build up of CO_2, CH_4 in the rumen)
- Failure to eructate and/or belch out

Make the animal belch by:

- Massaging the distended rumen through the abdominal wall
- Tickling the throat
- Making the animal walk around for about half an hour
- Giving a drench (drink) to the animal using the following:
    - Solution of sodium bicarbonate (cooking or baking soda) and water.
    - Small amount of kerosene (paraffin) in warm milk
    - Small bottle of peanut, soya, or linseed oil.
    - Very small amount of turpentine in either soya oil or peanut oil.
    - Use a commercially available machine
- Insert trocar through the hunger's hollow (left paralumbar fossa)

Regular Parasitic Control Program

- Parasitism - most commonly encountered disease in the rural areas; most important health problem considered by small farmers

Type of parasites:

1. Ectoparasites - found in the external surfaces of the animal's body
2. Endoparasites - inhabit the internal body organ (lungworms, liverflukes, kidney worms, intestinal worms)

Disease prevention procedures (Biosecurity measures)

1. Quarantine measures - isolation of animals upon arrival; strict selection of animals free from diseases
   Usual symptoms of diseases:
    - Fever, inappetence, coughing, sneezing, diarrhea, constipation, rapid breathing or thumping, recumbence, poor condition, rough hair coat, grunting, abortion, etc.

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2. Isolation of diseased animals

FEEDING MANAGEMENT

- Ruminants (like cattle, carabao, goat and sheep) can subsist on grasses and other fibrous feed materials. The compound stomach of ruminants enables it to utilize low quality feeds
  - Rumen - fermentation vat; microbial degradation of ingesta
  - Reticulum - microbial degradation; aids in regurgitation
  - Abomasum - further digestion
  - Omasum - glandular (true) stomach
  - Esophageal groove - serve as passageway of milk in young ruminants to flow directly to the omasum by-passing the other compartments thus preventing microbial action
  - Microbial action includes secretion of enzymes that converts carbohydrates and fiber in feeds to volatile fatty acids (VFAs like acetic acid, propionic acid and butyric acid)
  - VFA - primary source of energy in ruminants

Grazing systems

- Continuous grazing – animals are free to roam all over the entire pasture and selectively graze
- Leader-follower – the high milk producing cows graze 1st before the low producing cows
- Strip grazing – dividing pastures into smaller strips; smaller number of animals are allowed to graze a big paddock
- Rotational grazing – pastures are divided into paddocks; paddocks are rotated according to recommended grazing interval

Groups of feeds for ruminant feeding

1. Roughages – contain ≥ 18% crude fiber (CF) e.g. grasses, leaves of fodder trees like kakawate and ipil-ipil, rice straw, corn stover, etc.
   - *Brachiaria mutica* (Para grass) – ability to grow well under water—logged conditions and swampy areas
   - *Panicum maximum* (Guinea grass) – erect bunch grass which is good for grazing
   - *Brachiaria decumbens* (Signal grass) – high productivity under intensive management and can withstand heavy grazing

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- *Cynodon plectostachyus* (Star grass) – resistant to trampling and grazing
- *Pennisetum purpureum* (Napier grass) – most extensively used for cut-and-carry
- *Leucaena leucocephala* (Ipil-ipil)
- *Pueraria phaseoloides* (Puero or kudzu)

2. Concentrates - contain < 18% CF, high amount of digestible nutrients e.g. rice bran, copra meal, etc.
3. Feed additives and mineral supplements – added to improve feed utilization and balance the nutrients "shotgun treatment"
   - giving of mineral mixtures to prevent deficiency

- Ration - is the amount of feed provided to the animal in one day
- Nutrient requirements - is the amount of nutrients required by the animal to meet the targeted production performance or output
- Feeding strategy
  - Confinement feeding (zero grazing) - cut and carry feeding system
    - Lower energy expenditure for foraging
    - Prevents trampling on high yielding pasture
    - Lesser exposure to stresses imposed by weather changes, parasite infection
  - Tethering – restraining an animal with a rope or chain so as to allow limited movement
    - Practiced by small hold farmers
    - Amount of feed intake is affected by the length of rope and frequency of tethering which is being dictated by the amount of available forage in the area
  - Extensive grazing - animal freely forage feeds by itself in an area that may be enclosed by fences

Feed Consumption Capacity

- Daily dry matter (DM) consumption ranges from 2.5% of the body weight in mature animals to3.0% in younger or growing animals
- In practical feeding, additional 10-15% of the daily amount DM should be given to cover for animals feed preferences as not all the feeds will be consumed
- Feed wastage in the barn results in trampling of the animal and contamination with animal's excreta. Animals will not consume such conditions of feeds

Additional feed resource:

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- Leaves of leguminous fodder trees like ipil-ipil can supply up to 30% of the total feed requirement. Legume leaves are good protein source which can replace the use of expensive protein feeds

RECOMMENDED FEEDING PROGRAM (Amount per day)

- Calves/Yearling
    - Birth to 5 days – Colostrum: 10-12% of BW
    - 6 days to 3 months – Cow's milk: 10-12% of BW
        - Creep feed (NLT 20% CP):1.0-1.3% BW
        - Soilage/Roughage: Ad libitum
    - 3 months to 7 months – Soilage/Roughage: Ad libitum
        - Creep feed (NLT 20% CP): 1-2% of BW
- Growers/Stockers
    - Grower/stocker (0-0.25 kg ADG) – Soilage/Roughage: Ad libitum (approx. 2-3% of BW, DM basis)
        - Mineral-salt lick: free choice
    - Fattener/Feeder (200-300kg BW; 0.5-1.0 ADG) – Soilage/Roughage (60%-70% Ration): 2.5-3% of BW, DM basis
        - Concentrate mix, NLT 18% CP, (30-40% of ration): 0.75-1.5% of BW
        - Mineral-salt lick: free choice
- Breeder Animals
    - Pregnant cow – Soilage/Roughage: Ad libitum
        - Concentrate mix, NLT 16% CP: 0.75-1.0% of BW
        - Mineral-salt lick: Free choice
    - Lactating cow – Soilage/Roughage: Ad libitum
        - Concentrate mix, NLT 18% CP: 1-2kg.
        - Mineral-salt lick: free choice
    - Dry cow and heifer – Soilage/Roughage: Ad libitum (2-3% of BW, DM basis)
        - Concentrate mix, NLT 18% CP (for flushing): 3 weeks before breeding
        - Mineral-salt lick: Free choice
    - Bull – Soilage/Roughage: Ad libitum (2-3% of BW, DM basis)
        - Concentrate mix, NLT 18% CP: 1-2kg: add 2kg after each service
        - Mineral-salt lick: free choice

MILK PRODUCTION

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- After parturition, the level of milk production rises rapidly, until peak lactation is reached in 6-8 weeks, then declines slowly until drying off or towards the end of lactation
- Milk secretion declines more rapidly after the 5th month of pregnancy than in non-pregnant cows

Types of Milking parlor

- Herringbone parlor – cows stand in an elevated platform in an angled or herringbone fashion facing away from the operator area.
- Tandem parlor – cows are arranged head to tail in a row
- Parallel (side-by-side/ abreast) parlor – cows stand on an elevated parlor at 90° angle facing away from the operator area.
- Rotary parlor

HOUSING

The functions of animal shelter are as follows:

- Protection against climatic elements (rain, wind and temperature)
- Protection against losses due to theft, predators, infectious diseases and traumatic injuries

Housing provision is necessary to protect the animals from the extremes of climatic elements, as follows:

- Rain – heavy rain may penetrate the body covering of an animal and decrease its insulation value
- Wind - general min. requirement of 0. m/s velocity; up to 1.0 m/s when the temperature is nearing the upper critical
- Temperature - 10 to 20°C mean daily T° range "comfort zone" maintenance of homeothermy
- Radiation - heat load considerably increased by direct radiation from sun and those reflected from clouds or ground
- Humidity - varying abilities to sweat and in descending order as follows: horse, donkey, cattle, buffalo, goat, sheep and pig
  -poultry do not have sweat glands

Housing design must suit the desired performance and control of husbandry and breeding measures such as feeding, breeding and selection, reproductive events, rearing of the young and sanitation and hygiene.

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In feedlot cattle production

- Structure should be built with ample space for efficient health maintenance and manure management.

Items to be included:

1. Roofed, concrete floor squeeze for animal restraint
2. Flow - through configuration to minimize distance to be traveled by the animal
3. Durability of materials

Other farm structures

1. Loading ramp
2. Processing (routine operations: marketing, treatment deworming, spraying, etc.)
3. Feeding pens
    - floor space per animal 3-5 m^2
    - 1 side lined with feeding trough
        - 0.45 m (18 in) elevated from the floorto trough bottom
        - shallow trough floor (prevent feed decomposition)
        - 0.45-0.6 m width

Types of housing facility

- Conventional stall barn – type of housing facility wherein each cow is confined in stanchion or tie stalls
- Free-stall barn – type of housing facility wherein the cows are free to move among resting, feeding and watering areas.
    - In tropical countries, orientation of housing facility should be east-west

Working Facilities

- Curved Working Chutes
  -takes advantage of an animal's natural behavior to turn away from potential danger or unpleasant sites or sounds
  -prevent the animal from seeing the squeeze chute or truck until they are almost upon it. A facility with solid sides likely to require a catwalk

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- Minimum floor space requirement

Enterprise		Floor space
Feedlot	Shed area Loafing area	$4m^2$/hd $5m^2$/hd
Holding pen	Up to 270 kg 270-540kg	$1.3m^2$/hd $1.6m^2$/hd
Dairy cattle	Calves and young stock Cows	1.50-2.40 m x 0.55-1.10 m 2.40-2.60 m x 1.10-1.20 m

- Minimum feeding space requirement

Enterprise		Feeding space linear, mm/hd
Feedlot		750
Ranch		
Dairy cattle	Calves 3-6 mos. Calves 7-12 mos. Yearlings, Heifer, Milking and Dry cows; Cows in maternity stalls	450 500 750

Manure Management:

- Fresh manure
  - About 5% of a feedlot animal's live weight
  - Which is 2x its 2.5% DM daily ration
  - Contains 20-25% DM, 75-80% H20

Ideal parameters in dairy production

Parameter	**Ideal Value**
Lactation period	305 days
Milking cows in entire herd	40 to 48%
Milking cows in adult herd	60 to 74%

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Dry period	50 to 60 days
Age at 1st breeding	14 to 15 months
Age at 1st calving	24 months
Calving interval	12 to 13 months
Average days open	85 to 110 days
Average days open to 1st breeding	45 to 60 days
Cows open more than 120 days	<10%
Culling rate due to reproductive problems	<10%
Services per conception	<1.7
Conception rate	>50%
Abortion rate	<5%
Calf mortality	<5%
Cow mortality	<2%
Incidence of clinical mastitis	<5%

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Definition of terms in Dairy cattle production

- Artificial Vagina – used to collect semen
- Alveoli (S. Alveolus) – sac-like structure where milk is synthesized and secreted
- Chin-ball mating device – used in detecting animals in heat wherein a halter device is worn under the chin of a vasectomized bull or androgenized cow
- Cotyledon – placental side of the attachment point to the uterus
- Crop residues – parts of plants that remain in the field after harvesting the primary crop and is used as feeds for ruminants
- Dairy grade – offspring resulting from mating of purebred sire with a grade/native or between dairy animals not purebred
- Freemartin – female of a male-female twin calf
- Intensive/Indoor system – Production system wherein the animals are completely confined

- Longevity –Referred to as the length of productive life of a dairy cow and is usually measured in years of age or number of completed lactations
- Mammary gland involution – enables the body of the dam to replenish fat and protein that previous lactations may have depleted
- Mastitis – Inflammation of the udder
- Milk Ejection Reflex/Milk let-down – movement of milk from lumen of the alveoli into the major ducts and to the bottom part of the udder
- Progesterone – Hormone that is detected when using a bovine pregnancy test kit.
- Scorecard - a list containing a systematic, logical, brief and ideal description of the different parts of the animal with their numerical value
- Silage – Fermented roughage conserving its nutrient content and preserving the forage for a longer period of time especially during the dry season
- Superovulation – Treatment of a female with gonadotropins (typically FSH) to increase the number of oocytes that are selected to become dominant follicles and to ovulate

CARABAO

- Scientific Name of carabao: *Bubalus bubalis/ Bubalus bubalis carabanensis*
- Scientific Name of Buffalo: *Bubalus mindorensis*

The water buffalo or carabao remains important in the Philippine agriculture because:

- It is an integral part of crop production system
- It provides major source of draft power in almost all farming operations
- It also provides additional income from the sale of milk and live animals either for slaughter or breeding purposes.

Terminologies

- Carabao - Philippine water buffalo or swamp carabao
- Cara - prefix connected to the sex, class and meat of carabao as introduced by Dr. Valente Villegas
- Carabeef - meat of carabao
- Carabull - breeding bull at any age
- Caracow - mature female that already given birth
- Caraheifer - female carabao below 2 years of age and has not yet given birth
- Chevron - two diagonal stripes on the ventral side of the neck, one near the brisket and the other near the jaw
- Draft animal - carabao used for work

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- Silent heat - estrous behavior of carabao wherein the animal is physiologically "in heat" but does not show any visible signs
- Wallow - to roll in mud or water
- Yoke - a piece of arched or curved timber or wood fitted on the neck of draft carabao

Types of Water Buffalo/Carabao

- Swamp type- 2n=48
    - Breeds available are the Philippine carabao (*kalabaw*), Cambodian and Chinese buffaloes
    - Light gray to gray, white bands below the jaw and across the chest and legs
    - Sickle-shaped horns
    - 1.5-2kg of milk/day
- River type or Riverine- 2n=50
    - Breeds available are Bulgarian Murrah, Anatolian Buffalo (Turkey), Azeri or Caucasian (India), Bhadawari (India), Egyptian Buffaloes (Egypt), Jafarabadi (India), Mediterranean Italian Buffalo, and Murrah Buffalo (Most famous)
    - Black or dark gray in color
    - White tightly curled or drooping straight horn
    - Dairy type buffalo can produce 8-10 liters of milk

Breed Distribution (BAI, 2010)

- Philippine Carabao – 81.6%
- PhilMurrah Crossbreed – 16.7%
- Murrah – 1.7%

Trends in carabao population

- Total population is 3.11M head of carabao in 2002
- About 99.8% belongs to small hold farmers
- The decline (1980-1994) in population can be attributed to:
    - Low productivity
    - High extraction rate
    - High mortality rate
- The increase in inventory from 1995 onwards can be attributed to:
    - Carabeef importation
    - Increased government support

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Buffalo Production Systems

- Based on the number of animals raised in the farm, the carabao production system can be classified as:
    - Small holder or Backyard system→99.8%=backyard
    - Commercial system→ 0.2% commercial

Estimated Value of Domestic Carabao Production

- The estimated value of production in 2001 was PhP 13.7 Billion
    - 8.16 B in the form of draft
    - 5.19 B in the form of meat
    - 300 M in the form of milk
    - 95M in the form of hide

Changing role of the carabao

- In some areas of the country buffaloes are being displaced with small tractors; especially in:
    - Irrigation farms =15-35%
    - Rainfed areas=3-5%
    - Hilly land = 0%
- However, in small farms, because of the economy of scale, the use of buffalo as draft animal remains sustainable.

Laws concerning the carabao

- RA 7307, otherwise known as the Philippine Carabao Act of 1992 by Pres. Cory Aquino
    - Philippine Carabao Center (PCC) was created as an attached agency of the Department of Agriculture; became operational in 1993.
    - Conserve, promote and propagate the Philippine carabao as source of draft, meat, milk and hide, through Artificial Insemination and natural mating.
    - Avail of quality stocks
    - Benefit from technology transfer activities on the care and management of carabao and the processing of meat and milk
    - Encourage backyard dairy development in rural areas
    - Increase carabao population growth to keep pace with the growing population

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- RA 8435 or Animal Welfare Act was signed into law in 1997 by Pres. Ramos
    - To protect and promote the welfare of all animals by supervising and regulating the establishment and operations of all facilities utilized for breeding, maintaining, treating or training all animals
    - This law superseded the EO 626 or the carabao slaughter ban
    - Allows the slaughter of carabaos/ buffalos at any age provided that the animal is not pregnant

- EO 626 Carabao Slaughter Ban
    - EO No. 8 of 1954 prohibiting the slaughter of carabao regardless of age, sex and physical condition purposely to conserve work animals
    - Superseded by other EO's identifying a specific term period for slaughter and age limit.
    - EO 626 of 1980 locally known as 7-11, allows the slaughter of male buffalo 7 years old and above and female carabaos 11 yrs. old and above

Breeding

- The carabao can be bred throughout the year. Mating season usually takes place during the wet season
- High incidence of silent heat
- Estrous cycle usually occurs every 21 days (range from18-25days)
- Weight at breeding is about 250-300 kg
- Age at breeding is about 2.5 to 3 yrs
- Estrus period (sexual receptivity) is 18hrs (range from 5-36hrs)
- Ovulation takes place on the average of 15hrs after estrus

Signs of Heat

- The female stands still when mounted by a bull or another female (standing heat). This is the best sign of estrus
- Mounting other animals in the herd
- Swelling of the vulva and reddening of the vaginal epithelium
- Mucus discharge of varying viscosity from the vagina, it is clear and watery at first and becomes turbid and sticky towards the end of estrus
- Mounting other animals in the herd
- Animal becomes restless and lacks appetite

Note: Not all these signs may be manifested by a carabao in estrus

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When to breed or inseminate:

- Breeding can be done by natural mating or artificial insemination
- Best time to inseminate is towards the end of estrus, in the latter 2/3 of standing estrus or within few hours after estrus
- As a rule of thumb, apply a.m.-p.m. rule which is inseminate in the morning and repeat in the morning and repeat in the afternoon or inseminate in the afternoon and repeat in the morning.

Breeding Management

- Heat is checked every 18-24days, then 39-45days
- Conduct pregnancy diagnosis (PD) at 50-60 days
- Confirmatory PD at 80-90days
- Cows bred at least 30-60 days after calving, induced estrus if possible

Some Significant breeding observations:

- All first generation offsprings have chromosome number of 2n=49
- *Inter se*mating of F_1s resulted to offsprings (F_2) with 48, 49, and 50 chromosome numbers having 1:2:1 ratio.
- The unbalanced chromosome is due to the fusion of chromosome 4 and 9.

Gestation period and others

- Average gestation period is 320 day (range 295 to 339 days)
- Age at first calving is about 3 to 4 yrs. old
- Calving interval may range from 18-24 months

Benefits from crossbreeding

- Increase in milk production- from 1-1.5 to 4-6 liters per day
- Increase in growth rate – from 250-500g to 800-1200 g per day; meat quality is as good as beef
- Maintain draftability of crossbred
    - Due to heavier bodyweight, it can carry or pull heavier loads
    - It can plow as efficient as the PC

New developments in buffalo reproduction

- In vitro maturation oocyte

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- In vitro fertilization
- Embryo transfer
- Cloning

Act of giving birth (Caracalving)

- Normal duration
    - Labor stage- 2-6hrs
    - Expulsion of fetus- 0.5-3hrs
    - Expulsion of placenta- 0.5-8hrs
- If the placenta will not be expelled after 8-12 hours, then it called retained placenta and time to call the veterinarian
- Dystocia- means difficult birth

Signs of parturition

- Presence of milk in the mammary gland
- Swollen teats
- Presence of thick mucus
- Water bag appears first
- For normal delivery, the forelegs appear first, followed by head resting on the forefeet
- Expulsion of the fetus
    - Animal in standing position
    - Indication of normal delivery
    - Calf is fully covered with fetal membrane
- The cow will normally remove the membrane by eating it
- If the cow is weak, it needs our assistance

Post parturition

- Calf is fully groomed and need to suckle the colostrum immediately (within 1 hour of birth)
- Cut the navel cord about 2-3 inches from the base and dip in iodine solution

Calf Management immediately after calving:

- Clean the mucus that clogged the nostril;
- Remove any placental membrane and wipe to dry;
- Tie and cut navel cord and apply iodine solution
- Transfer the calf to dry-bedded pen.

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- Provide clean water

Pointers in calf feeding

- Protect the calf against infectious diseases thru passive immunity from colostrum
    - Right amount- 1.5-2L/feeding
    - Right frequency of feeding- 2 to 3 times/day
    - Right temperature- 38^0C
- Promote early rumen development
    - Milk is expensive
    - Mix feed (concentrate) and hay are cheaper and can hasten rumen development

Management of post-weaned calves: 3-6months

- Transfer to a new pen
- Feed with ration that can support 500g ADG
- Monitor health condition
- Deworm if needed
- Provide mineral lick

Management of heifer

- Buffalo heifers should be fed properly to reach a mature weight of 300-350kg
- Allows to interact with animals when it nears its pubertal age (16moths)
- Deworm if needed
- Provide mineral lick

Management of Pregnant Heifer and Cow: 1st and 2nd trimester

- Provide feed for body maintenance and for developing fetus
- Monitor the body condition score (BCS)

Feeding Practices

- Carabaos are raised very extensively
- The animals are allowed to graze by tethering in any available areas with native grasses or allowed to graze in rice paddies after rice harvest

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- It is common scenario to see a young boy riding a carabao to graze it along roads and waterways
- Feeding system:
  - All-roughages system
    - Forage in the form of pasture grasses soilage, silage, hay, etc.
    - Crop residues such as rice straw, corn stover, sugarcane tops, etc.
      - Rice straw feeding is common in rice producing areas, sugarcane tops is usually used in the Visayas.
      - Supplementation of rice bran, molasses, and salt is being practiced especially when the animal is being milked or during heavy work periods.
    - Cut and carry system or grazing/tethering.
  - Roughage-Concentrate system
    - Ratio 70:30
    - Addition of concentrate feeds in the ration e.g. grains (corn), oil meals (copra meal), root crops (cassava), milling and factory by-products (rice bran)
    - Use of backyard and feedlot fattening operations
  - Salt is sometimes added to drinking water.

Feeding Management

- Chopped Napier grass (45days)- ad libitum, year-round
- Flushing – feeding livestock with plenty of feeds to make them gain weight just before or during breeding season; can be done by giving up to 5kg per day per animal during the first 3 months post-calving depending on daily milk production (300g to 500g/L milk produced)
- Concentrate feed is given during milking
- Use brewer's spent grain- occasionally
- Water provided at all times- using drip system

House for feedlot fattening

- In an open confinement
  - Mature feeders = 4 to 4.7 sq. m.
  - Yearling = 2.8 to 3.7 sq. m.
- Pen for calves
  - Measures 1m x 1.5m
  - Elevated, about 1 foot high
  - Provided with hay bedding
  - Provided with waterer and pail for the concentrate
- Housing for mature carabaos

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- Each animal is allocated 6 to 8 sq. m of floor space
- With feeding trough, waterer and shower

Animal Health Management

- Deworming program
  - Calves
    - At 1 week of age
    - At 1 month of age
    - At 3moths of age
    - Then, every 4-6 months
  - Mature animals
    - Every 4-6 months
  - Dewormers
    - Albendazole, 1ml/10kg BW
    - Triclabendazole, 1ml/10kg BW
  - Vaccination program (FMD & Hemosep)
    - At 3 months of age (initial shot)
    - At 4 months of age (booster shot)
    - Then, every 6 months
  - Tuberculosis test
    - Once a year
  - California mastitis test (CMT) for milking animals
    - Once a year

Bloat or Tympany - is a distention of the abdomen due to gas trapped in the rumen

- Frothy bloat- caused by over eating legumes
- Remedy: to puncture on left paralumbar fossa or hunger hollow using a *trocar* to release the gas

Nutrients for Carabaos

Comparative composition of milk

Composition	Cattle	Carabao
Water	87.28	80.16
Fat	3.80	8.60
Protein	3.30	5.20

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Sugar	4.90	5.10
Mineral	0.72	0.94
Sugar	4.90	5.10

1. Water- Buffaloes are observed to take in more water than cattle per day, especially when fed relatively dry materials.
    a. 300kg average weight will need about 20-25L/day up to 35L at the height of summer (6-12% of BW)
    b. Lactating animals need up to 45 L/day
    c. Evaporation losses could amount to 20 L in summer
2. Energy
    a. The main sources of energy in the diets are carbohydrates, fats and protein.
    b. Fibers form a big component of carabao diet and their digestibility ranges from high in succulent and young plant parts to very low in residues and by-products
    c. Concentrates feed as grains, CHO are readily available, more expensive not affordable by small farmers
3. Proteins
    a. Are broke down into nitrogen and alpha keto acids
    b. N is incorporated by bacteria to form microbial protein which in turn passes through the rest of the digestive tract; broken down into amino acids and subsequently absorbed in the small intestines. Some by-pass proteins are similarly digested in the small intestine.
4. Vitamins and Minerals
    a. Microbial activity in the rumen makes it possible for the carabao to dispense with some vitamins such as B-vitamins, C and K on the diet, since these are normally synthesized by the microorganism.
    b. Vitamins A, D and E should be provided in the diet or if the feed does not supply enough, supplemental vitamins are needed.
    c. Mineral elements must be consistently included in the diet as they are constantly excreted from the animal's body
    d. Calcium and Phosphorus are constituents of skeletal parts and needed in bigger amounts
    e. Other minerals required include sodium, sulfur, chlorine, potassium, magnesium, iodine, iron, manganese, zinc, copper, cobalt and selenium.

Weaning

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- Under smallholder operation, weaning is 8 to 12 months
- Early weaning can also be done 4-5 months
- In commercial operation, weaning (from the dam) is done as early as 3-5days after calving and allow the calf to drink milk from the pail

Other management practices

- Branding and castration can be done at the age of 6months or after weaning
- Castration can be done by:
    - Slit method - this is done by making incision on the scrotal sac (more reliable and recommended)
    - Burdizzo pincher or the bloodless method - this is done by crushing the spermatic cord and blood vessels and that supply blood to the testicles
- Animal ID - Uses ear tags, tattoo or notch; Horn branding is also practiced

SMALL RUMINANTS

- In sheep and goats, the breeds are classified according to:
    - Primary purpose (meat, milk, or hair/wool)
    - Type of fibers it grows (fine, medium, long, or carpet, wool; or hair)
    - In sheep, color of their fleece (black, white, red, or mottled); and
    - Specific physical or production characteristics eg. Fat-tailed sheep; angora goat for mohair

Differences between sheep and goats

Parameters	**Sheep**	**Goat**
Body covering (generally)	Wool (fleece)	Hair (hair coat)
Tail posture	Hangs down	Stands up
Horn	Spiral growth (naturally polled)	Narrower, more upright, less curved (naturally horned)
Long hair growth	Mane	Beard
Social behavior	Flocking instinct	Curious/independent
Tolerance to pain	Amazingly high (don't show)	Low

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Habitat	Entirely domesticated	Abundant in the wild
Dietary differences		
Feeding habit	Grazer (biological lawnmower)	Browser (biological herbicide) bi-pedal stance
Pen feeding	Eat more low quality feed	Eat more high quality feed

GOAT

Scientific name: *Capra hircus*

Some attributes:

- "biological herbicide"
- browsing ability (browser)
- diet: 40% grasses and 60% forbs (shrubs, herds)
- bipedal stance
- ability to nibble barks of trees
- mobile lips (prehensile organ)
- walk long distances in search for food
- very selective in terms of feed preference (young shoot first is eaten)

Some breeds of goat in the country

Breeds	Origin	Characteristics
1. Anglo Nubian Anglo Nubian	England USA	Long wide pendulous ears Convex roman nose All purpose(meat, milk, hide) -proved better suited in the tropics -Low mortality and higher fertility than that of a local goat
2. Saanen	Switzerland	Cream to White Known to have the longest milking

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		period "queeny of dairy goat" Alert breed, upright ears -although able to produce very high milk yields under stall feeding husbandry, were not able to reach genetic potential otherwise
3. Alpine	France/Switzerland	Straight face; with clear pure white markings at each side of the belly; dairy type -shows erratic breeding and lactation performance when situated in humid tropics
4. Toggenburg	Switzerland	Erect ears carried forward; white ears with dark spot in the middle; two strips down the muzzle white from hock (hind legs) and knee (for legs) downward; oldest known dairy goat
5. La Mancha	Oregon/Spain	Straight face; very distinctive ears types (rudimentary); "gopher ear" (like squirrel or tortoise) and "elf ear"; dairy
6. Boar	South Africa	The name means farmer in Dutch to distinguish it from the Angora goats; meat goats; the first goat breed involved in meat production performance testing; horned with lop ears
7. Native goat		Small, stocky and lowest; believed to have originated from Katjang breed.

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- Tethering – goats are tied each to a rope about 6-10 meters long; to provide enough nutrients, the animal is transferred once or twice to areas where it will have access to fresh browsing.
- Extensive production – goats are usually kept in small herds of about 5 to 10 animals and let loose to browse on waste vegetation and hedges.
- Intensive production – either on cultivated fodder or pastures.
- Integration to plantation agriculture – this system can be either extensive, intensive or semi-extensive; it involves the integration of goat raising into established plantations of rubber, oil palm or coconut.

Sheep and goat production

- Goats tend to be prevalent on small scale farms and among the landless.
- This is because of the economics of the goat; goats require little initial capital to purchase and require little in the way of input cost
- As a small animal that has "a higher reproductive rate and higher digestive efficiency for cellulose than sheep" the goat is well suited to areas where the vegetation is coarse, such as many semi-arid and arid regions.
- Goats are inquisitive feeders, with a feed range from herbage and tree bark to skins and cloth
- In the Philippines, an estimated 99% of the goat population is in the hands of smallholders, with flock size of about 3 to 5.

Goat Breeds and Breeding

- Highly developed European breeds for production of milk and/or meat are in great demand in many tropical countries and have already been introduced in many places, including Malaysia, Indonesia and the Philippines.
- However, as purebreds, these goats either have difficulty adjusting to tropical climates or only excel in intensive systems similar to those found in Europe.
- Introduction of European breeds can improve productivity; however, within current systems, attention to the specific breed and careful study of cross-breeds is necessary.
- In many cases, the productive potential of local breeds has not yet been reached
- Introduction of new breeds may not be necessary and require changes in goat herding and management systems that are too extreme.
- There are approximately 300 breeds and types of goat, the majority of which live in tropics and sub-tropics

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- Little or no attention is paid to control of breeding or selection for performance.
- Smaller native breeds, such as the indigenous Kacang goat in Indonesia, can produce more meat than larger, faster growing foreign breeds

Housing

- Housing provision is necessary to protect the animals from the extremes of climatic elements, as follows:
  - Rain – heavy rain may penetrate the body covering of an animal and decrease its insulation value
  - Wind - general min. requirement of 0. m/s velocity; up to 1.0 m/s when the temperature is nearing the upper critical
  - Temperature - 10 to 20 °C mean daily T° range "comfort zone" maintenance of homeothermy
  - Radiation - heat load considerably increased by direct radiation from sun and those reflected from clouds or ground
  - Humidity - varying abilities to sweat and in descending order as follows: horse, donkey, cattle, buffalo, goat, sheep and pig
    -poultry do not have sweat glands

- The location of the goat/sheep house inside the farm should be:
  - Accessible
  - Protected by natural windbreaks; and
  - Easily supplied with potable water

- Minimum space requirement for goats

Category	**Zero Grazing**	**Daytime Grazing**
Breeding females	1.5	0.8
Breeding Bucks	2.0	1.5
Young stock	0.8	0.5

- Loafing area – an area with ample space allocated for goats to exercise outside its confinement and can also serves as gathering area before the goats are driven back to respective pens; a fenced lot adjacent to the goat

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shed that must be provided at about 2-3 sq.m. per head of goat complete with feeding rack and watering through

- Quarantine area – an area used to separate and restrict the movement of well animals who may have been exposed to a communicable disease to see if they become ill.

Practices necessary for a healthy goat herd

1. Start with a healthy breeder stocks (animal selection)
2. Good housing provision
3. Deworm newly arrived stocks effectively
4. Practice animal segregation
5. Practice vaccination
6. Check for presence of parasites and treat effectively
7. Provide salt lick and adequate amount of water
8. Confine animals during rainy days; graze when dews on the grasses already subsided
9. Check buck/ram (for infectious disease) prior to joining the breeding herd

SHEEP

Scientific Name: *Ovis aries*

Some attributes:

- Biological lawnmower
  - Essentially grazers
  - Flocking instinct- pecking order, a social behavior related to hierarchy of dominance.
  - Less selective of feed offered than goats- can utilize rice straw
  - Feed preference and digestive efficiency
    - Comparable to cattle
    - Less efficient than goats and water buffaloes in utilizing highly fibrous roughages

Some breeds of sheep

Breeds	Origin	Characteristics
1. Barbados Blackbelly	Barbados Island with African	Adapted to wide range of environment; high reproductive efficiency; hair type: sweet

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	Ancestry	mutton; black colors covers under parts completely extending up the neck and down the insides of the legs
2. Priangan	Indonesia	Primary for ram fighting and meat; Thin-tailed; often lacks external ears
3. Shropshire	England	Wool-type; believed to be as one of the ancestors of the Philippine sheep
4. Suffolk	England	Wool-type; meat, dark colored feces and legs
5. Merino	Spain	Finest wool producer; the other ancestor of Philippine sheep
6. Dorset	England	Reproductively very prolific; breeds year round; widely used to produce hothouse lambs; Mature ewe weighs 150 to 200 lbs; Rams weigh 225 to 275 lbs.
7. Katahin Sheep	Developed in USA	Mix of hardy, prolific breeds, including native hair sheep from the Virgin Islands; developed to be hardy in hot or cold climates, non-seasonal breeders, with multiple lambs.

Sheep Production in the Philippines

- Philippine Sheep (PS) introduction
  - By early Spanish explorers
  - Brought from Asiatic mainland

- Early importation (BAI Recorder Vol. 17, 1955. p.15)
  - 1908 – 10 Shropshire from USA
  - 1912 – 7 Merino, 9 Lincoln from Australia
  - 1918 – few Indian sheep from Bombay;
  - 1925, 1928, 1929 – Shropshire from USA;
  - 1938 – 20 Indian sheep from o. Africa;

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- 1947 – 9 Dumba sheep; 1 pair Blackhead Persian sheep from Pakistan; 35 Shropshire and 30 Southdown from USA

Housing: floor space requirement for sheep (m^2)

Category	Zero Grazing	Daytime Grazing
Breeding females	1.5	0.8
Breeding Bucks	2.0	1.5
Young stock	0.8	0.5

The estimation of the floor area of the goat house should be based on the number of breeding females with consideration of some of its parameters. This includes:

- conception rate at 80%
- kidding interval of 8 months;
- an average kidding size of 1.5 kids per kidding;
- a disposal age of one year for growing animals
- an annual replacement rate of 20% for female breeders
- 80% livability rate up to 1yr of age

However, the idea for possible expansion should be considered in the lay-out of the goat house.

For Example: 25-doe level

1. Breeder females (doe)→ 25 does x 1.5 m^2 per doe = 37.5 m^2
2. Breeder male (buck) → 1buck x 2.0 m2 per buck = 2m^2
3. Young stock
   a. 25 does x 80% conception rate x 1.5 kids/kidding x 1.5 kidding/yr x 80% livability rate = 36 offspring/yr x 0.8 m2 per animal = 28.8 m^2

Total floor area = 68.3m^2

Total number of goats = 62 head

Feeding space – 0.4m per animal

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Fool elevation – 1.0-1.5m above ground (for ease of manure collection disposal)

Fencing is needed to effectively control the animals in the pasture

- Goat- 1.5m high (made of hog wire)
- Sheep- 1.0 high (made of hog wire)
- Cattle – 1.0 high (made of 3 strands of barbed wire)

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Some infectious diseases of goats and sheep

Disease	Cause	Symptoms
Bacterial scours in kids	Multiple bacterial agents (enteropathogens)	Occurs as early as 24 hrs after birth; pasty yellowish white feces later becoming more liquid with fermented or pungent odor; kid weak with sunken eyeball unsteady gait and rough coat, mortality due to dehydration
Bacterial pneumonia	Multiple bacterial agents (pneumopathogens)	Fever, inability to suckle, nasal discharge, coughing and respiratory distress. Gradual emancipation may terminate as pneumonia-enteritis combination. Death common.
Infectious arthritis (big knee)	Multiple bacterial agents acting singly or together (*Erysipelorthrix, Corynebacterium, Chlammydia*	Swollen knee, lameness, pain if pressure applied of affected joint. Fever may be present. Joints involved are hock, carpus, elbow and stifle. Animal prefers recumbency, appetite

	Streptococcus, Straphylococcus, etc.) Predisposing: Dirty or unsanitary pens and areas of confinement	affected with gradual physical deterioration.
Blackleg	Bacteria (*Clostridium chauvoei)*	Sudden death in acute cases. Less acute; depression, fever, rapid respiration and suspended rumination. Typically, hot painful swelling in thigh and leg muscles. Cracking sensation on palpation and swelling due to gas in tissues.
Tetanus	Bacteria (*Clostridium tetani)*	Early stages characterized by rigidity and stiffness of muscles; slightly gait. Late stages with titanic convulsions, prolapsed of 3rd eyelid, stiff tail, head and neck thrown back; hyperexcitability
Mastitis	Multiple bacterial causes with *Streptococcus, Pasturella, Corybacteria and Eschericha mostly commonly noted. Also caused by Mycoplasma, Norchardia and Candida spp.*	Hot painful and swollen udder, may became red due to inflammation later changing to dark-reddish blue indicating necrosis of udder tissues. Milk maybe bloodstained, may contain flakes or clots. Fever, loss of appetite, depression and dehydration. Gait or movement of doe is affected
Contagious ophthalmia or infectious keratoconjunctivitis	*Chlamaldia and Mycoplasma spp.*	Sore eyes, congestion, lachrimation, photophobia, gluing of the eyelids. Corneal opacity, ulceration and pus formation. Blindness.
Contagious ecthyma or sore mouth	Virus	Scabby, lesions in the lips, muzzle, eyelids, udder, teats and feet. Warty growth develops in some areas which are deep. May become necrotic or ulcerous.

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Foot and Mouth Disease (FMD)	Virus. Types identified in the Philippines: A, O, C.	Fever. Vesicles and erosion and between hooves, coronary band (junction between skin and hoof), teats and udder, oral mucosa and tongue. Raw ulcerations follow rupture of vesicles; stringly or foamy salivation, smacking of the lips, difficulty in feed ingestion; staggering gait and lameness. Abortion in pregnant animals.
Foot rot	Bacteria *Spherophorusnecrophorus* Predisposing: wounds in the feet, wet and muddy ground	Sudden lameness when acute. Typical to see foul smelling ulcers, interdigital cleft swollen and painful and may worsen to cause fever and other systemic signs
Hemorrhagic septicemia	Bacteria (*Pasteurella multocida)* Predisposing: climatic stress, fatigue, transport, nutritional and parasitic stresses, etc.	High fever, loss of appetite. Respiratory distress, salivation, nasal discharge, swelling of throat and brisket congestion of mucus membrane, diarrhea becoming bloody later.
Anthrax	Bacteria (*Bacillus anthracis)*	Sudden onset of fever, depression and loss of appetite

Common Parasitic Disease of Goats and Sheep

Parasitic gastroenteritis	Various species of parasitic nematodes in the digestive tract. Young animals most susceptible.	Poor body condition, anemia, diarrhea, potbelly and weakness
Parasitic pneumonia	*Dictyocaulus spp.* (adult stages in the bronchioles	As in parasitic gastroenteritis for general signs. Specific symptoms include

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	of lungs)	persistent husky coughing, respiratory distress
Tapeworm infection	*Monies spp.*	Same as other internal parasitism; passage of tapeworm segment in the feces.
Liver fluke infection	*Fasciola gigantic and F. hepatica.* Requires intermediate hosts	Symptoms similar to parasitic gastroenteritis
Coccidiosis	Protozoa are generally not a primary condition but exist with other enteric diseases.	Common only l kids and yearlings. Diarrhea later becoming bloody and profuse dehydration and anemia
Cutaneous myasis or bowfly infestation	Maggots of calliphorine flies	Open wounds or sores with squirming maggots. May terminate toxemia gangrene or septic absorption.

Metabolic diseases of goats and sheep

Bloat	Occurs only under condition of management specially when feed lash herbage	Swollen left flank, which is resonant when tapped. Signs of colic such as uneasiness, difficult respiration, bloating and absence of rumen movements.
Milk fever	Actual: deficiency of calcium in the bloodstream Predisposing: grazing on young and rapidly growing forage, fasting or starvation in the last stages of pregnancy or during lactation, all forms of climatic and environmental stresses;	Initially, does show uncoordinated gait. There are fine tremors seen in the muscles. Weakness, staggering and finally recumbency. Doe becomes comatose and may lie in starched position with head twisted in the flank. Rapid breathing and frothing from nose and mouth. Death occurs within a few hours to a few days.

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	heavy concentrate feeding without calcium	
Grass tetany (hypomagnesia)	Actual deficiency of magnesium in the bloodstream together with physiologic disturbance mostly neuromuscular arising from combined defi-ciency and steroid action. Predisposing: lush pasture, overfertilization of pasture with K an N or confinement and overfeeding with concentrate deficient in Mg or low in energy; pregnancy and lactation; sudden management or environmental stresses	Initially, uneasiness, extreme sensitivity and muscular twisting. Sterna recumbency with rapid respiration, falling on side with paddling movements of the legs and stretching of the head. Animal is extremely alert or hyper excitable. Death may occur due to exhaustion form repeated attacks of convulsion.
Pregnancy toxemia (ketosis or acetonemia)	Actual acute glucose deficiency in the last stage of pregnancy due to twinning, large and rapid growing kid and sudden deficiency of feed. Predisposing: any sudden drop of intake attended (starvation due to feed withdrawal/isolation without feed, transport climatic changes) also poor health, parasitism and other concomitant	Sudden onset of depression or dullness; may appear uncoordinated. Neuromuscular disorders such as twitching and spasms especially of head muscles; hind limbs are spastic later becoming paralyzed. All are associated with concurrent pregnancy.

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	diseases.	
Urinary calculi	Mostly associated with faulty mineral nutrition, confinement or concentrate feeding, inadequate water intake, Vit. A deficiency and infections of the urinary tract with Mycoplasma and Ureaplasma	Commonly observed in adult male goats. Animal is uneasy or restless and shows stiff gait when moving or with hind legs under the body making frequent but unsuccessful urination. May show complete stoppage or dribbling of the urine due to obstruction in the urethra. Systemic signs may develop like fever, loss of appetite and increase respiration. Complete obstruction may cause uremia or death due to ruptured urinary bladder
Neonatal hypoglycemia	Direct or actual: low glucose level in the blood	Common in newly bon and young kids especially when unable to suck for approximately 36 hours. Lethargy, general weakness, recumbency, chilling. Low temp, coma and death. Nervous signs such as shivering, spasms or convulsions, champing of the jaw with frothy mouth, general in coordination. Older animals will show general wasting and weakness.
Acute indigestion (lactic acidosis or gain overload)	Sudden dietary changes such as exposure and feeding of concentrates after prolonged high roughage diet.	Signs appear from 10-36 hrs after dietary changes. Depression, loss of appetite, abdominal distention causing pain and discomfort. Diarrhea develops. Rapid respiration and pulse, incoordination, weakness, coma and death.

Some terms related to ruminant production

- Aftermath - forage grown following a harvest

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- Alternate stocking- the repeated grazing and resting of forage using two paddocks in succession
- Animal unit - one mature non-lactating bovine weighing 454 kg or 1000 kg and fed at a maintenance level, or the equivalent based on the average forage consumption of 12 kg DM/day.
- Animal Unit-day (AUD) – amount of forage required by an "animal unit" grazing for one day
- Animal Unit-week (AUW) – AUD x 7 days
- Animal Unit-month (AUM) – AUD x 30 days
- Browse - leaf and twig growth of shrubs, woody vines, trees, cacti and other non-herbaceous vegetation available for animal consumption
- Carrying capacity - the maximum stocking rate that will achieve a target level of animal performance, in a specified grazing method, that can be applied in over a defined time period without deterioration of the ecosystem
- Castration-done for purposes of preventing reproduction involving inferior bulls, improving carcass quality and for better temperament.
- Continuous stocking - a method of grazing livestock on a specific unit of land where animals have unrestricted and uninterrupted access throughout the time period when grazing is allowed
- Cutting interval – time elapsed between 2 successive cutting of forages.
- Daily dry matter intake or consumption - equal to 2.5-3.0% of body weight.
- Deferred grazing - the deferred grazing in a non-systematic rotation with other and units
- Disbudding – the process of stopping the growth of horns for economic and safety reasons.
- Forage - edible parts of plants, other than separated grain, that can provide feed for grazing animals, or that can be harvested for feeding
- Forbs - any herbaceous broad leaf plant that is not a grass and is not grass-like (sedges)
- Grazing pressure - the relationship between the number of animal units or forage intake units and the weight of forage dry matter per unit area at anyone point in time; an animal to forage relationship
- Grazing cycle - the time elapsed between the beginning of one grazing period and the beginning of the next grazing period in the same paddock where the forage is regularly grazed and rested.
- Grazing period - the length of times that grazing livestock or wildlife occupy a specific land area
- Rotational stocking - a grazing method that utilizes recurring periods of grazing and rest among two or more paddocks in a grazing management unit throughout the period when grazing is allowed.

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- Sward - a population of herbaceous plants, characterized by a relatively short habit of growth and relatively continuous ground cover, including above and below-ground parts
- Stocking density - the relationship between the number of animals and the specific unit of land being grazed at anyone point
- Stocking rate - the relationship between the number of animals and the grazing management unit utilized over a specified time period

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STATE UNIVERSITIES AND COLLEGES – ASSOCIATION OF COLLEGES OF AGRICULTURE IN THE PHILIPPINES (SUC-ACAP)

SELF-ASSESSMENT QUESTIONS

Animal Science

SUC-ACAP-LEA

2021

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PREFACE

Republic Act 8435 or the Agriculture and Fisheries Modernization Act of 1997, simply known as AFMA, was enacted on December 22, 1997. It is a policy instrument defining measures to modernize Philippine agriculture for the country to compete in the global market. A basic human resource development framework in modernizing agriculture under the Act is to enhance the professional competency of agriculturists by upgrading the standards of the agriculture and fisheries education that will ensure to similarly upgrade the quality and sustainability of, and promote global competitiveness, at all levels.

One strategy for professionalizing the agriculture sector is the Licensure Examination for Agriculturists (LEA). First implemented in 2003, LEA serves as a standardized evaluation tool to identify graduates of Agriculture and Agriculture-related degree programs who are academically and technically fit to be admitted in the agricultural profession. The three-day examination consists of multiple-choice type questions in the following component subjects: Crop Science, Soil Science, Crop Protection, Animal Science, Agricultural Economics and Marketing, and Agricultural Extension and Communication.

The COVID-19 pandemic has been a major concern of all higher education institutions. When review sessions for licensure board exams are already supposed to be grinding, SUCs are confronted by the reality that face-to-face review is undeniably not possible. Under the circumstance, universities face new pressures to provide an engaging, motivating, and effective teaching and learning experiences for both reviewees and reviewers.

Following this, the SUC-ACAP, with its goal of advancing the agriculture professional competency among their member-institutions, and the College of Agriculture and Food Science, UP Los Baños embarked on a joint initiative to enhance the capability of the SUC faculty particularly those engaged in the conduct of review classes in preparation for the Licensure Exam for Agriculture (LEA). To achieve this, the joint initiative conducted three online lectures, namely, 1) Innovative Teaching Methods and Tools Designed for Online Review Classes by no less than the Chancellor of the UP Open University, Dr. Melinda dP. Bandelaria. This online course aimed at introducing the faculty participants to the particularities of conducting review classes online with the use of innovative methods and tools; 2) Test Construction: the Multiple-Choice Type Test from a seasoned UP faculty, Dr. Blanda R. Sumayao, which gave the faculty-participants the opportunity to review the principles and rules for constructing multiple-choice type tests; and 3) Design and Development of Review Materials for LEA, again with Dr. Bandelaria with the objective of providing the faculty-participants the necessary knowledge and skills in designing and developing review materials and planning for the delivery of online review sessions.

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The Project initiative also gave the faculty-participants the opportunity to review their course syllabus viz-a-viz CMO #14 S. 2008 or the CHED-prescribed outline which enabled them to design and develop this harmonized syllabus for each of the six core subject matter area. The review materials are thus a product of many minds and talents, that is, of faculty-participants from the different SUCs. I would like to acknowledge all the writers/contributors who willingly presented themselves to the initiative and for strictly following deadlines in submitting drafts; to all University Presidents and the Deans of the Colleges of Agriculture who have been highly committed to the objective of this Project initiative and for the support provided to the faculty-participants; to all Cluster and Sub-Cluster Coordinators for the constant reminders and follow-ups; and to the SUC-ACAP Secretariat for the untiring administrative support.

And lastly, we are grateful to the College of Agriculture and Food Science, UP Los Baños for being true to its commitment of being a public service University, providing scholarly and technical assistance to its sister and brother SUCs. Specifically, we thank the CAFS faculty who served as Subject Area Coordinators/Specialists who unselfishly guided our faculty participants throughout their journey in the project, as well as the Technical Assistants and Administrative Staff who have lend full support to attaining the objectives of the joint initiative.

As I have always reiterated in many occasions, let us be reminded that student preparation for the licensure is just one of the factors that would determine performance. It is still incumbent upon us SUCs or higher educational institutions to ensure that we will be able to provide the necessary prerequisite services and institutional support in order for students to ultimately pass successfully and help advance the students' professional careers post licensure exam.

The preparation for publication of these review materials was supported by the SUC-ACAP.

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December 2021

FOREWORD

Unlike in *Lord of the Rings* and *Harry Potter* films, there are no wizards, Balrogs, Voldemort, precious rings, or elderly wands, on the journey to becoming a professional agriculturist. But there are narrow bridges and mazes that you must travel and cross over. One of these challenges is the Professional Licensure Exam.

In the current climate where globalization, changing natural conditions, production, and social composition; migratory movements, among others, have significantly impacted most agricultural landscapes in the developing world, more than ever, the Licensure Exam for Agriculture has become all the more relevant as it prepares graduates of Agriculture to embrace the tasks ahead of them with utmost professionalism by way of recognizing their expert knowledge and practice. The Licensure Examination for Agriculture was established to recognize the expert knowledge and practice of aspiring professional agriculturists.

This Review Material produced by the faculty participants of the joint initiative of the UPLB College of Agriculture and Food Science (UPLB-CAFS)- State Universities and Colleges-Association of Colleges of Agriculture in the Philippines (SUC-ACAP) is a comprehensive resource that covers all relevant subject areas in the field of agriculture which hopes to serve as a valuable review tool to students of SUC-ACAP member institutions.

I wish you the best in your journey and beginning your career as a professional agriculturist.



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GUIDE TO THE STUDENT

This module was prepared in practice exam format rather than a reviewer with textual content of information of all you have taken in your Animal Science courses. We believe that your textbooks and lecture syllabi will do a better job in providing you a deeper understanding of the concepts and fundamentals in animal science, and you do not need another material that repeats the same thing. With the student in mind, we learn best when we are asked questions that may mimic your licensure exam. This module may be good practice for you. The animal science faculty of different SUCs worked together in preparing these questions for you. Remember to read the explanation after the question regardless if you get the answer wrong or right. If you need more background, you can always refer to your textbooks and lecture notes. Have a strategy. If there are questions that you find difficult, you can use them as a guide to topics that you probably need to review. At the end, if you repeat answering this reviewer enough, you will be able to do well in the animal science section of the LEA and hopefully will help you pass the licensure exam

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Unit I

ANATOMY AND PHYSIOLOGY OF FARM ANIMALS

Instructions: Read each question carefully. Choose the best answer that satisfies the question or statement.

1. Two mechanisms by which the skin helps prevent heat loss when an animal is in a cold environment.

    a. Sweating and panting
    b. Shivering and sweating
    c. Shivering and vasoconstriction
    d. Sweating and vasoconstriction

    Answer: C

    Explanation: Shivering thermogenesis is the contraction of skeletal muscles to produce additional heat at the thoracic part of the body. Vasoconstriction is the reduction of peripheral blood capillary diameter to reduce heat loss from the skin.

2. Also called the superficial fascia, it consists of areolar tissue and adipose tissue that connect the skin to muscles.

    a. Dermis
    b. Epidermis
    c. Adipose tissue
    d. Subcutaneous tissue

    Answer: D

    Explanation: Subcutaneous tissue, also called the superficial fascia, consist of the areolar tissue and adipose tissue that connect the skin to the muscles.

3. Two neurons communicate via the ____.

    a. Axon
    b. Synapse
    c. Cell body
    d. Dendrites

    Answer: B

    Explanation: Synapse is the point of contact between two neurons.

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4. These are hairs that converge to form hair streams and vortices that are used as requisites in the registration of large animals.

    a. Cowlicks
    b. Fleece
    c. Fur
    d. Curls

    Answer: A

    Explanation: Cowlicks are hair on the body of the animal that converges to form hair streams and vortices that is used as requisites in the registration of large animals

5. A small muscle that makes the hair "stand up" when the animal is frightened or cold is called ____.

    a. Arrector pili muscle
    b. Stratum corneum
    c. Extensor muscle
    d. Erector muscle

    Answer: A

    Explanation: The arrector pili muscle is a tiny muscle connected to each hair follicle and the skin. When it contracts it causes the hair to stand erect, and a "goosebump" forms on the skin.

6. It is any changes in the environment of a nerve which if large enough will cause the nerve to produce an action potential and transmit a nerve impulse.

    a. Electrical current
    b. Sensation
    c. Signal
    d. Stimulus

    Answer: D

    Explanation: When nerve fiber is stimulated by a stimulus, it comes in a state of local excitation. The stimulus may be physical (heat, cold, pressure, etc.) mechanical or chemical or electrical in nature.

7. It is the part of the peripheral nervous system that innervates smooth muscles, cardiac muscles, and glands.

    a. Autonomic nervous system
    b. Parasympathetic nervous system
    c. Sympathetic nervous system
    d. Somatic nervous system

SUC-ACAP-LEA

Answer: A

Explanation: The ANS is the part of the nervous system that regulates activity in viscera and other structures not normally under voluntary control. These are the nerves that influence activity in smooth muscle, cardiac muscle, and glands.

8. It is the relay center connecting the cerebrum and cerebellum to the spinal cord that contains the control center for respiration; controls heart rate; controls blood supply and regulation of body heat and some digestive processes.

   a. Medulla oblongata
   b. Pons
   c. Brain Stem
   d. Spinal Nerve

   Answer: C

   Explanation: Brain stem (medulla oblongata and pons) is relay center connecting the cerebrum and cerebellum to the spinal cord; contains the control center for respiration; controls heart rate; controls blood supply and regulation of body heat and some digestive processes

9. The passage of nerve impulse is much faster in ___.

   a. Myelinated axon
   b. Sympathetic nerve
   c. Parasympathetic nerve
   d. Unmyelinated axon

   Answer: A

   Explanation: In myelinated nerve fiber, the medullary sheath is impermeable to ions. So action potential occurs only at some points called nodes of Ranvier. So, the action potential is conducted from node to node in a jumping manner. This conduction is 20 times faster than in non-myelinated nerve fiber.

10. Other than the nervous system, what system is also involved in regulating all body processes.

    a. Cardiovascular system
    b. Endocrine system
    c. Muscular system
    d. Urinary system

    Answer: B

    Explanation: The two systems involved in coordination are the nervous and endocrine systems. The first operates via electrical impulses along nerve fibers and the second by releasing special chemicals or hormones into the bloodstream from glands.

SUC-ACAP-LEA

11. It is the usual site of fertilization in the female genital system.

a. Ovaries
b. Uterus
c. Oviducts
d. Cervix

Answer: C

Explanation: Fertilization of the oocyte by the spermatozoa takes place in the uterine tubes called the oviducts (fallopian tubes). The oviducts then conducts the fertilized egg to the uterus where implantation occurs.

12. The funnel-shaped structure of the oviduct which picks up the egg when released by the ovary is called ____.

a. Uterine horn
b. Uterus
c. Fallopian tube
d. Infundibulum

Answer: D

Explanation: Infundibulum (or funnel) is located adjacent to the ovary and with long segments enclosing the ovary, the infundibulum collects the yolk after its release from the follicle as a funnel and directs it into the oviduct.

13. It is the site of implantation for the fertilized ovum in livestock.

a. Oviduct
b. Cervix
c. Ovaries
d. Horn of the uterus

Answer: D

Explanation: The horn of the uterus is the organ which serves as the site of implantation for the fertilized egg.

14. Which of the animals has bicornuate uterus?

a. Cattle
b. Sheep
c. Goat
d. Swine

Answer: D

SUC-ACAP-LEA

Explanation: Bicornuate uterus like swine have horns which are folded and convoluted and the body is short and not prominent while bipartite uterus (cattle, sheep, goat) has prominent uterine body.

15. Its opening closes when the animal gets pregnant to protect the uterine content.

a. Cervix
b. Ox cordi
c. Uterus
d. Os uteri

Answer: D

Explanation: The opening in the center of the cervix is called the internal os or the os uteri controlling what is allowed or not allowed to enter.

16. During cold weather, this muscle retracts the testicles against the body.

a. Retractor penis muscle
b. Ductus deferens
c. Epididymis
d. Cremaster muscle

Answer: D

Explanation: The tunica dartos and cremaster muscle are the thermoregulatory muscles of the testes. The cremaster assist in drawing closer to the body when the temperature is low.

17. ____ is the primary sex organ of the female animal. It is responsible for the production of the ovum and the female gonadal hormones.

a. Ovary
b. Vagina
c. Cervix
d. Testes

Answer: A

Explanation: Just like the testes, the ovaries are the primary organ of reproduction in female animals.

18. An animal whose testes remained inside the body and failed to descend to scrotum is called ____.

a. Capon
b. Cryptorchid
c. Castrate
d. Eunuch

SUC-ACAP-LEA

Answer: C

Explanation: A testis that fails to descend into the scrotum is called a cryptorchid testis, and the animal with such a condition is called a cryptorchid (Greek crypt, hidden; orcho, testi).

19. Accessory sex gland that provides nutrition to the spermatozoa.

a. Cowper's gland
b. Seminal vesicle
c. Prostate gland
d. Bulbous glands

Answer: B

Explanation: The seminal vesicles contains the sugar fructose, proteins, citric acid, inorganic phosphorus, potassium, and prostaglandins. It provides nutrition for protection and survival of the sperm.

20. The seat of spermatozoa production is ____.

a. Seminiferous tubules
b. Epididymis
c. Vas deferens
d. Vas efferentia

Answer: A

Explanation: The epididymis are the major sites for storage of spermatozoa while the seminiferous tubules which is found inside the testes are responsible for the production of spermatozoa.

21. Polyestrous animals are those that come in heat all throughout the year. Which of the following is an example of polyestrous animal?

a. Cattle
b. Goat
c. Horse
d. Bitch

Answer: D

Explanation: Animals that have multiple estrous cycles only during certain periods of the year are called seasonally polyestrous animals (horses, sheep, goats, deer, cats) while polyestrous animals are animals whose estrous cycle occurs all throughout the year (cattle, pigs, mites and rats).

SUC-ACAP-LEA

22. At what phase of estrus cycle does ovulation takes place?

a. Diestrus
b. Metestrus
c. Estrus
d. Proestrus

Answer: C

Explanation: Estrus which is the period of sexual receptivity wherein the female accepts the (standing heat) is the period when ovulation usually occurs.

23. The lifespan of ejaculated mammalian spermatozoa in the female reproductive part is about ___.

a. 24 hours
b. 5 days
c. 1 week
d. 2 weeks

Answer: A

Explanation: In most mammals the life span of ejaculated spermatozoa is 24 hours and about 14 days in chicken.

24. Part of the oviduct in chicken which is responsible for the secretion of albumin is called ____.

a. Uterus
b. Magnum
c. Isthmus
d. Fimbriae

Answer: B

Explanation: Secretion of albumin begins in the infundibulum but majority is secreted in the magnum.

25. Which of the hormone is responsible for the ovulation of mature follicle at the ovarian surface?

a. Progesterone
b. Luteinizing hormone
c. Estrogen
d. Follicle stimulating hormone

Answer: B

SUC-ACAP-LEA

Explanation: Estrogen stimulates production of luteinizing hormone which in turn causes LH surge necessary to induce ovulation of maturing follicles.

26. The site of fertilization in chicken.

a. Fimbriae
b. Uterus
c. Infundibulum
d. Isthmus

Answer: C

Explanation: The first part of the oviduct, the infundibulum (or funnel) is 3 to 4 inches long and engulfs the ovum released from the ovary is the part where fertilization is likely to occur.

27. Sexual receptivity which coincides with estrus in animals that exhibit estrous cycle is due to large amount of _____.

a. Testosterone
b. Estrogen
c. Progesterone
d. Relaxin

Answer: B

Explanation: Estrus, the period of sexual receptivity, is primarily initiated by the elevation in estrogens from mature follicles just prior to ovulation.

28. Estrogen is the female hormone secreted by the ovarian follicle, it is involved in the following functions except ____.

a. Stimulation of duct growth in the mammary glands
b. initiation of sexual receptivity
c. regulation of secretion of luteinizing hormone
d. promotion of the lobuloalveolar growth in the mammary gland

Answer: D

Explanation: Estrogen stimulates development of the duct system and progesterone stimulates the development of the lobulo-alveolar system.

29. The structural and functional unit of the kidneys is called ____.

a. Alveoli
b. Glomerulus
c. Nephron
d. Nephrite

SUC-ACAP-LEA

Answer: C

Explanation: The kidney is composed of over a million nephron. A nephron is the basic functional unit of kidney. Its chief function is to regulate the concentration of water and soluble substances like sodium salts by filtering the blood, reabsorbing what is needed and excreting the rest as urine.

30. It is the organ that regulates the volume and composition of internal fluid environment.

a. Heart
b. Endocrine glands
c. Kidney
d. Blood

Answer: C

Explanation: The kidney is essential in homeostasis maintaining a constant internal environment of the body's extracellular fluids.

31. Major site of gas exchange that secrete the pulmonary surfactant.

a. Pulmonary alveoli
b. Type I pneumocyte
c. Type II pneumocyte
d. Alveolus

Answer: C

Explanation: Type II pneumocyte or granular pneumocyte that secretes pulmonary surfactant (DPPC, dipalmitoylphoshatidylcholine, lipoprotein complex) that reduces surface tension.

32. Volume of air inspired or expired during normal breath

a. Inspiratory reserve volume
b. Expiratory reserve volume
c. Residual volume
d. Tidal volume

Answer: D

Explanation: Tidal volume. Inspiratory reserve volume is the amount of air that can be expelled from the lung following a normal expiration. Expiratory reserve volume is the maximal amount of air that can be expelled from the lung following a normal expiration. Finally, residual volume is the amount of air remaining in the lung after a maximal forced expiration.

SUC-ACAP-LEA

33. Muscle that raises ribcage during Inhalation/Inspiration.

a. External intercostals
b. Sternocleidomastoideus
c. Serratus dorsalis
d. Scalenus

Answer: A

34. Muscle that lifts sternum upward during inhalation/inspiration

a. External intercostals
b. Sternocleidomastoideus
c. Serratus dorsalis
d. Scalenus

Answer: B

Explanation: Serratus dorsalis, together with serratus ventralis, lift many ribs during inhalation. Scalenus lifts the first two ribs.

35. A strong expiratory effort made with glottis closed.

a. Valsalva's maneuver
b. Inspiratory capacity
c. Vital capacity
d. Functional residual capacity

Answer: A

Explanation: Inspiratory capacity is the maximal amount of air that can be inspired after a normal expiration. Vital capacity is the maximal amount of air that can be moved into and out of lungs with muscular effort. Functional residual capacity is the amount of air remaining in the lungs after a normal expiration.

36. This is the expiratory control center.

a. Upper pons
b. Dorsal medulla
c. Lower pons
d. Ventral medulla

Answer: D

Explanation: The ventral medulla is active during forceful expiration (forced breathing). The upper pons, lower pons and the dorsal medulla are the pneumotaxic center, apneustic center, and inspiratory center, respectively.

SUC-ACAP-LEA

37. Reflex for inflation and deflation that prevents overdistension and extreme deflation of lung chemical control of respiration.

a. Hering-breuer reflex
b. Sneeze reflex
c. cough reflex
d. Pneumatic reflex

Answer: A

Explanation: Sneeze reflex is the nasal defense mechanism. Cough reflex is the larynx/trachea/bronchi defense mechanism. There is no such thing as pneumatic reflex.

38. This animal has 50-60 tracheal rings.

a. Horse
b. Ox
c. Pig
d. Dog

Answer: A

Explanation: Ox – 50; Pig – 32-35; Dog – 40-45

39. The lungs of this animal are not divided into lobes by deep fissures.

a. Horse
b. Ox
c. Pig
d. Dog

Answer: A

Explanation: All of the other animals in the options are divided into lobes by deep fissures.

40. Considered to be the "voice box" of the respiratory system.

a. Trachea
b. Pharynx
c. Larynx
d. Bronchus

Answer: C

SUC-ACAP-LEA

41. Blood vessels that carry blood from the heart to the organs.

a. Arteries
b. Veins
c. Capillaries
d. Coronary Sinus

Answer: A

Explanation: Veins – blood vessels that carry blood from the organs to the heart. Capillaries – any of the minute blood vessels that form networks throughout the bodily tissues. Coronary sinus- -a collection of smaller veins that merge together to form the sinus (or large vessel), which is located along the heart's posterior (rear) surface between the left ventricle and left atrium.

42. A valve between the right atrium and the right ventricle.

a. Tricuspid valve
b. Bicuspid valve
c. Atrio-ventricular valve
d. Pulmonary semilunar valve

Answer: A

43. Amount of blood injected in the left ventricle in a minute.

a. Heart rate
b. Cardiac output
c. Cardiac cycle
d. Pulse rate

Answer: B

Explanation: Heart rate- one heartbeat in one cardiac cycle; Cardiac cycle- events that takes place in one heartbeat. Pulse rate- rate in pulse in one minute

44. A double walled sac around the heart.

a. Epicardium
b. Myocardium
c. Endocardium
d. Pericardium

Answer: D

Explanation: Epicardium- outer sac covering the heart, Myocardium- middle sac covering the heart, Endocardium- inner layer sac covering the heart.

SUC-ACAP-LEA

45. A major blood vessel shaped like a question mark and ascends to form an arch.

a. Pulmonary trunk
b. Superior vena cava
c. Posterior vena cava
d. Aorta

Answer: D

46. Major blood vessel of the heart that carry blood to and from the lungs.

a. Pulmonary trunk
b. Superior vena cava
c. Posterior vena cava
d. Aorta

Answer: A

Explanation: Superior vena cava- supplies blood from the upper parts of the body; Posterior vena cava- supplies blood from the lower parts of the body.

47. In heart circulation, blood from the body going to the right atrium will enter to ____.

a. Left atrium
b. Right ventricle
c. Pulmonary artery
d. Left ventricle

Answer: B

48. From the answer to number 7, blood will enter to ____.

a. Left atrium
b. Right ventricle
c. Pulmonary artery
d. Left ventricle

Answer: C

49. Blood from the answer to number 8, will then enter to the left ventricle via

a. Left atrium
b. Right ventricle
c. Pulmonary vein
d. Left ventricle

Answer: C

SUC-ACAP-LEA

50. Phase of cardiac cycle that occurs during ventricular diastole with all the valves closed.

a. Isovolumetric Relaxation
b. Rapid atrial ejection
c. Isovolumetric contraction
d. Ventricular ejection

Answer: A

Explanation:
Rapid atrial ejection – atria is ejecting, there is filling; atrio ventricular valves are open, semi lunar valves are closed.
Isovolumetric contraction - all valves are still closed
Ventricular ejection - full ventricular systole opens semilunar valves.

51. The only one direct function of vertebrate muscles is ____.

a. Relaxation
b. Contraction
c. Extension
d. Formation

Answer: B

Explanation: Vertebrate muscle has only one function: contraction. "Relaxation" is a passive process, a lack of contraction. Thus, the varied functions of the muscular system are all based on the contraction (or shortening) of muscle fibers. Layers of smoothness in the walls of the stomach and intestines contract to mix and propel food along the gastrointestinal tract; smooth muscle layers in the walls of blood vessels control the distribution of blood, which is propelled by the contraction of the cardiac muscle of the heart.

52. What is the hardest working muscle in the vertebrate animal?

a. Masseter
b. Soleus
c. Tongue
d. Heart

Answer: D

Explanation: The hardest working muscle is the heart. It pumps out 2 ounces (71 g) of blood at every heartbeat. Daily the heart pumps at least 2,500 gallons (9,450 liters) of blood. The heart can beat over 3 billion times in a person's life.

SUC-ACAP-LEA

53. The type of muscle that is most familiar as the meat of our domestic animal is ____.

a. Skin muscle
b. Skeletal muscle
c. Cardiac muscle
d. Smooth muscle

Answer: B

Explanation: The bulk of the muscle in the body is skeletal muscle, and it is responsible for producing the voluntary movements of the limbs, trunk, and head. is also the muscle tissue with which are most familiar as the meat of our domestic animals.

54. What type of muscle tissue found in the system of the body with automatic functions, which are major components of the wall of organs of the digestive system, urogenital system, and most blood vessels?

a. Intestinal muscle
b. Skeletal muscle
c. Cardiac muscle
d. Smooth muscle

Answer: D

Explanation: Smooth muscle is found in the system of the body with autonomic function. Thus, smooth muscle is a major component of the wall of organs of the digestive and urogenital system and most blood vessels.

55. A science which deals with the study of the Muscular System is known as ____.

a. Biology
b. Neurology
c. Myology
d. Osteology

Answer: C

Explanation: Myology is the scientific study of muscles.

56. The kind of toxin which produces spasm of the masseter muscles (lockjaw) followed by other muscles was coming from bacteria called ____.

a. *Clostridium tetani*
b. *Clostridium botulinum*
c. *Clostridium histolyticum*
d. *Clostridium sordellii*

Answer: A

SUC-ACAP-LEA

Explanation: The disease caused by the toxin from *Clostridium tetani*, which produces a spasm of the masseter muscle (lockjaw) followed by a spasm of other muscles, is also specifically referred to as tetanus.

57. This is the chief muscle of inspiration, that separates thoracic and abdominal cavities, and creates a partial vacuum in the thorax, causing air to enter the lungs.

a. Sphincters
b. Diaphragm
c. Alveoli
d. Blood vessels

Answer: B

Explanation: The diaphragm is the chief muscle of inspiration. It is a dome-shaped sheet of muscle separating the thoracic and abdominal cavities.

58. Muscles that tend to pull a limb toward the median plane are adductors, and those that tend to move the limb away from the median are ____.

a. Adductors
b. Abbducers
c. Abductors
d. Adoptors

Answer: C

Explanation: Muscles that tend to pull a limb toward the median plane are adductors, and those that tend to move the limb away from the median plane are abductors.

59. What nutrient is the major chemical component of muscles?

a. Carbohydrates
b. Protein
c. Minerals
d. Water

Answer: B

Explanation: Protein is the main component of muscles, bones, organs, skin, and nails. Excluding water, muscles are composed of about 80% protein, making this nutrient especially important for athletes.

SUC-ACAP-LEA

60. Muscles that encircle an opening, whether they are striated or smooth which control the passage of food from the stomach is known as ____.

a. Pyloric valve
b. Intestine valve
c. Sphincters
d. Diaphragm

Answer: C

Explanation: Muscles that encircle an opening, whether they are striated or smooth, are sphincters, that control the passage of food from the stomach.

61. Used primarily for acquiring, cutting and/or crushing, and mixing food with saliva.

a. Mouth
b. Teeth
c. Lips
d. Tongue

Answer: A

Explanation: Mouth is used primarily for acquiring, cutting and/or crushing, and mixing food with saliva.

62. _____ common passage of food and air.

a. Larynx
b. Pharynx
c. Esophagus
d. Trachea

Answer: B

Explanation: Pharynx is a common passage of food and air, it is caudal to oral and nasal cavities, and is lined by mucous membrane and surrounded by muscles.

63. The esophagus joins the stomach at the _____.

a. Pylorus
b. Cardia
c. Sphincter
d. None of the above

Answer: B

Explanation: The esophagus joins the stomach at the cardia, a part of the stomach so named because of its proximity to the heart.

SUC-ACAP-LEA

64. Reticulum and rumen are often collectively called as _____.

a. Reticulumen
b. Rumenuticulum
c. Ruminoreticulum
d. Reticulorumeno

Answer: C

Explanation: Because of their functional and anatomic relatedness, the reticulum and rumen are often collectively called ruminoreticulum.

65. The most cranial compartment of the ruminant forestomach.

a. Reticulum
b. Rumen
c. Omasum
d. Abomasum

Answer: A

Explanation: The reticulum is the most cranial compartment of the forestomach.

66. Compartment of the ruminant forestomach also called as the "honeycomb".

a. Reticulum
b. Rumen
c. Omasum
d. Abomasum

Answer: A

67. _____ is a spherical organ filled with muscular laminae that lie in sheets, much like the pages of the book.

a. Abomasum
b. Omasum
c. Rumen
d. Reticulum

Answer: B

Explanation: The omasum is a spherical organ filled with muscular laminae that lie in sheets, much like the pages of the book giving the omasum its colloquial name, *book stomach*.

SUC-ACAP-LEA

68. It is the first glandular portion of the ruminant digestive tract also called the "true stomach".

a. Omasum
b. Abomasum
c. Reticulum
d. Rumen

Answer: B

Explanation: The abomasum (true stomach) is the first glandular portion of the ruminant digestive system.

69. The longest part of the small intestine.

a. Duodenum
b. Jejunum
c. Ileum
d. Cecum

Answer: B

Explanation: The jejunum is the longest part of the small intestine.

70. Largest gland in the body.

a. Skin
b. Liver
c. Pancreas
d. Heart

Answer: B

Explanation: The liver is the largest gland in the body constituting 1-2% of total adult body weight.

71. The act of bringing food into the mouth.

a. Prehension
b. Mastication
c. Chewing
d. Regurgitation

Answer: A

Explanation: The act of bring food into the mouth is prehension. The teeth, lips, and tongue are used as prehensile organs.

SUC-ACAP-LEA

72. Salivary enzyme with antibacterial actions.

a. Amylase
b. Hydrolase
c. Oxidase
d. Lysozyme

Answer: D

Explanation: Lysozyme is a salivary enzyme with antibacterial actions.

73. Starch-digesting enzyme present in the saliva of omnivores but absent in ruminants.

a. Lysozyme
b. Amylase
c. Hydrolase
d. Oxidase

Answer: B

Explanation: Amylase the starch-digesting enzyme is present in the saliva of omnivores (pigs) and to a limited degree in horse but absent in ruminants and carnivores.

74. Are produced by fermentation of carbohydrates consumed by ruminants, including carbohydrates produced by the actions of microbial cellulase.

a. Volatile fatty acids
b. Non-volatile fatty acids
c. Cholesterol
d. Triglycerides

Answer: A

Explanation: Volatile fatty acids are produced by fermentation of carbohydrates consumed by ruminants, including carbohydrates produced by the actions of microbial cellulase.

75. A rounded mass of food.

a. Ingesta
b. Cud
c. Bolus
d. Food mass

Answer: A

SUC-ACAP-LEA

76. _____ is a protective response to remove potentially harmful ingesta from the stomach and upper small intestine.

a. Swallowing
b. Regurgitation
c. Rumination
d. Emesis

Answer: D

Explanation: Emesis or vomiting is a protective response to remove potentially harmful ingesta from the stomach and upper small intestine.

77. The process of returning food to the mouth from the forestomach.

a. Rumination
b. Regurgitation
c. Remastication
d. Resalivation

Answer: B

Explanation: Regurgitation is the process of returning food to the mouth from the forestomach.

78. Refers to the combination of substances secreted into the stomach lumen by gastric glands.

a. Gastric juice
b. Enzymes
c. Ingesta
d. Chylomicrons

Answer: A

Explanation: Gastric juice refers to the combination of substances secreted into the stomach lumen by gastric glands.

79. A mushy, semisolid mixture of food, water and gastric juice.

a. Ingesta
b. Cud
c. Chyme
d. Bolus

Answer: D

Explanation: After the food is chewed and mixed with saliva, a bolus (rounded mass of food) forms and is move to the upper surface of the tongue.

SUC-ACAP-LEA

80. The primary digestive function of the liver is to provide _____ which facilitate the enzymatic digestion of lipids.

a. Enzyme
b. Stones
c. Bile
d. Bile salts

Answer: D

Explanation: The primary digestive function of the liver is to provide bile salts which facilitate the enzymatic digestion of lipids.

81. _____ are responsible for bile formation.

a. Hepatocytes
b. Enzymes
c. Liver
d. Small intestine

Answer: A

Explanation: Hepatocytes or liver cells are responsible for bile formation.

82. Small droplets formed in the intestinal chyme that contain lipids, bile salts, and products of lipid digestion.

a. Mecilles
b. Micelle
c. Chyme
d. Chylomicrons

Answer: B

Explanation: Micelle is a term for small droplets formed in the intestinal chyme that contain lipids, bile salts, and products of lipid digestion.

83. Finger-like projections that increase the surface area for nutrient absorption.

a. Lumen
b. Lacteal
c. Villi
d. Crypt of Lieberkuhn

Answer: C

Explanation: Villi are finger-like projections that increase the surface area of the small intestine for nutrient absorption.

SUC-ACAP-LEA

84. Salivary gland located ventral to the ear in relation to the caudal border of the mandible.

a. Mandibular
b. Parotid
c. Sublingual
d. None of the above

Answer: B

Explanation: Parotid salivary gland located ventral to the ear in relation to the caudal border of the mandible.

85. Consists of muscular tube lined with mucous membrane that is continuous with the external skin at the mouth and at the anus. Its primary functions are prehension, mastication, digestion, and absorption of foods, and elimination of undigested/ unabsorbed wastes.

a. Circulatory system
b. Nervous system
c. Respiratory system
d. Digestive system

Answer: D

Explanation: Digestive system consists of muscular tube lined with mucous membrane that is continuous with the external skin at the mouth and at the anus. Its primary functions are prehension, mastication, digestion, and absorption of foods, and elimination of undigested/ unabsorbed wastes.

SUC-ACAP-LEA

Unit II

ANIMAL NUTRITION

Instructions: Read each question carefully. Choose the best answer that satisfies the question or statement.

1. It is the breakdown of feed particles into suitable products for absorption that includes mechanical forces, chemical action and enzyme activity.

    a. Digestion
    b. Absorption
    c. Metabolism
    d. Nutrition
    e. None of the above

    Answer: A

2. The series of processes by which an organism takes in and assimilates food for promoting growth and replacing worn-out or injured tissues.

    a. Digestion
    b. Absorption
    c. Metabolism
    d. Nutrition
    e. None of the above

    Answer: D

3. It is the cheapest and most abundant nutrient that makes up 65 – 85 % of the animal's body weight at birth and 45 – 60% of the body weight at maturity

    a. Carbohydrates
    b. Fats
    c. Proteins
    d. Minerals
    e. Water

    Answer: E

4. The dietary requirement of this nutrient is highest in young growing animals and decline gradually to maturity

    a. Carbohydrates
    b. Fats
    c. Proteins
    d. Minerals
    e. Water

SUC-ACAP-LEA

Answer: C

5. A classification of nutrient that makes up approximately ¾ of plant dry weight and thus the largest part of the animals feed supply

   a. Carbohydrates
   b. Fats
   c. Proteins
   d. Minerals
   e. Water

   Answer: A

6. It is an organic nutrient required in small quantities necessary for regulating metabolic processes, but does not become an actual component of body structures

   a. Vitamins
   b. Lipids
   c. Water
   d. Minerals
   e. Proteins

   Answer: A

7. A class of nutrients that yields 2.25 times more energy than carbohydrates or proteins

   a. Vitamins
   b. Fats
   c. Water
   d. Minerals
   e. Proteins

   Answer: B

   Explanation: Fats or lipids are a group of compounds with C, H, and O just like carbohydrates; however, there is much more of carbon and hydrogen than oxygen. Fats are insoluble in water, but soluble in organic solvents like petroleum ether, hexane, etc. they are high energy materials, supplying 2.25 more energy than carbohydrates and proteins. Some oil-bearing plants include coconut, oil palm, rapeseed, sunflower, soybean, etc. Pork fat is called lard, and beef fat is tallow.

SUC-ACAP-LEA

8. It is determined as the loss in weight that results from drying a known weight of feed to constant weight at 100°C

   a. Crude protein
   b. Ash
   c. Ether extract
   d. Crude fiber
   e. Moisture content

   Answer: E

9. It is determined by ignition of a known weight of the feed at 550°C until all carbon has been removed.

   a. Crude protein
   b. Ash
   c. Ether extract
   d. Crude fiber
   e. Moisture content

   Answer: B

10. It is calculated from the nitrogen content of the feed and determined by a modification of a technique originally devised by Kjeldahl over 100 years.

    a. Crude protein
    b. Ash
    c. Ether extract
    d. Crude fiber
    e. Moisture content

    Answer: A

11. This fraction is determined by subjecting the food to a continuous extraction with petroleum ether for a defined period.

    a. Crude protein
    b. Ash
    c. Ether extract
    d. Crude fiber
    e. Moisture content

    Answer: C

12. It is determined by subjecting the residual food from the ether extraction to successive treatments with boiling acid and alkali of defined concentration. It contains cellulose, lignin and hemicelluloses.

SUC-ACAP-LEA

a. Crude protein
b. Ash
c. Ether extract
d. Crude fiber
e. Moisture content

Answer: D

13. Its fraction includes starch, sugars, fructans, pectins, organic acids and pigments.

a. Crude protein
b. Ash
c. Nitrogen free extract
d. Crude fiber
e. Crude fat

Answer: C

14. All are examples of essential amino acids EXCEPT for _____

a. Alanine
b. Arginine
c. Leucine
d. Lysine
e. Isoleucine

Answer: A

Explanation: Essential amino acids are those amino acids that must be supplied in the diet because the animal cannot synthesize or cannot synthesize them fast enough to meet its requirement. Examples of essential amino acids: threonine, valine, tryptophan, isoleucine, lysine, leucine, phenylalanine, methionine, histidine, arginine

15. It is considered the first limiting amino acid in swine nutrition

a. Alanine
b. Arginine
c. Leucine
d. Lysine
e. Isoleucine

Answer: D

SUC-ACAP-LEA

16. It is considered the first limiting amino acid in poultry nutrition

    a. Alanine
    b. Arginine
    c. Leucine
    d. Lysine
    e. Methionine

    Answer: E

17. All are examples of non-essential amino acids EXCEPT for _____

    a. Cysteine
    b. Glycine
    c. Proline
    d. Serine
    e. Isoleucine

    Answer: E

    Explanation: Non-essential amino acids are amino acids which are essential to the animal but are normally synthesized or sufficient in the diet and need not be supplemented. Examples of non-essential amino acids: alanine, asparagine, aspartic acid, cysteine, cystine, glutamic acid, glutamine, glycine, hydroxyproline, proline, serine, tyrosine.

18. It is an example of a trace mineral.

    a. Cu
    b. Zn
    c. Co
    d. F
    e. All of the above

    Answer: E

    Explanation: Microminerals (trace minerals) are normally present at low levels in the animal body or needed in very small amounts in the diet. Examples are: Co, Fe, Cu, I, Mn, Mo, Se, Zn, F, Ni, Cr

19. It is an example of a macro mineral.

    a. Ni
    b. Zn
    c. I
    d. Ca
    e. F

    Answer: D

SUC-ACAP-LEA

Explanation: Macrominerals are normally present at greater levels in the animal body or needed in relatively large amounts in the diet. Examples are: Mg, P, Ca, Cl, Na, K, S

20. It is the most abundant mineral element in the animal body.

    a. Ni
    b. Zn
    c. I
    d. Ca
    e. F

    Answer: D

21. It is a disease caused by carbohydrates deficiency.

    a. Marasmus
    b. Kwashiorkor
    c. Avitaminosis
    d. all of the above
    e. none of the above

    Answer: A

22. It is an example of a fat-soluble vitamin

    a. Cobalamin
    b. Retinol
    c. Biotin
    d. Thiamine
    e. Ascorbic Acid

    Answer: B

    Explanation: Examples of Fat-soluble vitamins are: (Vit. A) Retinol, (Vit. D) Ergosterol, (Vit. E) Alpha-tocopherol, (Vit. K) Phylloquinone

23. It is an example of a water-soluble vitamin.

    a. Inositol
    b. Vitamin B1
    c. Vitamin B2
    d. Vitamin B6
    e. all of the above

    Answer: E

SUC-ACAP-LEA

Explanation: Examples of water-soluble vitamins: Vit. B1(Thiamine), Vit. B2 (Riboflavin), Vit. B6 (Pyridoxine), Vit. B12 (Cobalamin), Niacinamide, Pantothenic Acid, Vitamin H (Biotin), Folic Acid, Vit. C (Ascorbic Acid)

24. It is a part of the stomach that is primarily mucus producing cells.

a. Cardiac region
b. Pyloric region
c. Peptic gland region
d. Duodenum
e. Ileum

Answer: B

25. It is the last part of the small intestine that is involved in nutrient absorption.

a. Cardiac region
b. Pyloric region
c. Peptic gland region
d. Duodenum
e. Ileum

Answer: E

26. The first section of the large intestine that is relatively large in the horse and rabbit

a. Cecum
b. Colon
c. Rectum
d. Duodenum
e. Ileum

Answer: A

27. A favorable condition in the rumen for the microorganisms.

a. Anaerobic environment
b. Moisture
c. Removal of toxic end-products
d. Constant warm temperature
e. All of the above

Answer: E

SUC-ACAP-LEA

28. It can be described as controlled vomiting.

    a. Esophageal groove
    b. Rumination
    c. Eructation
    d. Belching
    e. none of the above

    Answer: B

    Explanation: Rumination is a controlled set of contractions of the esophagus, reticulum and rumen that allows the ingesta to be regurgitated back up to the esophagus where fluids are swallowed again and additional re-mastication and re-swallowing of solids occurs.

29. It is a Law also known as the "Livestock and Poultry Feeds Act", which governs manufacture, importation, labelling, advertising, distribution, and sale of livestock and poultry feeds and feedstuff

    a. Administrative Order No.35 s. 1975
    b. Administrative Order No.40 s. 1976
    c. Republic Act 1556 as amended by Presidential Decree No.7
    d. Republic Act No 3720 as amended by Executive Order No 175
    e. Republic Act No 6675

    Answer: C

30. This organ serves as an out-pocketing of the esophagus that provides storage for consumed food.

    a. Proventriculus
    b. Crop
    c. Gizzard
    d. Gastric Stomach

    Answer: B

31. This organ receives digestive enzymes and bicarbonate from the pancreas and bile from the liver.

    a. Duodenum
    b. Jejunum
    c. Ileum
    d. Large Intestine

    Answer: A

SUC-ACAP-LEA

32. It is a feed ingredient commonly added to chicken feed that aids in digestion.

a. Marble
b. Saw dust
c. Rice hull
d. Grits

Answer: D

33. It is a flexible tube that connects the mouth with the rest of the digestive tract.

a. Pharynx
b. Esophagus
c. Trachea
d. Ceca

Answer: B

34. What is the process of removing the gases in ruminants?

a. Farting
b. Eructation
c. Bloating
d. None of the above

Answer: B

Explanation: Eructation is the process of belching of gas which allows removal of large volumes of gases produced in the rumen. In this process, contractions of the upper part rumen force the gas up the esophagus and from there, the gas penetrates into the trachea and lungs.

35. It is the regurgitation of ingesta from the reticulum, followed by remastication and reswallowing.

a. Eructation
b. Defecation
c. Rumination
d. All of the above

Answer: C

Explanation: Rumination is the process of movement of ingesta back-up the esophagus to the mouth for additional mechanical breakdown – "chewing the cud".

SUC-ACAP-LEA

36. This organ produces hormones most notably glucagon and insulin, which are the primary regulators of blood sugar levels.

a. Pancreas
b. Liver
c. Kidney
d. Spleen

Answer: A

37. The part of an organ where bile salts from the liver and pancreas are released for fat digestion.

a. Ceca
b. Duodenum
c. Jejunum
d. Ileum

Answer: B

Explanation: Duodenum is an active site of digestion that receives secretions from the pancreas, liver and intestinal walls.

38. This mineral is needed for hydrochloric acid production in the stomach.

a. Chlorine
b. Potassium
c. Choline
d. Magnesium

Answer: A

39. The following are examples of hexoses except:

a. Ribose
b. Galactose
c. Glucose
d. Xylose

Answer: A

40. It is classified as less than ten simple sugars

a. Monosaccharides
b. Disaccharides
c. Oligosaccharides
d. Polysaccharides

SUC-ACAP-LEA

Answer: C

41. This carbohydrate is obtained from the hydrolysis of starch.

    a. Glucose
    b. Mannose
    c. Fructose
    d. Galactose

    Answer: C

42. It is a nitrogen-containing polysaccharide which is a principal constituent of the exoskeleton of insects, crustaceans and arachnids.

    a. Pectin
    b. Dextrin
    c. Chitin
    d. Cellobiose

    Answer: C

43. Its structure contains an aldehyde or a ketone group in their structure

    a. Carbohydrates
    b. Proteins
    c. Minerals
    d. Fats

    Answer: A

44. The principal function of this nutrient is to provide the amino acids needed for the synthesis of body cells and tissues

    a. Carbohydrates
    b. Proteins
    c. Minerals
    d. Fats

    Answer: B

45. The following are group of carbohydrates except:

    a. Starch
    b. Fiber
    c. Legume
    d. Sugar

    Answer: C

SUC-ACAP-LEA

46. This nutrient deficiency in humans can lead to the development of beriberi.

    a. Vitamin A
    b. Vitamin B1
    c. Vitamin D
    d. Vitamin E

    Answer: B

47. A mineral mainly found in the skeleton and the remainder in soft tissues and body fluids.

    a. Manganese
    b. Potassium
    c. Calcium
    d. Magnesium

    Answer: D

48. The water content from feeds ranges between

    a. 5-10%
    b. 8-30%
    c. 18-30%
    d. 5-8%

    Answer: B

49. The following are starch except:

    a. Amylose
    b. Amylopectin
    c. Dextrin
    d. Glycogen

    Answer: C

50. All of the B vitamins contain nitrogen except:

    a. Thiamine
    b. Riboflavin
    c. Nicotinamide
    d. Inositol

    Answer: D

SUC-ACAP-LEA

51. The major movement of food through the GI tract is called

a. Peristalsis
b. Segmentation
c. Hostile churning
d. Chyme propulsion

Answer: A

Explanation: The primary functions of the GI tract include prehension of food and water; mastication, salivation, and swallowing of food; digestion of food and absorption of nutrients; maintenance of fluid and electrolyte balance; and evacuation of waste products. These functions can be broadly characterized as: motility, secretion, digestion, absorption, blood flow, and metabolism. Normal GI tract motility involves peristalsis, muscle activity that moves ingesta from the esophagus to the rectum; segmentation movements, which churn and mix the ingesta; and segmental resistance and sphincter tone, which retard aboral progression of gut contents. In ruminants, these movements are of major importance in normal forestomach function.

52. Which of the following enzymes digests protein?

a. Pepsin
b. Lipase
c. Trypsin
d. Pepsin and Trypsin

Answer: D

53. How does bile aid in digestion?

a. It breaks down proteins in chemical digestion to prepare for absorption.
b. It neutralizes the acid from the stomach.
c. It breaks down fats to smaller particles in chemical digestion to prepare for absorption.
d. It breaks down carbohydrates into smaller molecular chains in chemical digestion to prepare for absorption.

Answer: C

Explanation: Digestion is the process of mechanically and enzymatically breaking down food into substances for absorption into the bloodstream. Bile is fluid that is made and released by the liver and stored in the gallbladder. Bile helps with digestion by breaking down fats into fatty acids, which can be taken into the body by the digestive tract.

SCC-ACAP-LEA

54. What would result if a disaccharide went through hydrolysis?

a. You would get two disaccharides.
b. You would get a disaccharide and a monosaccharide.
c. You would get a polysaccharide.
d. You would get two monosaccharides.

Answer: D

Explanation: Carbohydrates are also known as sugars or saccharides. Hydrolysis is a reaction with water. Hydrolysis reactions are also referred to as hydrolytic reactions. Acid hydrolysis of disaccharides produces monosaccharides by breaking the glycosidic links (ether bonds) between monomer units in the structure of the molecule.

55. Which of the following is the proper order of structures that food must pass through on its way to the stomach?

a. Mouth, oropharynx, laryngopharynx, esophagus, stomach
b. Mouth, oropharynx, esophagus, stomach
c. Mouth, nasopharynx, oropharynx, laryngopharynx, esophagus, stomach
d. Mouth, laryngopharynx, oropharynx, nasopharynx, stomach

Answer: A

56. How does food move through the digestive tract?

a. By gravity
b. By wavelike muscle contractions
c. By cilia
d. By chemical absorption

Answer: C

57. What is the scientific name of Madre de agua?

a. *Lantana camara*
b. *Trichanthera gigantea*
c. *Gliricidia sepium*
d. *Centrocema pubescens*

Answer: B

58. The main functions of the digestive system are

a. Ingestion and digestion
b. Propulsion and secretion
c. Absorption and elimination
d. all of the above

SLC-ACAR-LEA

Answer: D

59. All of the following are organs of the digestive system *except* the

a. Nasopharynx
b. Liver
c. Tongue
d. Esophagus

Answer: A

60. Which selection includes *only* accessory digestive organs?

a. Salivary glands, thyroid gland, pancreas, liver
b. Stomach, duodenum, pancreas, gallbladder
c. Gallbladder, liver, pancreas, salivary glands
d. Liver, thyroid gland, gallbladder, spleen

Answer: C

61. Which list proceeds from the superior to the inferior end of the stomach?

a. Pylorus, fundus, cardia, body
b. Cardia, body, fundus, pylorus
c. Cardia, fundus, body, pylorus
d. Body, cardia, pylorus, fundus

Answer: C

62. Which sequence lists the regions of the large intestine in order, from the end of the ileum to the anus?

a. Cecum, rectum, anal canal, colon
b. Colon, rectum, anal canal, cecum
c. Cecum, colon, rectum, anal canal
d. Colon, cecum, rectum, anal canal

Answer: C

63. Which of the following proteins is the major component of hairs, nails, hooves, horns, and teeth.

a. Albumin
b. Collagen
c. Elastin
d. Keratin

SIC-ACAP-LEA

Answer: D

64. Rickets, osteomalacia and osteoporosis are condition associated with deficiency in _____.

a. Calcium
b. Iodine
c. Iron
d. Magnesium

Answer: A

65. Micromineral essential for the formation of hemoglobin related to oxygen transport and utilization.

a. Manganese
b. Iodine
c. Iron
d. Zinc

Answer: C

66. Nutrients must be broken down into simple molecules before it will be absorbed. The basic unit of protein is _____.

a. Amino acid
b. Fatty acid
c. Glycerol
d. Simple sugar

Answer: A

67. It works as a biological antioxidant and as a detoxifying agent.

a. Ascorbic acid
b. Cyanocobalamin
c. Menadione
d. Tocopherol

Answer: D

68. A vitamin that promotes healthy immune system, heals wound and aids in absorption of iron.

a. Ascorbic acid
b. Cyanocobalamin
c. Retinol
d. Tocopherol

SUC-ACAP-LEA

Answer: A

69. It is a by-product of oil extraction from dried coconut meat.

a. Coco meat
b. Canola meal
c. Copra meal
d. Hominy

Answer: C

70. Full-fat soybean meal is derived from _____

a. Drying soybean
b. Extracting of oil from soybean
c. Grinding whole fresh soybean
d. Grinding whole dried soybean

Answer: C

71. Limestone, sodium chloride and monocalcium phosphate are examples of ____.

a. Feed additives
b. Feed Supplements
c. Feedstuff
d. Ration

Answer: B

72. _____refers to a chemical produced by microorganisms which can be added to a ration in a sub-therapeutic level.

a. Antioxidants
b. Antimold
c. Antibiotics
d. Surfactants

Answer: C

73. Live beneficial microorganisms added to the diet to improve the microbial balance of the animals is called

a. Antibiotics
b. Prebiotics
c. Probiotics
d. Toxin binders

Answer: C

SUC-ACAP-LEA

74. Flubendazole, mebendazole and piperazine are examples of

a. Antibiotics
b. Pigmenters
c. Dewormers
d. Flavorings

Answer: C

75. The following are classified as protein feed except

a. Soybean meal
b. Tallow
c. Copra meal
d. Fish meal

Answer: B

76. The following are feed materials containing more than 18% crude fiber and are generally low in energy, except ____

a. Hay
b. Silage
c. Soilage
d. Rice straw

Answer: C

77. Poor feed quality and adulterated feed ingredients are the two major problems in the feed milling industry, thus a careful feed quality evaluation is required. If the evaluation is made with the naked eye or with the use of hand lens or compound microscope, it is called _____

a. Chemical Analysis
b. Microscopy
c. Sensitivity Analysis
d. Visual Evaluation

Answer: B

78. It is a process by which different feed ingredients are proportionally combined to give the animals the required amount of nutrients they need.

a. Feed analysis
b. Feed formulation
c. Feed milling
d. Feed adulteration

Answer: B

SUC-ACAP-LEA

79. In commercial feed milling, formulating ration is accomplished through ____

a. Linear programming
b. Trial and Error
c. Pearson Square
d. B & C

Answer: A

80. A practice of increasing doubling the amount of feed for about 7-14 days prior to insemination in sows or feeding concentrates to ruminants.

a. Finishing
b. Flushing
c. Conditioning
d. All of these

Answer: B

81. Increased bulk density, reduced dustiness, prevent animal from sorting feed ingredients, improved feed efficiency and digestibility are some of the advantages of

a. Mash
b. Crumbles
c. Pellets
d. Whole grain

Answer: C

82. The recommended moisture content of most mixed ration is _____%. This is to prevent microbial growth, chemical change, and freshness.

a. 8-10
b. 10-12
c. 12-14
d. 14-16

Answer: B

83. A feed mixing equipment characterized by a U-shape bin which contains a central mixing blade mounted on a rotating shaft.

a. Horizontal mixers
b. Rotary mixer
c. Micro mixer
d. Vertical mixer

Answer: A

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84. It is used to crash or powder feedstuff to improve ingestion, uniform mixing and digestibility.

a. Mortal and pestles
b. Grinders
c. Hammer mill
d. Chopper

Answer: C

85. The common silage' additives use to promote microbial growth by supplying energy for them to produce acid that serve as preservative

a. Salt
b. Inoculant
c. Molasses
d. Urea

Answer: C

86. To facilitate drying of hay crops after harvesting, it is recommended to pass it through a set of rollers to crack or open the stems. This practice is called__

a. Breaking
b. Crashing
c. Conditioning
d. Squeezing

Answer: C

87. The best time to harvest forage for ensilaging, hay making and soiling is during___.

a. Early vegetative
b. Late flowering
c. Late vegetative
d. Early flowering

Answer: B

88. Paragrass is scientifically known as _____. `

a. *Brachiria decumbens*
b. *Brachiria mutica*
c. *Brachiria humidicola*
d. *Brachiaria ruziziensis*

Answer: B

SUC-ACAP-LEA

89. A swine producer wants to balance a 100-kg diet for sows that contains 16% CP with the following ingredients: 20% ground hay (fixed); 10% oats (fixed), 5% vitamins and mineral premix (fixed), corn, and soybean. The compositions of the feed ingredients are as follows:

Feed Ingredient	**% Crude Protein**
Hay	6.0
Oats	12.0
Premix	0.0
Corn	10.0
Soybean Meal	44.0

Solution:

20% ground hay 6% CP
10% oats (fixed) 12% CP
5% vitamins and minerals 0
Corn 10% CP
Soybean meal 44% CP

20 ´ (6/100) = 1.2
10 ´ (12/100) = 1.2
5 ´ (0/100) = 0
--- ---
35 2.4% CP of the fixed ingredients

100 – 35 = 65
16 – 2.4 = 13.6%

Adjusted % CP requirement = (13.6/65) ´ 100 = 20.92

Corn 10 23.08
Soybean meal 44 10.92

34

(23.08/34) ´ 100 = 67.88%

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(10.92/34) ´ 100 = 32.12%

To Check:

(67.88/100) ´ 65 = 44.12
(32.12/100) ´ 65 = 20.88

44.12 ´ 10/100 = 4.41 Corn
20.88 ´ 44/100 = 9.19 Soybean meal
1.20 Ground hay
1.20 Oats

16% CP

90. A beef producer found a good buy on wheat and desires to balance a diet that will contain wheat and grain sorghum at a 60:40 ratio in the grain mix. The producer wants to use hay fixed at 20% of the diet and use Cotton Seed Meal (CSM) as the other ingredient. The producer desired a diet balanced for 14% CP. The grain mix will be 60% wheat (13% CP) and 40% Sorghum (9% CP). The hay will be fixed at 20% of the total diet and contains 8% CP and CSM contains 41% CP.

Solution:

Hay 20% (fixed) 8% CP
CSM 41% CP

Grain mix :
Wheat 60% 13% CP
Sorghum 40% 9% CP

20 ´ 8/100 = 1.6% CP in Hay
14 – 1.6 = 12.4

100 – 20% (Fixed hay) = 80%

Adjusted % CP requirement = 12.4/80 ´ 100 = 15.5% CP

Protein content:
60% Wheat ´ 13 = 7.80%
40% Sorghum ´ 9 = 3.60%

11.14% Grain

Grain 11.14 25.5

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CSM 41.00 4.1

29.6

Grain 25.5/29.6 ´ 80 = 68.92
CSM 4.1/29.6 ´ 80 = 11.08
Wheat 68.92/100 ´ 60 = 41.35
Sorghum 68.92/100 ´ 40 = 27.57

To Check:

20.00 ´ 8/100 = 1.60 Hay
11.08 ´ 41/100 = 4.54 CSM
41.35 ´ 13/100 = 5.38 Wheat
27.57 ´ 9/100 = 2.48 Sorghum

14.00%

91. Stella wanted to produce 50-kg ration of pre-starter mash with 20% CP. Given the available raw materials in the locality of Barili, this includes: Yellow corn (YC) with 8.5% CP, Meat and Bone meal (MBM) 50% at 50.6% CP, Soybean Meal (SBM) 44% at 43% CP, and Soybean hull (SBH) at 9% CP.

Solution:

Energy Source:
100 ´ 8.5/100 = 8.5 YC

Protein Feed:
33.33 ´ 50.6/100 = 16.86 MBM
33.33 ´ 43.0/100 = 14.33 SBM
33.33 ´ 9.0/100 = 3.00 SBH

34.19

Yellow Corn 8.50 14.19
Protein Feed 34.19 11.50

25.69

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14.19/25.69 ´ 100 = 55.24 Yellow Corn
11.50/25.69 ´ 100 = 44.76 Protein Feed

100.00

44.76 ´ 33.33/100 = 14.92

To check:

55.23 ´ 8.5/100 = 4.69
14.92 ´ 50.5/100 = 7.55
14.92 ´ 43.0/100 = 6.42
14.92 ´ 9.0/100 = 1.34

20.00

92. Balance a ration with 16.80% CP given the following feed ingredients and their respective ratios:

Energy Source Ratio 3:1:2	**Protein Source Ratio 2:1**
White Corn – 8.5%	Copra Meal - 18%
Camote Meal – 3.5%	Fish Meal (imported) – 60%
Corn Gluten Feed - 17%	

Solution:

Energy source: 3:1:2

White Corn	8.5 ´ 3 =	25.5
Camote Meal	3.5 ´ 1 =	3.5
Corn Gluten Feed	17 ´ 2 =	34.0
	-----	--------
	6 parts	63

SUC-ACAP-LEA

63/6parts = 10.5 Energy source

Protein Source: 2:1

Copra Meal	18 ´ 2 =	36.0
Fish Meal (imported)	60 ´ 1 =	60.0
	3 parts	96

96/3 parts = 32 Protein source

Energy Source	10.5	15.2	
Protein Source	32		6.3
			21.5

15.2/21.5 ´ 100 = 70.70
6.3/21.5 ´ 100 = 29.30

To check:

White Corn	70.70 ´ 3/6 =	35.35	´ .085 = 3.00
Camote Meal	70.70 ´ 1/6 =	11.78	´ .035 = 0.41
Corn Gluten Feed	70.70 ´ 2/6 =	23.57	´ .170 = 4.01
Copra Meal	29.30 ´ 2/3 =	19.53	´ .180 = 3.52
Fish Meal (imported)	29.30 ´ 1/3 =	9.77	´ .600 = 5.86
			16.80

SUC-ACAP-LEA

Unit III

GENETICS AND LIVESTOCK IMPROVEMENT

Instructions: Read each question carefully. Choose the best answer that satisfies the question or statement.

1. All of the choices describe animal breeding except one. Which one is it?

   a. It is the application of animal genetics principles.
   b. Its goal is to genetically improve farm livestock.
   c. It aims to improve the quantity, efficiency, quality, and aesthetic value of farm animals and their products.
   d. All of the choices
   e. None of the choices

   Answer: E

   Explanation: All of the above choices describe animal breeding. Animal breeding is the application of the principles of genetics in order to improve farm livestock. With the principles and concepts of heredity and variation, animal breeders are able to manipulate the genetic composition of animals resulting to a desired genetic improvement, which encompasses a wide range of aspects including the quantity, efficiency, quality, and aesthetic value of farm animals along with their products.

2. Genetics deals with the study of heredity and variation. The unit or factor which is of prime importance to studying heredity is the gene. Which of the following is not true about the gene?

   a. It is the basic unit of heredity.
   b. Genes are passed on from one generation to another.
   c. A gene is a segment of DNA.
   d. Both A and C
   e. None of the choices

   Answer: E

   Explanation: None of the given choices is correct. In other words, choices A, B, and C are all true about the gene. It is the basic unit of inheritance. It is well-known that it is a segment of the DNA, which is being passed on from one generation to another, i.e., from parents to offspring.

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3. In 1909, this Danish botanist coined the term "gene" to refer to the particulate factor which was hypothesized by Mendel as the basic unit heredity.

   a. William Bateson
   b. Wilhelm Ludvig Johannsen
   c. Francis Crick
   d. James Watson
   e. Gregor Mendel

   Answer: B

   Explanation: Wilhelm Ludvig Johannsen coined the term "gene" to describe the particulate factor which was hypothesized by Mendel as the basic unit of heredity. Apart from that, he also introduced the terms "phenotype" and "genotype". William Bateson, on the other hand, was an English biologist who, in 1906, showed that Mendelian laws applied to animals by producing the first evidence of inheritance through experiments with chickens. He provided the classical definition of genetics as a field of study, and he coined the terms "homozygote," "heterozygote," and "allelomorph". Francis Crick and James Watson were recognized for having described the chemical structure of the chromosome as a double helical structure of DNA. Meanwhile, Gregor Mendel did not yet know the term "gene" during his time, hence, he only used the term "factor" in his common garden pea breeding experiment.

4. The factors influencing the expression of a phenotype of an individual for a particular trait are the genotype, environment, and their interaction. The genotype by environment interaction is equated to zero if

   a. Individuals are raised in the same environment.
   b. Individuals are of the same breed.
   c. Individuals given the same diet are raised in the same pen.
   d. Different genotypes receive different management.
   e. Different genotypes respond similarly in the same environment.

   Answer: E

   Explanation: The G x E interaction is the interaction between the genotype of the individual and the environment under which it is raised. When (GxE)=0, this means that any of the two is zero (i.e., either same genotype responds similarly in different environments, or different genotypes raised in the same environment show no difference). Hence, there is no interaction effect observed. For choice "a", the genotype of the individuals was not specified which makes it incomplete. For "b", these individuals were of the same genotype; description of the environment was not mentioned. Choice "c" is incorrect because what is only described is the environment (i.e., diet and pen); no description of the genotype mentioned. Lastly, when different genotypes receive different management, then there is a possibility of GxE interaction (i.e., (GxE)≠0 because the response of the genotypes could vary across management.

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5. Who among the following scientists believed that superior parents were more likely to produce superior progeny that were inferior parents?

   a. William Bateson
   b. Gregor Mendel
   c. Robert Bakewell
   d. Jay Lush
   e. None of the choices

   Answer: C

   Explanation: Robert Bakewell, the father of animal breeding, believed that "like begets like" because from his observations on his animal breeding experiments, superior parents were more likely to produce superior progeny than were inferior parents. Jay Lush, on the other hand, believed that "like does not always beget like."

6. Which of the following is not a function of the basic unit of inheritance?

   a. Store and transmit information from one cell to another and from parent to offspring
   b. Replicate itself with great consistency and precision
   c. Does not undergo mutation or error during replication
   d. Both A and B
   e. Both B and C

   Answer: C

   Explanation: The basic unit of inheritance is the gene, a segment of the DNA or the genetic material. As a unit of heredity, the gene is responsible in the storage and transmission of genetic information from one cell to another and from parent to offspring as well as copy and replicate itself with great consistency and precision. Along with these functions, the gene undergoes mutation or "error" during replication which would subsequently be copied and replicated. Without the occurrences of mutation or errors, there would be no observed genetic differences among animals within populations. Hence, the gene has to undergo this error, although this is rare, because mutation is the ultimate source of variation, a very important raw material for selection to occur and consequently for evolution to take place.

7. Which of the following is true about heredity?

   a. Heredity is the transfer of genetic material from parents to offspring.
   b. It is only possible through the process of reproduction.
   c. It is made possible through meiosis and fertilization.
   d. A and B only
   e. Choices A, B, and C are true about heredity.

   Answer: E

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Explanation: Heredity, or the transfer of genetic material from parents to offspring, is only made possible through the process of reproduction, and this process of reproduction is achieved through two specific processes called meiosis and fertilization. The reproductive cells, or otherwise known as gametes, undergo meiotic division to produce haploid cells each carrying only half of the genetic complement of the individual. Through the process of fertilization, the gametes (i.e., sperm and egg) unite which then subsequently restores the diploid constitution of the fertilized egg.

8. All of the following are phenotypes except _____.

   a. litter size
   b. coat color
   c. drooping ears
   d. polled
   e. A and B

   Answer: E
   Explanation: A phenotype is that which is observed or measured on an animal or an organism for a particular trait. An example of a trait is coat color. The possible phenotypes for coat color could be roan, red, or white. Also, litter size is a trait, wherein the phenotypes that could be recorded for this trait could be the number of piglets born in one farrowing, such as 11, 12, 14, and so on. Hence, among the choices, the phenotypes are drooping ears and polled. Litter size and coat color are examples of traits.

9. The genetic information is located in which part of the cell?

   a. ribosomes
   b. nucleus
   c. cytoplasm
   d. cell membrane
   e. DNA

   Answer: B

   Explanation: It is in the nucleus where DNA, the genetic material, is located in eukaryotic organisms. More specifically, the chromosomes or the thread-like structures made of an organized package of DNA along with protein known as histones providing structural support, are altogether located inside the nucleus.

10. Hen-day egg production, parity, and calving interval are examples of _____ traits.

    a. sex-linked
    b. sex-limited
    c. sex-influenced
    d. qualitative
    e. quantitative

    Answer: B

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Explanation: Sex-limited traits are determined by genes located on autosomes and are expressed only in one sex (Lee, 2016). In this case, the genes for a sex-limited trait are present in both sexes, however, expression is only in one sex, that could either be expressed in males only or females only, not both. Examples of which are hen-day egg production, parity, and calving interval. All of which can be determined in females only. Males do not produce eggs and do not give birth, hence these traits are not observed in them. Moreover, the answer cannot be D because among the traits, only 'parity' is considered a qualitative trait while hen-day egg production and calving interval are both quantitative traits. Sex-linked traits are those traits which are linked to or located on the sex chromosome. In other words, when a gene for a particular trait is present on the X chromosome, the mode of inheritance is X-linked, and when a gene for a trait is on the Y chromosome, this is usually called Y-linked. However, since the Y chromosome is relatively smaller in size, it contains fewer genes, making it difficult to detect Y linkage. Hence, sex-linkage generally refers to the mode of inheritance for X-linked traits. The answer to the question cannot be letter "a" because X-linked traits can be expressed in both sexes, although it is usually males who are affected because they have a single copy of X chromosome that carries the mutation. In females, the effect of the mutation may be masked by the second healthy copy of the X chromosome (Sex Linked (genome.gov). In the case of sex-influenced traits, these are autosomal traits influenced by sex. For instance, if a male has one recessive allele, he will show that trait, but it will take two recessives for the female to show that same trait (Sex-Linked, Limited, and Influenced, n.d.).

11. Which of the following is true about the equation P=G+E+(GxE), where P refers to the phenotype, G is the genotype, and E, the environment?

    a. The factors affecting the expression of the phenotype of an individual are genotype and environment.
    b. Environment refers to the totality of the non-genetic factors affecting the phenotype of an individual.
    c. The interaction of the G and E could be equated to zero when individuals of different genotypes respond similarly to the same environment.
    d. All of the choices
    e. None of the choices

    Answer: D

    Explanation: Genotype and environment play very important roles in the expression of a phenotype for a particular trait. The genotype x environment interaction (GxE) is the interplay of genetic and nongenetic effects causing differential relative performances of genotypes in different environments. For example, a genotype × environment interaction may be a change in the relative performance of a character of two or more genotypes measured in two or more environments (Osei et al., 2018). Genotype refers to the genetic composition of an individual while environment encompasses all non-genetic factors that influence the phenotype. This could include management factors such as production and management practices.

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12. Non-random mating occurs when the probability that two individuals in a population will mate is not the same for all possible pairs of individuals. Which of the following is an example of non-random mating?

    a. Roan coat-colored males mated to red coat-colored females only
    b. Roan coat-colored males mated roan coat-colored females only
    c. All males of different coat colors have equal chances of mating the opposite sex
    d. Both A and B
    e. Both A and C

    Answer: D

    Explanation: Non-random mating means that mate selection is influenced by phenotypic differences based on underlying genotypic differences (Lecture 17: Biology 102, n.d.). It occurs when rules dictating the mating strategy are imposed (Bondoc, 2008). Hence, based from choices A and B, mating is preferred to particular phenotypes (and its corresponding genotypes) only, i.e., roan to red and roan to roan mating only. For letter C, there is equal chances of being selected as a mate, hence, this is considered random. In particular, non-random mating could be assortative or disassortative. Assortative mating describes situations in which individuals mate with other individuals similar to themselves, while disassortative mating (also called negative assortative mating) describes situations in which individuals mate with other individuals different from themselves (Singh et al., 2021). In other words, this type of mating is based on phenotypes.

13. Gregor Mendel's breeding experiment on the common garden pea (*Pisum sativum*) was not much recognized until these three scientists independently rediscovered Mendel's works in 1900.

    a. James Watson, Francis Crick, and Rosalind Franklin
    b. William Bateson, Hugo de Vries, and Carl Correns
    c. Wilhelm Johannsen, William Bateson, and Carl Correns
    d. Hugo de Vries, Carl Correns, and Erich von Tschermak-Seysenegg
    e. Robert Bakewell, Jay Lush, and Ronald Fisher

    Answer: D

    Explanation: It was in 1865 when Gregor Mendel published his works on the common garden pea, particularly the Laws of Particulate Inheritance. However, Mendel's work remained unnoticed until Hugo de Vries, Carl Correns, and Erich von Tschermak-Seysenegg independently rediscovered the laws of Mendel in 1900 (Corcos & Monaghan, 1990). The three scientists are European botanists in particular. Correns was a German, de Vries was a Dutch, and Tschermak-Seysenegg was an Austrian.

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14. Who among the following scientists applied Mendelian laws on chickens?

a. William Bateson
b. Wilhelm Johannsen
c. Gregor Mendel
d. Carl Correns
e. Jay Lush

Answer: A

Explanation: William Bateson was an English biologist who demonstrated that Mendel's laws can also be applied to animals. He provided the first evidence of inheritance through experiments with chickens. Using crossbreeding and hybridization experiments, Bateson demonstrated that both plant and animal populations exhibit Mendelian principles (William Bateson (1861-1926) | The Embryo Project Encyclopedia, n.d.). Wilhelm Johannsen, on the other hand, coined the term "gene" and proposed the distinction between genotype and phenotype in the study of heredity while working in Denmark in 1909 (Peirson, 2012). Gregor Mendel studied the transmission of the genetic material in plants, specifically the common garden pea (Pisum sativum). Carl Correns was one of the three scientists who independently rediscovered Mendel's laws. Lastly, Jay Lush is regarded as the father of modern animal breeding, and he made significant contributions to livestock breeding.

15. The genetic material of all cells which is situated primarily in the chromosomes is called

a. Deoxyribonucleic acid
b. Ribonucleic acid
c. Gene
d. Chromatin
e. Nucleotide

Answer: A

Explanation: The genetic material of all cells is the deoxyribonucleic acid, or DNA (Alliance & Screening Services, 2009). It is located in the nucleus primarily in the chromosomes, although DNA is also present in the mitochondria of eukaryotes. It is also important to note that DNA does not directly travel from the nucleus to the site of protein synthesis (i.e., ribosomes) but has to be converted first to RNA before it is translated to proteins. It must be noted, however, that the processes of replication, transcription, and translation do not occur in a chronological manner. This means that replicated DNA does not always undergo transcription and subsequent translation. It depends on the need of the organism. The primary role of ribonucleic acid or RNA is the conversion of the information stored in the DNA into proteins. It acts as a messenger between DNA and protein synthesis through complexes known as ribosomes, forms vital portions of ribosomes, and serves as an essential carrier molecule for amino acids to be used in protein synthesis (Rao & Arora, 2020). Meanwhile, the gene is just a segment of the DNA, which can have various forms called allelomorphs or simply alleles. Genes contain the code for a specific protein. Since genes are segments of DNA, they are also situated in the chromosomes. The chromatin consists of the unraveled condensed structure of DNA for packaging into the nucleus. The

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chromatin is different from the chromosome in that the chromosome consists of the highest condensed structure of the DNA double helix for the proper separation of the genetic material between daughter cells during cell division (Panawala, 2017). Lastly, nucleotide is the basic building block of nucleic acids. DNA and RNA are made of long chains of nucleotides and each nucleotide consists of a sugar molecule attached to a phosphate group and a nitrogen-containing base (Nucleotide, n.d.).

16. Which of the following statements is true about the action of genes?

a. Genes may be active only when they occur in pairs of alleles during the haploid phase.
b. Genes may be active only when they occur in pairs of alleles during the diploid phase.
c. In the additive type of gene action, neither allele is considered dominant.
d. In the additive type of gene action, both alleles are dominant.
e. Both B and C

Answer: E

Explanation: Genes may be active only when they occur in pairs of alleles during the diploid phase. The term "gene action" refers to the way in which genes exert their effects on the body. The action of genes may be detected only from the phenotype. There are two general types of gene action, namely additive and non-additive gene action. In the additive type of gene action, the phenotypic effect of one gene adds to the phenotype of its own allele or other genes in the genotype, which affect the expression of the trait. In other words, each of two alleles contributes equally to the production of phenotypes, i.e., neither allele is dominant. On the other hand, non-additive gene action could be dominance or epistasis, wherein the phenotypic expression of one gene does not add to the phenotypic expression of the other (Bondoc, 2008).

17. All of the following are examples of a trait except one. Which one is it?

a. Ear orientation
b. Comb type
c. Presence of horns
d. Plumage color
e. Silky feathers

Answer: E

Explanation: Ear orientation, comb type, presence of horns, and plumage color are all examples of a trait while a silky feather is considered a phenotype. A trait is a specific characteristic of an individual. It is something that is observable and quantifiable. In contrast to a trait, a phenotype is that which is being recorded and measured. Say, for instance, the trait is ear orientation; the phenotypes could be drooping ears or erect ears. Phenotype is more specific. It characterizes the trait. The possible phenotypes for presence of horns could be polled and horned.

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18. In the equation P=G+E+(GxE), the portion that is passed on from parents to offspring is the G or genotype. Which of the following is not an example of an environment (E) affecting the expression of the phenotype P of an individual?

    a. Type of diet
    b. Housing system
    c. Breed of the animal
    d. Both A and B
    e. Both B and C

    Answer: C

    Explanation: As previously mentioned, E encompasses the non-genetic factors or elements that affect the expression of a phenotype of an individual. Among the choices, only the breed of the animal is the genetic factor. The rest – type of feed and housing system – and other production and management practices are all non-genetic factors.

19. Gene P codes for polledness in cattle. Heifers and bulls carrying the PP and Pp genotypes are polled. Which of the following describes the type of gene action exhibited?

    a. P is dominant over p.
    b. The type of gene action exhibited is additive.
    c. Individuals with Pp genotypes can still become horned because they carry a recessive allele.
    d. Epistasis is observed.
    e. None of the choices is correct.

    Answer: A

    Explanation: The type of gene action exhibited is dominance, wherein there is interaction of alleles to give such a phenotypic effect. More specifically, the P allele masked the effect of p allele, hence, what is expressed is the dominant one, which is the polled phenotype. Since there is masking or concealing, then the gene action cannot be additive. Choice C is incorrect because the presence of two recessive p alleles is required for the expression of the horned phenotype. Also, epistasis is not observed in this situation because only one locus is considered, i.e., P-p locus.

20. This refers to the interaction between genes at two or more loci.

    a. Dominance
    b. Co-dominance
    c. Epistasis
    d. Linkage
    e. Both A and B

    Answer: C

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Explanation: The correct answer is epistasis. Epistasis means "non-allelic interaction". Meaning to say, the alleles that interact to express the phenotype for a particular trait are in different loci. Oftentimes, epistasis is called simply as "interaction" to denote the interaction of non-allelic genes. In contrast to epistasis, what interacts to produce a particular phenotype in the case of dominance are the alleles that are in the same locus. In linkage, non-allelic genes affecting two traits are present in the same chromosome. They do not interact in a way that they mask, conceal, or block the expression of the alleles of another gene but rather they tend to be inherited together as single group.

21. All of the following statements describe the unit of heredity except one. Which one is it?

a. It is located on the chromosome.
b. It lies outside the nucleus.
c. One of its forms may conceal the effect of its alternative form.
d. The concealed version is called recessive.
e. It is active in the gametic phase.

Answer: E

Explanation: Before an answer is made, it is important to identify that the unit of heredity being referred to in the question is the gene, so that from there, each of the choices can then be evaluated if they describe the unit of heredity. All of the choices except E are true about the gene. It is situated in the nucleus particularly on the chromosome (i.e., choice A is true). There are also a few genes present in the mitochondria, i.e., situated outside the nucleus, making choice B true. A gene can have various forms called allelomorphs or simply alleles. An allele could be dominant (conceals the effect of its alternative form). The alternative form, the one that is concealed, is the recessive allele. Choice E is left among the choices given, and this is not true about the gene. The gene is not active in the gametic phase but rather it is inert.

22. In the absence of dominance, the genotypes BB, Bb, and bb _____.

a. have the same phenotype.
b. have different phenotypes.
c. interact with one another.
d. are not expressed unless there is another gene present.
e. are linked.

Answer: B

Explanation: In the absence of dominance, the genotypes BB, Bb, and bb have different phenotypes. The type of gene action being exhibited is incomplete dominance, meaning to say, each genotype has a distinct phenotype. In particular, codominance pertains to the genetic phenomenon in which gene products from the two alleles in a heterozygote are produced in roughly equal amount, where gene products refer to either different transcripts from the two alleles, different proteins from cellular processing of the transcripts, or different metabolites specifically associated with the enzymatic activity of the allele-specific transcripts or proteins (Xia, 2013).

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23. Genes governed by additive gene action contribute to the phenotypic expression of many economically important traits in farm animals. Among the economically important in pigs and cattle are _____ and _____, respectively.

a. egg number; parity
b. litter size; egg number
c. litter size; calving interval
d. milk yield; litter size
e. egg number; calving interval

Answer: C

Explanation: The correct answer is C. Litter size is a trait in pigs which is of economic importance. This is the number of piglets born in one farrowing. Meanwhile, calling interval refers to the period from one calving to the next calving (Gaines and Palfrey, 1931). Egg number is a trait specific to poultry and milk yield is to dairy cattle.

24. Which of the following is NOT true?

a. Instead of segregating and recombining in a random manner during gamete formation, the genes are inherited together when they are linked.
b. In testing for the presence of linkage groups, it is important to backcross a homozygous dominant animal for both pairs of genes to a double heterozygous animal.
c. If the genes are not linked, the phenotypes of the offspring are in equal ratios after backcrossing.
d. Genes that are more closely linked tend to be inherited together more often than those that are located further apart.
e. The farther the distance between genes in the chromosome, the more likely that crossover will take place between them.

Answer: B

Explanation: It is important to backcross a homozygous recessive animal for both gene pairs to a double heterozygous animal to determine the presence of linkage groups. If no linkage is involved, four phenotypes will appear in the offspring in approximately equal proportions. Otherwise, only two phenotypes will be expressed and also in equal numbers (Bondoc, 2008).

25. All of the choices are examples of non-Mendelian inheritance except one. Which one is it?

a. linkage
b. pleiotropy
c. multiple alleles
d. dominance
e. sex-related traits

Answer: D

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Explanation: Non-Mendelian inheritance refer to inheritance patterns which do not follow Mendel's laws of inheritance. Examples of which include linkage, pleiotropy, multiple alleles, and sex-related traits. These result to variations in phenotypic ratios.

26. Which of the following is true about the action of genes?

    a. It influences phenotypic values.
    b. It cannot be detected from the phenotype.
    c. It can be observed even in the haploid phase.
    d. Both A and B
    e. Both A and C

    Answer: A

    Explanation: The action of genes may be detected only from the phenotype, hence, it influences the phenotypic values. It must be noted, however, that genes are active only in the diploid phase (Bondoc, 2008).

27. Which of the following is not a gamete of the genotype DdEEGg?

    a. DEG
    b. DEg
    c. Deg
    d. dEG
    e. dEg

    Answer: C

    Explanation: Following the law of segregation of Mendel, members of a gene pair separate into gametes. Hence, the gametes of the genotype DdEEGg are DEG, DEg, dEG, and dEg.

28. Which of the following genotypes is not an offspring of the cross AABb x AaBb?

    a. AABb
    b. aaBB
    c. AAbb
    d. Aabb
    e. AaBb

    Answer: B

    Explanation: The genotypes of the offspring of the cross AABB x AaBb are AABB, AABb, AaBB, AaBb, AAbb,and Aabb.

29. A black wool of sheep is due to a recessive allele b and a white wool to its dominant allele B. The mating between a black ram and a white ewe produced a black lamb. The genotypes of the ram and ewe are ____ and ____, respectively.

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a. BB; Bb
b. bb; BB
c. Bb; Bb
d. Bb; bb
e. bb; bb

Answer: B

Explanation: The possible genotypes of the parents are bb (black) and B_ (white). However, the best answer among the choices is letter B, i.e., bb and BB for ram and ewe, respectively.

30. Ear orientation is a qualitative trait in pigs which can be used as a breed characteristic such as in the case of Large White (they have erect ears) and Landrace (they have drooping ears). It is known that the erect orientation of ears is governed by the dominant allele A. Suppose a population of 1800 pigs consists of 27 individuals with drooping ears while the rest have erect ears. What is the frequency of the homozygous recessive genotype? Note that this is governed by complete dominance.

a. 0.015
b. 0.15
c. 1.5
d. 0.105
e. 0.51

Answer: A

Explanation: The number of individuals with the homozygous recessive genotype is 27. To get the frequency, get the proportion of the homozygous recessive genotype, i.e., 27/1800=0.015.

31. All of the statements are true except _____.

a. Jay Lush is the father of animal breeding.
b. Jay Laurence Lush is a British animal breeder.
c. Philamin cattle was developed before the Second World War.
d. Pekin duck is scientifically known as *Cairina moschata*.
e. William Bateson provided the classical definition of genetics.

Answer: B

Explanation: Jay Lush was an American animal breeder.

32. Animal breeding industries have undergone series of development over the years. However, in the case of developing countries such as the Philippines, selection of farm animals has been very limited and less effective. Why is this so?

a. Marketing systems are not well-established.
b. Breeding objectives are well-constructed.

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c. Breeding farms ensure large population sizes.
d. There is a detailed economic assessment conducted.
e. Both A and C
f. All of the choices

Answer: A

Explanation: All of the choices except A favor the effective selection of farm animals. With proper identification of breeding objectives, breeder farms have a goal in mind which trait to improve. Breeding farms must also ensure a large population size to avoid the negative effects of inbreeding.

33. Wilhelm Johannsen coined the following terms except

a. gene
b. genetics
c. genotype
d. phenotype
e. none of the choices

Answer: B

Explanation: Wilhelm Johannsen coined the terms gene, genotype, and phenotype. William Bateson provided the classical definition of genetics.

34. William Bateson showed that Mendelian laws can also be applied to animals by producing the first evidence of inheritance through experiments with what animal?

a. *Bos taurus*
b. *Coturnix coturnix*
c. *Anser anser*
d. *Gallus gallus*
e. *Anser cygnoides*

Answer: D

Explanation: William Bateson applied the laws Mendel to chickens, scientifically known as *Gallus gallus*.

35. This field of study aims to improve animals by changing their genetic abilities for economically important traits.

a. animal science
b. animal breeding
c. animal genetics
d. animal production
e. animal nutrition

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Answer: B

Explanation: Animal breeding aims to apply the principles of animal genetics in order to genetically improve farm livestock (Bondoc, 2008).

36. Which of the following is a qualitative trait?

a. Skin color
b. Shank color
c. Growth rate
d. Milk yield
e. Egg production

Answer: B

Explanation: Qualitative traits are those whose phenotypes can be described in terms of classes, kinds, or categories. An example of which is shank color, such as those yellow shank and black shank of chickens.

37. It is based on the hypothesis that many genes contribute to the expression of traits.

a. Mendelian genetics
b. Population genetics
c. Quantitative genetics
d. Animal breeding
e. Hardy-Weinberg law

Answer: C

Explanation: Quantitative genetics deals with the study of inheritance of quantitative traits, the expression of which are influenced by many genes, or sometimes called polygenes.

38. All of the following scientists were quantitative geneticists except

a. Fisher
b. Falconer
c. Mendel
d. Galton
e. Pearson

Answer: C

Explanation: Mendel led the foundations of Classical Genetics, which is sometimes called Mendelian genetics. This branch of genetics deals with the behavior or mode of transmission of the units of heredity.

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39. When did animal breeding start?

a. before the domestication of animals
b. after Neolithic Period
c. 1993
d. none of the choices
e. all of the choices

Answer: A

Explanation: The history of animal breeding started even before the domestication of animals was ever recorded (Bondoc, 2008).

40. Which of the following is NOT true about the gene?

a. The word "gene" was introduced by Gregor Mendel.
b. The gene is inherited across generations.
c. The gene is the basic unit of life.
d. Both A and B
e. Both A and C

Answer: A

Explanation: The word "gene" was introduced by Wilhelm Johannsen.

41. Offspring resemble their parents more than they resemble unrelated individuals. This is a concept of

a. variation
b. heredity
c. selection
d. genetic improvement
e. all of the choices

Answer: B

Explanation: Heredity is the transmission of genetic materials from parents to offspring. Hence, this explains why offspring tend to resemble their parents more than they resemble unrelated individuals.

42. Which of the following is not true about animal domestication?

a. Genetic changes in wild animals are brought about through a selection process imposed by man.
b. It is not an evolutionary process.
c. The sheep was domesticated even before the cattle
d. None of the choices
e. All of the choices

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Answer: B

Explanation: Animal domestication is an evolutionary process. Domestication is an evolutionary process by which animals are artificially selected and undergo huge phenotypic behavioral and physiological alterations (Trut et al., 2009).

43. All of the following was developed by the father of animal breeding except

a. Dolly the Sheep
b. Old Longhorn cattle
c. Shire horses
d. Leicester sheep
e. None of the choices

Answer: A

Explanation: Dolly the Sheep was the first mammal cloned from a cell from an adult animal. The cloning process was conducted by Ian Wilmut and his colleagues (Griffin, 1999). Meanwhile, the father of animal breeding in the name of Robert Bakewell developed the old Longhorn cattle, Shire horses, and Leicester sheep.

44. Which of the following was developed before World War II?

a. Los Baños Cantonese chickens
b. Berkjala pigs
c. Philamin cattle
d. All of the choices
e. None of the choices

Answer: D

Explanation: The Los Baños Cantonese chickens, Berkjala pigs, and Philamin cattle were popular breeds of farm livestock developed in the Philippines before the World War II through the leadership of Dr. Bienvenido Gonzales (Bondoc, 2008).

45. The chromosome number of *Numida meleagris* and *Meleagris gallopavo* are ___ and ___, respectively.

a. 78; 82
b. 82; 78
c. 27; 28
d. 72; 87
e. 72; 88

Answer: A

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Explanation: Numida meleagris is a guinea fowl with a chromosome number of 2n=78 while a Meleagris gallopavo is commonly known as turkey having a chromosome number of 2n=82.

46. Poultry species generally have higher chromosome number because they have a

a. greater number of nucleotides
b. group of microchromosomes
c. longer DNA strand
d. lower generation interval
e. polyploid condition

Answer: B

Explanation: Poultry species such as the chicken have a group of microchromosomes. In fact, the typical avian karyotype is composed of a few macrochromosomes and around 60 indistinguishable small microchromosomes (Fillon, 1998). This is the reason why poultry species have relatively higher chromosome number than mammals.

47. Genes that are located in the sex-chromosomes are referred to as

a. sex-linked
b. sex-limited
c. sex-influenced
d. sex-related
e. all of the choices are possible

Answer: A

Explanation: Sex-linked genes are those which are located in the sex-chromosomes. It could either be in the X or Y, however, since there are relatively fewer genes in the Y chromosome, sex-linked genes are more commonly referred to as X-linked.

48. Which of the following is true about mutation?

a. It occurs constantly in all cells.
b. It gives rise to variation among individuals of a population.
c. It does not change the genetic composition of an individual.
d. Both A and B
e. Both A and C

Answer: D

Explanation: Mutation is the change in the sequence of nucleotides, which thereby gives rise to genetic differences across individuals in a population. It occurs constantly in all cells of the body, though in rare cases (Bondoc, 2008).

SUC-ACAP-LEA

49. The location of a gene on the genetic material is called

a. chiasma
b. chromatin
c. centromere
d. telomere
e. locus

Answer: E

Explanation: The gene, or its alternative forms called alleles, are located specifically in the locus of a chromosome. Its plural form is loci. Chiasma is the cross-connection between non-sister chromatids, which indicate the locations of completed genetic exchanges (Stahl, 2013). It is a structure that forms between a pair of homologous chromosomes by crossover recombination and physically links the homologous chromosomes during meiosis (Hirose et al., 2011). Chromatin is a highly organized complex of DNA and proteins and is a principal component of the cell nucleus. This becomes distinct bodies during cell division and during which they are called chromosomes (Ramirez et al., 2019). Centromere, on the other hand, is a permanent well-defined region of the chromosome where force is exerted during the separation of dividing chromosomes (Ramirez, 2019). Meanwhile, telomeres are specialized nucleoproteins which cap and protect the ends of linear chromosomes from degradation (Losso and Bansode, 2004).

50. These are animals that can produce progeny like themselves when they are mated *inter se*

a. hybrids
b. crossbreds
c. purebreds
d. strains
e. varieties

Answer: C

Explanation: Purebreeding animals, or simply purebreds, can produce progeny like themselves when they are mated inter se (Bondoc, 2008). Inter se mating means that the individuals mate among themselves. Hybrid is the other term used to describe the crossbreds, which are animals produced from the mating between two different breeds.

51. The mating of two purebreds that differ for a single character is called a

a. coupling linkage
b. test cross
c. monohybrid cross
d. dihybrid cross
e. *inter se* mating

Answer: C

SUC-ACAP-LEA

Explanation: In a monohybrid cross, both parents are heterozygous (or a hybrid) for a single (mono) trait. Using monohybrid crosses, Mendel observed that although different alleles could influence a single trait, they remained indivisible and could be inherited separately. This is the basis of Mendel's First Law, also called The Law of Segregation, which states: during gamete formation, the two alleles at a gene locus segregate from each other; each gamete has an equal probability of containing either allele (Monohybrid Crosses and Segregation, 2020).

52. Which of the following best describes Mendel's law of segregation?

a. Members of a gene pair separate into gametes formed by an individual.
b. There is independent segregation taking place between two gene pairs.
c. Segregation at one locus does not affect the segregation at another.
d. Allelic pairs separate equally at the metaphase plate.
e. Crossing over always takes place during the segregation of alleles at one locus.

Answer: A

Explanation: The Law of Segregation is referred to as the law of purity of gametes. During the formation of male and female gametes alleles responsible for a particular character separate and are passed into different gametes. These gametes can unite randomly in different possible combinations during fertilization and produce the genotype for the traits of the progenies (Pierce, 2017).

53. If a given gene obscures the presence of another gene that is its allele, the first gene is said to be dominant relative to the second. This is called a/an _____.

a. allelic interaction
b. non-allelic interaction
c. linkage
d. epistasis
e. pleiotropy

Answer: A

Explanation: Allelic interaction deals with the interaction of members of a gene pair to produce a particular phenotype. Examples of which include complete dominance, incomplete dominance, or co-dominance, and overdominance. Non-allelic interaction refers to the interaction of non-allelic genes to give a phenotypic effect. A common example used is epistasis.

54. DNA can be found in which part of the animal cell?

I. Nucleus
II. Mitochondria
III. Ribosomes
IV. Chloroplast

SUC-ACAP-LEA

a. I only
b. II only
c. III only
d. I and II only
e. I, II, and III only

Answer: D

Explanation: The DNA can be found in the nucleus, mitochondria, and chloroplast. However, animal cells do not have a chloroplast, hence, the correct answer is D (I and II only).

55. Which of the following best describes the deoxyribonucleic acid (DNA)?

a. It is found in the nucleus of prokaryotic cells.
b. Few DNA can be found in the ribosomes.
c. DNA molecules never leave the nucleus.
d. None of the choices
e. All of the choices

Answer: C

Explanation: Prokaryotic DNA is not contained within a nucleus, but rather is attached to the plasma membrane and contained in the form of a nucleoid, an irregularly-shaped region that is not surrounded by a nuclear membrane (Characteristics of Eukaryotic DNA, 2020). Also, there is no DNA found in the ribosomes because DNA molecules never leave the nucleus.

56. The use of molecular information in animal improvement through breeding is accomplished by employing which of the following?

a. natural selection
b. selection relative to performance of contemporaries
c. performance selection
d. genomic selection

Answer: D

57. Hens have the ____ sex chromosome pair.

a. XX
b. XY
c. ZZ
d. ZW
e. None of the choices

Answer: D

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Explanation: Each sex chromosome in poultry species is labeled as Z or W. Females have the ZW sex chromosome pair while the males have the ZZ (Lambio, 2010).

58. The chromosome number of *Cygnopsis cynoides* is ____.

a. 76
b. 78
c. 80
d. 82
e. 84

Answer: C

Explanation: Goose (*Cygnopsis cynoides*) has a diploid number of 80 (2n=80) chromosomes (Lambio, 2010).

59. The following are examples of traits governed by additive gene action, except:

a. shank length
b. eggshell thickness
c. body weight
d. eggshell color
e. plumage color

Answer: E

Explanation: Inheritance of plumage color in chickens is an example of a trait governed by epistatic gene action. The interaction of the C and O genes results in colored plumage. However, the presence of the dominant gene I inhibits the expression of colored plumage (Lambio, 2010).

60. The gene action when the phenotypic expression of the trait in the heterozygous individual is similar to that of the homozygous dominant individual.

a. epistasis
b. additive
c. overdominance
d. complete dominance
e. non-additive

Answer: D

Explanation: Complete dominance gene action is when the phenotypic expression of the trait in the heterozygous individual is similar to that of the homozygous dominant individual (Lambio, 2010).

SUC-ACAP-LEA

61. The following are examples of quantitative traits, except:

a. feathering rate
b. feed efficiency
c. growth rate
d. age at sexual maturity
e. livability

Answer: A

Explanation: Feathering rate is an example of a qualitative trait. Qualitative traits are easily classified into distinct phenotypic categories (Lambio, 2010).

62. This selection method based on records works successfully for traits that are highly heritable.

a. progeny test selection
b. sib selection
c. family selection
d. individual records
e. records of relatives

Answer: D

Explanation: The selection method based on individual records works successfully for traits that are highly heritable. Significant improvement could be attained through the use of this method of selection (Lambio, 2010).

63. The following statements are TRUE about inbreeding, EXCEPT.

a. The major genetic effect of inbreeding is to increase the number of pair of genes that are homozygous.
b. Inbreeding showed adverse effects on traits related to reproductive performance and fitness which are lowly heritable.
c. Inbreeding results in a large improvement in the average performance of the F1 progeny over that of the parents.
d. Inbreeding can be utilized to increase the prepotency of an individual.
e. Poultry breeders resort to inbreeding to establish pure line stock.

Answer: C

Explanation: Crossbreeding results in a large improvement in the average performance of the F1 progeny over that of the parents. This phenomenon is called heterosis or hybrid vigor (Lambio, 2010).

SUC-ACAP-LEA

64. Egg-laying capacity, calving ease and mothering ability are examples of _____ traits.

a. Sex-influenced
b. Sex-limited
c. Sex-linked
d. Sex-related
e. Sex-linkage

Answer: B

Explanation: Sex-limited traits have a phenotypic expression that is limited to only one sex. These traits cannot be expressed in both sexes under normal conditions (Bondoc, 2008).

65. The following are factors affecting the genetic composition of a population, except:

a. Random mating
b. Selection
c. Small population
d. Migration
e. Mutation

Answer: A

Explanation: The properties of a population are derived from the Hardy-Weinberg Law, which states that "for a large random mating population, the gene and genotypic frequency remains constant from generation to generation in the absence of mutation, migration and selection." Nonrandom mating without selection alters genotypic frequencies but not necessarily the gene frequencies from one generation to the next (Bondoc, 2008).

66. It is considered a major force that causes change in gene frequencies.

a. Random mating
b. Selection
c. Small population
d. Migration
e. Mutation

Answer: B

Explanation: Selection causes or allows certain individuals to produce the next generation. It is considered a major force that causes change in gene frequencies (Bondoc, 2008).

SUC-ACAP-LEA

67. This factor causes directional changes in gene frequency, EXCEPT:

a. Random mating
b. Selection
c. Small population
d. Migration
e. Mutation

Answer: C

Explanation: In small populations, fluctuations in gene frequency may occur from one generation to the next simply because of chance variation. Unlike migration, mutation and selection, which cause directional changes in gene frequency, fluctuations due to small population size are random in both direction and magnitude (Bondoc, 2008).

68. It is the ultimate source of new genetic material for evolutionary process.

a. Random mating
b. Selection
c. Small population
d. Migration
e. Mutation

Answer: E

Explanation: Mutation is the sudden heritable change in the genetic material and the information it carries. It is the ultimate source of new genetic material for evolutionary processes and selection (Bondoc, 2008).

69. If a batch of *Meleagris gallopavo* eggs was set in August 27, the eggs are expected to hatch on:

a. September 22
b. September 23
c. September 24
d. September 25
e. September 26

Answer: C

Explanation: The incubation period of turkey (*Meleagris gallopavo*) eggs is 28 days (Lambio, 2010).

SUC-ACAP-LEA

70. All statements are true, EXCEPT:

a. Ewes would be homogametic sex.
b. Boars would be the homogametic sex.
c. Ducks would be the heterogametic sex.
d. Hens would be the heterogametic sex.
e. Buck would be the heterogametic sex.

Answer: B

Explanation: In mammals, the female is XX and the male is XY. In poultry, females have the ZW sex chromosome pair while the males have the ZZ (Lambio, 2010). Boars would be the heterogametic sex.

71. The G x E effects is best described in the following situations:

a. Many trials in the same location
b. Single trial in the same location
c. Many trials in different locations
d. Single trial in different locations
e. Both c and d

Answer: C

Explanation: The GxE represents interaction between genotype and environment, suggesting that certain genotypes perform better under certain environments than other genotypes (Bondoc, 2008).

72. ___ carry the genetic material, but do not determine the sex of the individual.

a. Y chromosomes
b. X chromosomes
c. Sex chromosomes
d. Autosomes
e. Alleles

Answer: D

Explanation: Autosomes carry genetic material but do not determine the sex (Bondoc, 2008).

73. The five-carbon sugar compound present in DNA molecule is:

a. Erythrose
b. Deoxyribose
c. Ribose
d. Ribulose
e. Pentose

SUC-ACAP-LEA

Answer: B

Explanation: The sides of DNA are composed of two strands of deoxyribose (a five-carbon sugar molecule) (Flanders and Gillespie, 2016).

74. Which of the following is sex-linked trait in poultry?

a. Barring plumage pattern
b. Color-blindness
c. Albumen height
d. Semen characteristics
e. Comb pattern

Answer: A

Explanation: Barring plumage pattern in poultry is known to be sex-linked (Bondoc, 2008).

75. The mating of unrelated animals within a particular breed.

a. Outcrossing
b. Linecrossing
c. Crossbreeding
d. Grading-up
e. Species hybridization

Answer: A

Explanation: Outcrossing is the mating of unrelated animals within a particular breed (Bondoc, 2008).

76. The following are advantages of using artificial insemination, except:

a. The producer may use a sire of higher quality than he or she could otherwise afford.
b. Producers do not have to keep the male.
c. It allows the producer to select the type of sire needed for a particular group of females
d. The likelihood of disease is lessened.
e. It allows the production of many offspring from a superior female.

Answer: E

Explanation: The use of embryo transfer allows the rapid advancement of genetics from the dam. Just as artificial insemination allows the production of many offspring from a superior male, embryo transfer allows the production of many offspring from a superior female (Herren, 2007).

SUC-ACAP-LEA

77. What is the average length of estrus cycle in sow?

a. 16 days
b. 18 days
c. 19 days
d. 20 days
e. 21 days

Answer: E

Explanation: The average length of estrus cycle in sow is 21 days (Herren, 2007).

78. What is the average length of gestation in a mare?

a. 336 days
b. 305 days
c. 281 days
d. 245 days
e. 355 days

Answer: A

Explanation: The average length of gestation in a mare is 336 days (Herren, 2007).

79. The diploid number of chromosomes in cattle is ___.

a. 22
b. 27
c. 30
d. 32
e. 39

Answer: C

Explanation: The diploid number of chromosomes in cattle is 30 (Flanders and Gillespie, 2016)

80. This method of selection provides added accuracy in evaluating animal and for lowly heritable traits.

a. Progeny Test Selection
b. Sib Selection
c. Family Selection
d. Individual Records
e. Records of Relatives

Answer: C

SUC-ACAP-LEA

Explanation: Family selection is preferred when heritability of the trait is low (Bondoc, 2008).

81. The mating of animals from different inbred lines.

a. Outcrossing
b. Linecrossing
c. Crossbreeding
d. Grading-up
e. Species hybridization

Answer: B

Explanation: Linecrossing is the mating of animals from different inbred lines (Bondoc, 2008).

82. The mating of crossbred females with one of the parent purebreds.

a. Backcrossing
b. Topcrossing
c. Crisscrossing
d. Two-breed crossing
e. Four-breed terminal cross

Answer: A

Explanation: Backcrossing is the mating of crossbred females with one of the parent purebreds (Bondoc, 2008).

83. Gene is a segment of ___ which determines the base sequence of ____ that makes up the code for a certain ___.

a. mRNA – protein - DNA
b. mRNA – DNA – protein
c. DNA-mRNA-protein
d. DNA-protein-mRNA
e. protein-mRNA-DNA

Answer: C

Explanation: From the Watson-Crick hypothesis, the gene may be defined as a segment of the DNA which determines the base sequence of nucleotides in the mRNA that makes up the code for protein (Lasley, 1978).

SUC-ACAP-LEA

84. Animal breeding generally aims to improve the following except _____

a. quantity of production of farm animals and of their products per unit of time
b. efficiency of reproduction of farm animals and their products
c. quality of farm animals and their products.
d. efficiency of production of farm animals and their products.
e. aesthetic value of farm animals and their products.

Answer: B

Explanation: The following are the general objectives of animal breeding (Lasley, 1978): improve the quantity of production of farm animals and of their products per unit of time, improve the efficiency of production of farm animals and their products, improve the quality of farm animals and their products and improve the aesthetic value of farm animals and their products.

85. The nucleic acid present in virus

a. DNA
b. RNA
c. Either DNA or RNA
d. Both DNA and RNA
e. None of the choices

Answer: C

Explanation: Viruses contain nucleic acid, either DNA or RNA (Lasley, 1978).

86. The following are examples of interspecific hybrids, except:

a. Mule
b. Zebroid
c. Mulard
d. Hinny
e. Pien niu

Answer: C

Explanation: The mulard is a hyrid between two different genera of domestic duck: Muscovy duck (*Cairina moschata*) and the domestic duck (*Anas platyrhynchos*). The cross between the jack of the ass species (*Equus asinus*) and the mare of the horse species (*Equus caballus*) to produce the mule, with the reciprocal called a Hinny. A zebra-horse hybrid (*Equus zebra* x *Equus caballus*), also called a zebroid. Pien niu is produced by crossing cattle and a yak from Tibet (*Bos taurus* x *Bos grunniens*). (Bondoc, 2008).

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87. This a term used to refer to an interbreeding group of animals within a species with some identifiable common appearance, performance, ancestry, or selection history.

a. breed
b. population
c. variety
d. strain
e. line

Answer: A

Explanation: A breed is an interbreeding group of animals within a species with some identifiable common appearance, performance, ancestry, or selection history (Oldenbroek and van der Waaij, 2014).

88. This is a process by which certain individuals in a population are preferred over others for the production of the next generation progenies.

a. culling
b. selection
c. breeding
d. upgrading

Answer: B

Explanation: Culling is the act of removing unproductive animals from a group. Breeding involves the deliberate manipulation of the genetic composition of a population for the attainment of a desired characteristics. Upgrading is the mating system where the purebred sire of a given breed is mated to the native dam.

89. One of the best selection methods which uses certain formula in computing the heritability of a family is called

a. selection index
b. independent culling level
c. tandem
d. family selection

Answer: A

Explanation: Independent Culling Level Method - have already predetermined standards. Tandem Method - selection of a trait one at a time. Family Selection Method - selection based on the performance of the family.

SUC-ACAP-LEA

90. The changes in the mean performance of population due to selection is known as

a. selection index
b. genetic change
c. progeny testing
d. culling

Answer: B

91. Reproductive performance and vigor ____ when practicing inbreeding in the flock.

a. are temporarily improved
b. decrease with time
c. remain the same
d. are improved as the coefficient of inbreeding decreases

Answer: B

92. ___ leads to random genetic drift because of chance variation or sampling effects or errors

a. migration
b. non-random mating
c. selection
d. small population size

Answer: D

93. Inbreeding depression in most reproductive traits in farm animals is caused by _____.

a. outcrossing
b. species hybridization
c. parent-offspring mating
d. upgrading

Answer: C

94. In Mendel's Law of Segregation, the following are stated except one. Which one is it?

a. Genes determine characters.
b. Genes occur in pairs.
c. Only one of the gene pairs is transmitted by a particular gamete.
d. Segregation and recombination are random.

Answer: A

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95. His work focused on population genetics for animal breeding.

a. Jay Lush
b. Robert Bakewell
c. William Bateson
d. Lanoy Nelson Hazel

Answer: A

Explanation: Robert Bakewell - breeding work on horses, sheep and cattle. William Bateson - demonstrated Mendelian inheritance. L.N Hazel - selection index.

96. This is the mating of unrelated animals within a breed.

a. Crossbreeding
b. Inbreeding
c. Outcrossing
d. Crisscrossing

Answer: C

Explanation: Crossbreeding – The mating of animals of two or more different breeds.
Inbreeding – Mating of closely related individuals.
Crisscrossing – It consists of crossing two or more breeds and alternately mating the hybrid females over several generations with males of the original breeds.

97. Asexual reproduction technology in animals which completely gives rise to an exact copy of its parent.

a. Artificial insemination
b. Test tube baby
c. Embryo transfer
d. Cloning

Answer: D

98. Characteristic of breeder animals that must not be culled

a. A cow which calves every year
b. Produces large amount of milk
c. Heifers which come to heat
d. All of the choices

Answer: D

SUC-ACAP-LEA

99. Sexual receptivity which coincides with estrus in animals that exhibit estrous cycle is due to large amounts of

a. testosterone
b. progesterone
c. estrogen
d. relaxin

Answer: C

100. A phase of the estrous cycle which is characterized by sexual receptivity of the female

a. proestrus
b. estrus
c. metestrus
d. diestrus

Answer: B

101. Milk production in cows and egg production in quails are examples of economically important traits known as

a. maternal effect
b. sex-influenced
c. sex-limited
d. sex-linked

Answer: C

Explanation: Maternal effect - Influence of dams (mothers) on performance of progenies Sex-influenced - Sex of individual influences the phenotypic expression of certain genes. Sex-linked – X-linked in mammals, Z-liked in birds (avian species).

102. This gene action happens when the phenotypic expression of heterozygotes is similar to the homozygotes

a. dominance
b. complete dominance
c. partial dominance
d. overdominance

Answer: B

Explanation: Dominance - There is an interaction of genes within a locus. Partial dominance - Phenotypic expression of Aa is between AA and average of AA and aa. Overdominance - Aa is better (not necessarily higher) than either AA or aa individuals.

SUC-ACAP-LEA

103. If an animal has a genotype Aa, what is the probability of the gametes being an A?

a. ½
b. 1
c. ¾
d. 0

Answer: A

104. When a gene suppresses the expression of its allele, the former is called the dominant while the latter is referred to as the ____ gene.

a. homozygous
b. recessive
c. regulator
d. structural

Answer: B

105. It is the notation used to refer to the heterogametic sex chromosomes of a hen.

a. XX
b. XY
c. ZZ
d. ZW

Answer: D

106. In terms of chromosome number, which of the following does not belong to the group?

a. *Anas platyrnchos*
b. *Meleagris gallopavo*
c. *Coturnix coturnix japonica*
d. *Cairina moschata*

Answer: C

Explanation: Options a, b, and d are animals with a chromosome number of 2n=80, option c has 78 only.

107. He is credited with the first systematic use of 'progeny testing'.

a. Robert Bakewell
b. Jay Laurence Lush
c. Ronald Fisher
d. Charles Roy Henderson
e. Alan Robertson

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Answer: A

108. This reproductive technology speeds up genetic change because it allows in principle a global choice of semen and a much greater dissemination of the genetics of particular males who have the desired traits.

a. Artificial selection
b. Artificial insemination
c. Embryo transfer
d. Cloning
e. Multiple ovulation embryo transfer

Answer: B

109. This assisted reproductive technology speeds up the dissemination of female as well as male genes.

a. Artificial selection
b. Artificial insemination
c. Embryo transfer
d. Cloning
e. None of the choices

Answer: C

110. Which of the following is not a goal of animal breeding?

a. increase interval between births
b. greater fertility
c. earlier reproductive maturity
d. longer productive lifetime of breeding females
e. none of the choices

Answer: A

111. These are traits that increase an animal's biological and economic efficiency not by higher outputs of products but by reduced costs of production.

a. production traits
b. functional traits
c. sex-linked traits
d. carcass traits
e. product quality traits

Answer: B

SUC-ACAP-LEA

112. Typical functional traits are those that contribute to

a. decreasing the number of young who die early or are born dead.
b. increasing the longevity of breeding pigs, dairy cows and ewes.
c. improving health by breeding for better resistance to parasites, to infections and to production and genetics-related diseases such as lameness
d. none of the choices
e. all of the choices

Answer: E

113. Functional traits are sometimes labelled as _____ traits.

a. health and welfare
b. carcass quality
c. product quality
d. production
e. fertility

Answer: A

114. Selection of broiler chickens has focused on the following except _____

a. growth rate
b. percentage of breast meat
c. feed conversion efficiency
d. egg production
e. none of the choices

Answer: D

SUC-ACAP-LEA

Unit IV
SLAUGHTERING, PROCESSING, AND MARKETING OF FARM ANIMALS

Instructions: Read each question carefully. Choose the best answer that satisfies the question or statement.

1. It refers to the meat coming from cattle less than 1 year old.

   a. Beef
   b. Veal
   c. Mutton
   d. Chevon

   Answer: B

   Explanation: Beef refers to the meat coming from cattle more than 1 year old while mutton and chevon refer to those that come from sheep and goat, respectively.

2. It refers to the liquid milk prepared by mixing together the milk powder and water to approximate the original composition of fresh milk.

   a. Recombined milk
   b. Reconstituted milk
   c. Filled milk
   d. Condensed milk

   Answer: B

   Explanation: Recombined milk is prepared by mixing together skim milk powder or butter milk powder, butter oil and water. Filled milk is milk reconstituted with vegetable oil such as coconut and corn. Condensed milk refers to concentrated milk to which sugar has been added.

3. The following are the basic requirements in slaughtering EXCEPT

   a. Cleanliness of meat produced
   b. Hygiene of production
   c. Efficiency of meat inspection
   d. None of the above

   Answer: E

   Explanation: All the above-mentioned are the basic requirements in slaughtering.

SUC-ACAP-LEA

4. Listed below are the recommended ages for fresh meat retailing and meat processing, EXCEPT

   a. 4-8 months for swine
   b. 3 years or younger for cattle
   c. 5 years or older for carabao
   d. 1 year old for goat

   Answer: C

   Explanation: The recommended ages for fresh meat retailing and meat processing for cattle and carabao is 3 years or younger.

5. The following are the requirements for the accreditation of slaughterhouses set by National Meat Inspection Services, EXCEPT

   a. Must have good drainage and ventilation
   b. Must have facilities for hanging carcasses and refrigeration units
   c. Must have separate area for handling meat and offals
   d. Must be at least 10 meters away from residential areas

   Answer: D

   Explanation: One of the requirements for the accreditation of slaughterhouses set by National Meat Inspection Services is that it MUST be located at least 100 meters away from residential houses.

6. Sticking must be done within _____ after stunning.

   a. 3 minutes
   b. 10 minutes
   c. 15 minutes
   d. 20 minutes

   Answer: A

   Explanation: Stunning is done to make the animal unconscious, followed by sticking. Sticking is a term for "bleeding" and it must be done within 3 minutes after stunning. If done later, the animal may regain consciousness or the animal may die.

7. The following are the basic principles to follow in cutting carcasses EXCEPT

   a. Thick portions must be separated from thin ones
   b. Tender parts must be separated from the less tender portions
   c. The high priced cuts are separated from the less expensive parts
   d. Cuts must be made along the muscle fiber

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Answer: D

Explanation: The following are the basic principles to follow in cutting carcasses,
1. Thick portions must be separated from thin ones
2. Tender parts must be separated from the less tender portions
3. The high priced cuts are separated from the less expensive parts
4. Cuts must be made across the muscle fiber (not along the muscle fiber!)

8. The temperature of the scalding water must be _____.

a. 30 to 50ºC
b. 54 to 82ºC
c. 85 to 95ºC
d. 90 to 100ºC

Answer: B

Explanation: The temperature of the scalding water must be maintained at 54 to 82ºC. Too hot may cause the setting of hairs while too cold water cannot effect loosening of hairs.

9. _____ refers to cutting carcasses into standard wholesale and retail cuts.

a. Evisceration
b. Fabrication
c. Splitting
d. Shrouding

Answer: B

Explanation: Evisceration refers to the removal of the visceral organs from the opening of the carcass up to the complete removal of the internal organs from the body cavity. Splitting is dividing the carcass into two along the backbone/vertebral column. Shrouding refers to wrapping of the carcass with cheesecloth previously soaked with lukewarm water. The shrouding cloth absorbs the remaining blood at the surface of the carcass, smoothens the external fat covering and prevents excessive shrinkage and oxidation. Shrouding is usually done when slaughtering cattle and carabao.

10. No form of stress should be given to animals prior to slaughter. The following are the disadvantages of stress EXCEPT

a. High temperature of carcasses
b. high water holding capacity of resulting meat
c. Low aroma, flavor, texture and juiciness scores
d. Loss of muscle glycogen

Answer: B

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Explanation: The disadvantages of stress include high temperature of carcass, loss of muscle glycogen, low water holding capacity of resulting meat, low aroma, flavor, texture and juiciness scores.

11. It means that round, elongated fusiform muscle, (supracoracoid muscle or deep pectoral) found on each side of the keel bone

    a. Breast fillet
    b. Trimmed breast
    c. Half breast
    d. Wish bone

    Answer: A

12. Beef carcasses are chilled to attain an internal temperature of _____ prior to fabrication:

    a. 3°C
    b. 2°C
    c. 1°C
    d. 0°C

    Answer: A

13. It is unchilled weight after harvest and removal of the hide, head, gastrointestinal tract, and internal organs:

    a. Hot meat
    b. Hot carcass weight
    c. Live weight
    d. Dressing percentage

    Answer: B

14. Stunning equipment that if not properly applied may result in excessive hemorrhage in the muscles or spinal fractures.

    a. Gas
    b. Electrical
    c. Percussion
    d. Shotgun

    Answer: B

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15. An examination that requires observation of all parts of the carcass, dressing procedures, equipment, and facilities to prevent contamination of edible parts:

    a. Processing inspection
    b. Ante-mortem
    c. Post-mortem
    d. All of these

    Answer: B

16. Example of disease that must be tested during ante-mortem inspection since it is unsafe for human consumption:

    a. Leptospirosis
    b. Tuberculosis
    c. Brucellosis
    d. All of these

    Answer: D

17. Result for pigs that are susceptible to high environmental temperatures and humidity:

    a. Stress
    b. Heart failure
    c. Heat stroke
    d. Mild stroke

    Answer: C

18. Ideal pH of meat after 24 hours of slaughter:

    a. 6.2
    b. 6.2 or lower
    c. 6.2 or higher
    d. None of these

    Answer: B

19. An act to promote animal welfare in the Philippines, otherwise known as:

    a. "The Animal Welfare Act of 1998"
    b. "The Animal Behavior Act of 1998"
    c. "The Animal Well – being Act of 1998"
    d. "The Animal Health Act of 1998"

    Answer: A

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20. The following are some of the major objectives of ante-mortem inspection except:

    a. To not screen all animals destined to slaughter
    b. To ensure that animals are properly rested and that proper clinical information, which will assist in the disease diagnosis and judgment is obtained.
    c. To reduce contamination on the killing floor by separating the dirty animals and condemning the diseased animals if required by regulation
    d. To ensure that injured animals or those with pain and suffering receive emergency slaughter and that animals are treated humanely.

    Answer: A

21. Salt level in meat (%) which water binding capacity is back to the same level as in fresh unsalted meat.

    a. 10%
    b. 11%
    c. 15%
    d. 1.5 to 3%

    Answer: B

22. The substance has a notable effect on raw red meat, in particular on the hygienically very sensitive minced meat, as it can reverse dark brownish colors of over-stored products causing them to lighten and reddening in color.

    a. Formalin
    b. Chlorine
    c. Hydrogen peroxide
    d. Sulfur dioxide

    Answer: D

23. Meat products contain mixes of lower-grade muscle trimmings, fatty tissues, head meat, animal feet, animal skin, blood, liver and other edible slaughter by-products.

    a. Raw-cooked
    b. Pre-cooked
    c. Cured meat cuts
    d. Fresh processed

    Answer: B

24. The following are signs of the abnormalities which are checked on ante-mortem examination, except:

    a. Abnormalities in respiration
    b. Abnormalities in behavior

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c. Abnormalities in gait
d. Abnormalities in temperature

Answer: D

25. A carcass or parts of a carcass food animal which was slaughtered in an unregistered establishment and has not undergone the required inspection.

a. Fresh meat
b. Hot meat
c. Meat
d. Meat inspection

Answer: B

26. Any procedure or tests conducted try a competent person on live animals for the purpose of judgment of safety and suitability and disposition.

a. Anti - Mortem
b. Ante - Mortem
c. Post - Mortem
d. Slaughter

Answer: B

27. Animals plainly showing symptoms of a disease or condition that would cause condemnation of their carcasses on post-mortem are marked "condemned" and are killed by the official establishment in a manner prescribed for condemned carcasses. The following are showing symptoms in animals, except:

a. Rabies
b. Tetanus
c. Anaplasmosis
d. None of the above

Answer: D

28. The physical parameters and normal meat color of a chevon:

a. Light red to bright red
b. Bright cherry red to dark red
c. Grayish pink to grayish red
d. None of the above

Answer: A

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29. They were the first to record that meat was preserved by salting and sun drying.

a. Indians
b. Malays
c. Egyptians
d. All of the above

Answer: C

30. The one who introduced hermetic sealing of foods to preserved meat. He is also the one who started the canning of foods.

a. Nicolas Appert
b. Louis Pasteur
c. Peter Durand
d. None of the above

Answer: A

31. The following are the problems encountered of the meat industry in the Philippines, except:

a. Lack of peoples concern on meat quality
b. Increasing cost of equipment used in processing and slaughtering
c. Lack of trained people
d. Exportation of meat and processed meats

Answer: D

32. Concentrated source of calories (2.25 x than CHO), aid in the absorption of fat soluble vitamins (ADEK)

a. Volatile Fatty Acids
b. Lipids
c. Protein
d. Fats

Answer: D

33. Considered as an excellent source of B – complex vitamins (thiamine, riboflavin, niacin, pyridoxine and cyanocobalamin) and are poor in the fat-soluble groups and Vitamin C. Liver is the greatest single source of vitamin A, D & K.

a. Multi-vitamins
b. Minerals
c. Fats
d. Vitamins

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Answer: D

34. Establishment where specified animals are slaughtered and dressed for human consumption and that is approved, registered and or listed by the competent for such purposes:

a. Slaughter House
b. Abattoir
c. Cold Slaughter
d. None of the above

Answer: B

35. The progressive separation of the body of an animal into a carcass and other edible and inedible parts

a. Disinfection
b. Dressing
c. Defect
d. None of the above

Answer: B

36. Removal of internal organs from the abdominal and thoracic cavities of a carcass:

a. Flaying
b. Evisceration
c. Stunning
d. Skinning

Answer: B

37. Refers to the withdrawal of feeds prior to slaughter but water is given *ad libitum*:

a. Pasting of animals
b. Fasting of animals
c. Animals should be fasted
d. None of the above

Answer: B

38. Poor transportation can have serious deleterious effects on the welfare of livestock and can lead to significant loss of quality and production. Which of the following is a good example of this:

a. Heart Burn
b. Broken Hearted

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c. Stress
d. Toxic

Answer: C

39. The carcass of male cattle that has been castrated before the full development of the secondary sex characteristics

a. Heifer carcass
b. Cow carcass
c. Steer carcass
d. Bull carcass

Answer: C

40. The fundamental operation in applying cold to meat to reduce its temperature quickly.

a. Freezing
b. Cold Storage
c. Chilling
d. None of these

Answer: C

41. Refers to the process of exposing meat to a temperature range of (-32ºF) resulting in crystallization of water in the tissues.

a. Freezing
b. Cold Storage
c. Chilling
d. None of these

Answer: A

42. The storage period from the time of freezing for as long as the product maintains its organoleptic and nutritive characteristics and is suitable for human consumption or for further processing.

a. PSL (Practical Storage Life)
b. PSL (Practice Storage Life)
c. PSL (Philippine Super Liga)
d. PSL (Practical Storing Life)

Answer: A

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43. _____ is another critical phase in the freezing process as it involves a change from ice crystals to melted water, which is reabsorbed and microbial reactivation.

a. Thawing
b. Throwing
c. Tawning
d. None of the above

Answer: A

44. The following are the methods of quick freezing, except:

a. Still air
b. Plate freezing
c. Cold air blast
d. None of the above

Answer: D

45. The classification of Carabao less than 1 year old:

a. Caraveal
b. Carabeef
c. Corned beef
d. None of these

Answer: A

46. It is toxin of *Penicillium vividicatum*

a. Aflatoxin
b. Ochratoxin
c. *Clostridium botulinum*
d. *Staphylococcus*

Answer: C

47. It is law that facilities must be adequate, clean and sanitary and the animal must free from pain, suffering and deprivation:

a. RA 8485
b. RA 7394
c. RA 9296
d. All of these

Answer: A

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48. Cutting or fabrication room should be maintained at:

a. 10°C
b. 7°C
c. 0°C to 4°C
d. -0 °C

Answer: A

49. Immobilization of large ruminants is still carried out through the use of a sharp and pointed knife.

a. Sticking knife
b. Skinning knife
c. Puntilla
d. None of these

Answer: C

50. The following are the dressing percentage of slaughtered livestock animals, except:

a. Cattle/Carabao (60%)
b. Calves (62%)
c. Hogs (75%)
d. Sheep/Goat (50%)

Answer: C

51. The practical storage life of meat and meat products especially for pork carcass.

a. 6-12-15
b. 6-12-16
c. 6-12-17
d. 6-12-18

Answer: A

52. The following are the abnormalities in structure or conformation, except:

a. Swellings
b. Enlarged joints
c. Umbilical swelling
d. Enlarged loin

Answer: D

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53. DFD stands for:

a. Dark, Firm and Dry
b. Dark, Firmness and Dry
c. Dark, Firm and Dryness
d. Darkness, Firm and Dry

Answer: A

54. The red skin due to urine soakage

a. Transit erythema
b. Transit tetany
c. Transit or shipping fever
d. Fatigue and exposure

Answer: A

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Unit V

POULTRY PRODUCTION

Instructions: Read each question carefully. Choose the best answer that satisfies the question or statement.

1. The following are the environmental factors affecting laying hen behavior except:

   a. Endocrine activity
   b. Housing temperature
   c. Humidity
   d. Airflow

   Answer: A

   Explanation: Endocrine activity is one of the factors affecting laying hen behavior but not an environmental factor. Environmental factors include proper housing temperature, humidity and airflow.

2. The following are the advantages of caging pullets except:

   a. Expedites culling
   b. Eliminates broodiness
   c. Houses more birds in a given floor space
   d. Higher investment per pullet for housing

   Answer: D

   Explanation: Letter D is a disadvantage of caging pullets since it requires higher cost of investment in purchase of housing materials.

3. How many percent protein is the composition of layer rations?

   a. 23-24%
   b. 20-21%
   c. 19-20%
   d. 17-18%

   Answer: D

   Explanation: Layer ration will generally contain a protein level ranging from 15 to as high as 18%.

4. The regular frequency of feeding the layers is__?

   a. Once a day
   b. Two times a day
   c. Three times a day

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d. Four times a day

Answer: B

Explanation: Feed layers two times a day to induce greater feed consumption, stirring the feed at least once or twice a day

5. What is the recommended amount of grits to feed the layers per week?

a. 227 g (0.5 lb) per 50 layers
b. 227 g (0.5 lb) per 100 layers
c. 136 g (0.5 lb) per 50 layers
d. 136 g (0.5 lb) per 100 layers

Answer: B

Explanation: There is experimental evidence that feeding grit to caged layers will improve egg production although the amount of grit necessary is small, 227 g (0.5 lb) grit per 100 layers per week.

6. Which of the following statements is NOT TRUE in preparing the pullets for laying?

a. If the house has been void with birds for some time and has been cleaned, it will need new litter.
b. Add litter about 3 in. (10 cm) during colder months.
c. Nest should be in the laying house, and open, about a week before the first eggs are laid.
d. Open the nest at night, even before eggs are produced. It is a good management habit to allow pullets to remain in the nests overnight.

Answer: D

Explanation: The statement is False. The correct statement is: Close the nest at night, even before eggs are produced. It is a bad management habit to allow pullets to remain in the nests overnight.

7. Which of the following sets of characteristics are the signs of laying?

a. Large, moist, and puckered vent
b. Hard, round and dry vents
c. Scabs were sloughed off from the wattles and combs
d. Combs and wattles are cold to touch

Answer: A

Explanation: A pullet that has begun to lay has a large, moist vent, showing a wide, dilated condition and looseness, whereas a bird that is not laying has hard, round, dry, puckered vent. Other signs are: the comb and wattles become enlarged and reddened; the bird

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becomes more active and happier in disposition as shown by her crackling a great deal. Small pieces of scabs may be seen sloughing of the comb wattles. These wattles and the earlobes become warm to the touch. The whole abdomen is dilated so that the pelvic arc becomes widespread and the keel forced away from it. The greater the number of eggs a bird is going to lay in successive days, the greater will be the size of the abdomen.

8. The following are the basis of selection for a healthy and vigorous layer except?

   a. Bright prominent eyes, glossy feathers
   b. Active, alert and lively
   c. Has full crop upon retiring for the night
   d. Crow head, phlegmatic, and masculine

   Answer: D

   Explanation: Letter D is selection based on head development, the rest of the choices are selection for vigorous and healthy layers such as bright prominent eyes, glossy feathers, and deep yellow shanks (for breed with yellow shanks). They are active, alert and lively. They should have ravenous appetite and have always full crop upon retiring for the night.

9. The regular amount of feed consumption of a mature laying hens:

   a. 40-90 g/day
   b. 90-140 g/day
   c. 140-180 g/day
   d. 150-200 g/day

   Answer: B

   Explanation: Generally, a mature laying hens will consume about a slow as 90 g to as high as 140 g a day. One year consumption varies from 40-42 kg.

10. Water consumption of layers during normal days.

    a. 4 gallons per 100 heads
    b. 5 gallons per 100 heads
    c. 6 gallons per 100 heads
    d. 7 gallons per 100 heads

    Answer: D

    Explanation: One hundred layers can consume about 7 gallons of water during a normal day and increases during the hotter days. It is still best to provide them a fresh drinking water all the time so that egg production rate will not decline.

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11. What is the recommended lighting hours for layers per day?

a. 10-11 hours/ day
b. 11-12 hours/ day
c. 12- 14 hours/ day
d. 14-16 hours/ day

Answer: D

Explanation: The recommended light hours for layers are 14-16 hours per day.

12. The common age of egg type chicken to start their egg laying.

a. 18 weeks old
b. 20 weeks old
c. 22 weeks old
d. 24 weeks old

Answer: C

Explanation: The age at which the first egg is laid depends upon the breeds or strain and the kind of management given. However, egg type breeds should come into production when they are about 22 weeks (5 and 1/2 months).

13. What is the recommended space allowance for cage layers?

a. 0.75-1.0 sq.ft./bird
b. 1.0-1.5 sq.ft/bird
c. 1.5-2.0 sq.ft./bird
d. 2.0-2.5 sq.ft./bird

Answer: A

Explanation: Litter floor - 2.25 sq.ft./bird
Slat floor pens - 1.5-2.0 sq.ft./bird
Cages - 0.75-1.0 sq.ft./bird

14. What is the recommended roofing style for larger poultry units?

a. Semi-monitor type
b. Monitor type
c. Single span
d. Double span

Answer: B

Explanation:
Monitor – Desirable for big establishment. It has an outlet of air at the top of roof.

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Semi-monitor – like a monitor type, except the outlet on top is only one side of the roof.
Shed or Single Span – this is the simplest type of roof. Usually used for a small construction.
Gable or Double Span – this is the most conventional system of roof construction.

15. How many days is the incubation period of a pheasant?

a. 28- 35 days
b. 26-28 days
c. 22 -24 days
d. 21-22 days

Answer: C

Explanation: The following are the incubation period (days) of different poultry species:

Chickens	- 21 days	Quails	- 16 -18
Muscovy ducks	- 35-37 days	Guinea fowls	- 26 - 28
Mallard ducks	- 28 days	Pea fowls	- 28
Turkeys	- 28 days	Pheasants	- 22 - 24
Geese	- 30-34 days	Ostriches	- 42
Pigeons	- 18 days		

16. In broiler raising, one important factor to consider is the quality of chicks. The quality chicks should include the following, except?

a. Dry and fluffy feather
b. Bright eyes
c. Active appearance
d. Unhealed navel

Answer: D

Explanation: Choices a, b, and c are correct quality of chicks

17. In selecting best strains in broiler raising, what is/are parameters that should not be considered?

a. Carcass recovery of the bird
b. Fast growing strain
c. High converter efficiency of feed intake
d. Dressing percentage

Answer: D

Explanation: Choices a, b and c were correct while D must be with correct measurement. Dressing percentage statement has no exact amount, where it should include the word High in order to consider as one of the choices when selecting the best strains.

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18. Broiler housing is a necessity for broiler raising, which of the following is/are proper guide when constructing broiler house?

    a. Provide the essential requirements for comfort, biosecurity, and protection of the chickens against elements of nature.
    b. Provide convenience and ease of work for the caretakers.
    c. Provide strength and durability to allow maximization of investments.
    d. Use locally available and cheap construction materials without sacrificing durability and comfort.
    e. All of the above

    Answer: D

19. This scheme of broiler raising is applied when all birds are of the same age during purchase, and of the same age at market.

    a. All-out-all-in system
    b. All-in-all-out system
    c. Multiple age system
    d. Multiple range operation

    Answer: B

20. This scheme of broiler operation needs separate housing, whereas there will be more batches raised and marketed in one year.

    a. All-out-all-in system
    b. All-in-all-out system
    c. Multiple age system
    d. Multiple range operation

    Answer: C

21. There were two types where broilers can be raised, which of the following is the best advantage of utilizing separate brooder and separate grower system.

    a. The equipment for brooding is centralized in one area
    b. The utilization of brooding facilities will minimize
    c. The chicken will grow in the same house until market age
    d. The chicken will not grow in the same house until market age

    Answer: D

22. Brooding is the most critical stage of broilers chick's life, during brooding stage the temperature requirement of broiler chicks for the first week is:

    a. 35.5-39°C
    b. 32.2-35°C

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c. 37-39°C
d. 30.2-32.2°C

Answer: B

23. In a brooder pen proper temperature should be strictly monitored, what should be the behavior of the chicks if the temperature is in proper condition.

a. The chicks huddle together
b. Chicks were piling
c. The chicks open their wings and pant
d. The chicks roam around freely

Answer: D

24. Cannibalism among broilers may occur, but can be barred by:

a. Overcrowded space
b. Poor ventilation
c. Adequate feeders and drinkers
d. Imbalance and poor diet

Answer: C

25. This can be the best control of cannibalism among broilers by;

a. Provide adequate floor space
b. Debeaking
c. Sufficient feeding and drinking space
d. Providing proper ventilation

Answer: B

Explanation: Choices a, b and c were not a control but can be a proper management

26. The Crude protein requirement of the chick booster at high is?

a. 20%
b. 21%
c. 22%
d. 23%

Answer: D

27. The Crude Protein requirement of the broiler at finisher stage.

a. 16%
b. 17%

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c. 18%
d. 19%

Answer: C

28. There are various forms of broiler feeds, what form is recommended during the first week?

a. Crumbles
b. Mash
c. Pellets
d. Crumb's pellet

Answer: B

29. Pelleting of feeds for broilers is advantageous to?

a. Increase the growth rate of broilers
b. Increase water consumption of birds
c. Lessen feed cost
d. Lessen feed wastage

Answer: D

Explanation: Pelleting of feed will not increase growth rate but will only increase feed consumption resulting in increased growth, it will not increase water consumption but decreases, it will not lessen feed cost but increases.

30. Which of the following statement is true about brooding broiler;

a. Brooding takes about 3-4 weeks during cold season
b. Brooding may take about 2-3 weeks during cold season
c. Brooding should be at least 2 weeks
d. Brooding is important during cold season

Answer: C

Explanation: Choices a, b and d were wrong statement

31. How many estimated grams of feeds should be consumed by 10-day old chicks for a day?

a. 25 grams
b. 50 grams
c. 75 grams
d. 100 grams

Answer: A

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32. What management practices should be done in order to minimize feed wastage

a. Maximizing feed filling on the feeder
b. Feeders should be provided according to size of birds
c. Both a and b
d. B only

Answer: D

33. The route administration for the NCD (B1 strain)

a. Ocular
b. Intranasal
c. Intramuscular
d. Mass method

Answer: B

34. This is the cause of a viral infection of poultry that is characterized by formation of scab-like lesions on the skin of the legs.

a. Fowl pox
b. Avipox virus
c. Coronavirus
d. Newcastle's disease virus

Answer: B

35. How many kilograms of feed is needed in order to produce a 1.75 live weight of broiler taking a 2.0 conversion ratio at 35 days.

a. 3.0 kg
b. 3.5 kg
c. 2.5 kg
d. 4.5 kg

Answer: B

Explanation: FCR=Feed consumption/gain in weight =3.5/1.75= 2.0

36. Which of the following poultry species is raised in the largest scale of production in the world?

a. Turkey
b. Duck
c. Quail
d. Chicken

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Answer: D

Explanation: Chicken is the most popular poultry species. According to the report of FAO as cited by the USDA (2019) chicken ranks second (33%) to pork (36%) as the most widely-eaten meat in the world where beef ranks third (24%) and goat/sheep ranks fourth (5%).

37. What breed of chicken is widely used in commercial egg production?

a. White Leghorn
b. Sussex
c. Ancona
d. Minorca

Answer: A

Explanation: Typically, the white leghorn is widely used commercial egg-laying industry as these birds are intended to produce eggs at a higher rate annually. For backyard producers who have just started their egg business, this breed is not recommended as this breed can be loud, nervous, and occasionally aggressive towards their environment (Akers, 2002).

38. A group of chicken within a breed that possesses the same plumage color and type comb.

a. Breed
b. Variety
c. Strain
d. Class

Answer: B

Explanation:
Breed- a group of chicken (fowls) possessing a certain conformation or shape of the body that distinguishes them from other chicken
Variety- a group of chicken within a breed that possesses the same plumage color and type comb (Leghorn- white, single comb)
Strain- a group of chickens within a variety of breeds which has under a constant specific section of certain traits by specified breeders for about 5 to 8 years

39. What province is the top producer of eggs in the year 2020?

a. Batangas
b. Pampanga
c. Cebu
d. Bukidnon

Answer: A

Explanation: Based on the report of PSA (2020), Batangas province is the largest egg producer in 2020, producing 151.88 million tons among the provinces. They are closely

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followed by Pampanga (69.72 thousand tons) and Cebu (45.99 thousand tons). The other major producing provinces of the country in 2020 were Bukidnon, Bulacan, South Cotabato, Davao City, Rizal, Tarak, and Negros West. The total production of these ten egg-producing provinces accounted for 69.2% of the country's total egg production.

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Unit VI

SWINE PRODUCTION

Instructions: Read each question carefully. Choose the best answer that satisfies the question or statement.

1. It is said to be the primary reason for castrating male pigs before reaching sexual maturity.

   a. For better feed efficiency
   b. To make growing male pigs be docile for ease of handling
   c. To avoid a "boar taint odor" in pork
   d. To prevent unwanted reproduction

   Answer: C

   Explanation: The main purpose of castration in pigs is to prevent boar taint, which is present in the meat of some entire males once they reach puberty.

2. The following are the reasons why Iron (Fe) supplementation in baby pigs is important, EXCEPT?

   a. Low body storage of Fe in the Newborn pig
   b. Low Iron content of colostrum and milk
   c. Rapid growth rate
   d. None of the Above

   Answer: D

   Explanation:
   Low body storage in the newborn pig. Pigs are born with only 40 to 50 mg of Fe and it depletes very quickly since pigs need 7 to 16 mg of absorbed Fe/day
   Low Fe content of colostrum and milk. The Fe content of colostrum is only about 2 ppm and that of milk for only 1 ppm. Thus pigs receive no more than 1 mg/day from milk versus the requirement of 7 - 16 mg.
   Rapid Growth. Pigs will double their weight during the 1st week and weigh up to 5 times or more by 4th week. This will result in a rapid increase in the Fe requirement because of the corresponding increase in plasma volume.

3. This represents the largest operating cost for most commercial animal raisers.

   a. Feed cost
   b. Labor cost
   c. Housing
   d. Water and Electricity

   Answer: A

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Explanation: Feed is the largest single cost item for livestock and poultry production, accounting for 60%–70% of the total cost.

4. This major breed of hogs is black with a white band around the body at the shoulder including the front legs and feet.

   a. Berkshire
   b. Yorkshire
   c. Hampshire
   d. Pietrain

   Answer: C

   Explanation: The most striking characteristics of the Hampshire are the white belt around the shoulder and the body including the foreleg.

5. It is extremely important to have a good boar since it contributes half the quality of the herd. The following are areas to be considered in selecting breeding boar, which among is not?

   a. Boars that have at least 12 nicely placed rudimentary teats.
   b. Selected from sows, which wean 8 -10 or more piglets per litter.
   c. Boar having a sound body conformation.
   d. Boar must be at least 5 months ready to use for breeding.

   Answer: D

   Explanation: Boar must be 7 1/2 to 8 months old at first service. Some boars show sexual development as early as 5 months of age. However, they are not ready for regular service. Sexual development is a gradual process. Age is a better indication of sexual maturity than body size. As age increases, semen volume and sperm production gradually increase.

6. The purebred is known to be the "longest breed" among purebreds.

   a. Duroc
   b. Landrace
   c. Large white
   d. Pietrain

   Answer: B

   Explanation: Landrace are known for their length of body, high percentage of carcass weight in the ham and loin, and ideal amount of finish

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7. This breed is black with white band around the heart girth including the front legs.

   a. Duroc
   b. Landrace
   c. Hampshire
   d. Yorkshire

   Answer: C

   Explanation: The Hampshire pig is a domestic swine breed characterized by erect ears, a black body, and a white band around the middle, covering the front legs.

8. Surgical operation during which the primary sex organs are removed.

   a. Caponizing
   b. Vasectomy
   c. Castration
   d. Slaughtering

   Answer: C

   Explanation: Castration (also known as orchiectomy or orchidectomy) is any action, surgical, chemical, or otherwise, by which an individual loses use of the testicles: the male gonad.

9. Baby pig anemia is brought about by a deficiency of:

   a. Ca
   b. P
   c. Fe
   d. Na

   Answer: C

   Explanation: As the name implies, iron deficiency anemia is due to insufficient iron. Without enough iron, your body can't produce enough of a substance in red blood cells that enables them to carry oxygen (hemoglobin). As a result, iron deficiency anemia may leave you tired and short of breath.

10. Which of the following is an important management practice to prevent navel ill.

    a. Cut needle teeth
    b. Inject iron dextran
    c. Cut umbilical cord
    d. Wipe the mucus

    Answer: C

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Explanation: Cutting the umbilical cord of any newborn farm animal is an important management practice for the prevention of navel ill. The navel of the newborn is basically a hollow tube which leads directly into the abdominal cavity and liver of the animal. It remains moist for several days after birth, making it easy for bacteria to gain entrance into the body.

11. The S-shape structure in the penis of the boar is called:

a. Scrotal sac
b. Seminal vesicle
c. Sigmoid flexure
d. Semen

Answer: C

Explanation: The boar has a fibroelastic penis with a sigmoid flexure (S-shape) along its shaft and a characteristic corkscrew-shaped glans penis. The boar's penis is retracted into the prepuce while at rest.

12. The average gestation period (days) in swine:

a. 113 days
b. 114 days
c. 115 days
d. 116 days

Answer: B

Explanation: Gestation length of the swine is 114 days, so average farrowing will occur 114 days after breeding. An easy way to remember this is "3 months (90 days), 3 weeks (21 days), 3 days" (90 + 21 + 3 = 114).

13. Reasons for castrating pigs.

a. Fast growth rate
b. Better feed efficiency
c. Remove the boar taint
d. All of the above

Answer: D

Explanation: The main purpose of castration in pigs is to prevent boar taint, which is present in the meat of some entire males once they reach puberty. Other benefits of castration include: fast growth rate, better feed efficiency, reducing aggressive and mounting behavior with their resulting injuries, and the possibility of producing a higher-quality end-product.

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14. The milk is produced in the first few days after parturition.

a. Fresh milk
b. Milk
c. Colostrum
d. A & C

Answer: C

Explanation: The fluid your breasts produce in the first few days after birth is called colostrum. It's thick and usually a golden yellow color.

15. The weight at 8 months that the gilt can be culled.

a. <100 kg
b. <110 kg
c. <120 kg
d. <130 kg

Answer: A

16. The system of pig keeping wherein they are supplied with high quality feeds, optimal utilization of housing facilities.

a. Semi-intensive
c. Intensive
b. Extensive
d. Semi-extensive

Answer: A

Explanation: The farmer confines all the pigs within a building and does not allow them to move out. The animals were provided with feeds, water and veterinary services, routinely, in adequate measure. The feeding efficiency is high, thus, the growth rate is also very high.

17. The physical characteristics of a replacement boar should be:

a. Two big testicle
b. Strong legs
c. Long body
d. All of the above

Answer: D

Explanation: Conformation and physical soundness of the boar is important therefore particular attention should be given to the boar's legs and feet as they are subjected to considerable strain during mating.

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18. This is the period of pig production cycle between weaning and the next service called.

    a. Rearing period
    b. Dry period
    c. Gestation period
    d. Lactation period

    Answer: B

    Explanation: The time interval from weaning to service is known as the dry period where the sow is unproductive. It is important that the weaning to service interval be kept as short as possible to increase sow productivity.

19. A white swine breed which is characterized by its long side square, relatively short legs and medium to drooping ears most often covering the face and eyes. This breed is also known as the longest breed of swine.

    a. Landrace
    b. Berkshire
    c. Yorkshire
    d. Duroc

    Answer: A

    Explanation: Landrace hogs are white in color. They are long-bodied and their ears lop forward and down. Landrace sows are noted for their mothering ability and its prolificacy under Philippine conditions.

20. This breed is called "Large White" and also known as the "Mother" breed because its mothering ability is exceptionally good.

    a. Duroc
    b. Yorkshire
    c. Tamworth
    d. Pietrain

    Answer: B

    Explanation: The color of Yorkshire is white. The ears are erect and the face slightly dished. Yorkshires have large litters, high feed efficiency and rapid growth. It is called the mother breed because its superior mothering ability is exceptional with high milk production during lactation.

21. This swine breed is black colored with six (6) white points, at the forehead, the forelegs and switch.

    a. Berkshire
    b. Duroc

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c. Hampshire
d. Yorkshire

Answer: A

Explanation: The Berkshire is a medium-sized hog that produces an acceptable carcass. The animal is black with six white points. Four white points are found on the feet. There is also some white on the face and the tail. The head is slightly dished and the ears are erect.

22. A sow disease which refer to the inflammation and infection of the uterus which often occurs after farrowing, dystocia or abortion is called

a. Mastitis
b. Metritis
c. Agalactia
d. Brucellosis

Answer: B

Explanation: Metritis is part of the MMA syndrome so frequently met as a clinical entity in the pig practice. Improper manual extraction of conceptus during farrowing can increase incidence in metritis. This can also be a result from retained fetuses, laceration, infection and dystocia.

23. Puberty is the beginning of the reproductive capability or sexual maturity of the animal. Swine reaches puberty as early as

a. 5 months
b. 6 months
c. 7 months
d. 9 months

Answer: A

Explanation: Swine reaches puberty at about five months of age. However, at this stage boars and gilts are not yet ready for breeding. Breeding at this stage usually retards the growth and development of the animals.

24. Shelf life of a processed boar semen for artificial insemination can be prolonged by storing at this temperature at approximately 40 hours

a. 20°C
b. 2 - 8°C
c. 10 - 14°C
d. 16 - 17°C

Answer: D

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Explanation: Extended/processed semen can be stored at an ideal temperature of 16 – 17°C to preserve the potency of the semen.

25. Creep feeding is a feeding practice of suckling piglets to give a good start before weaning. This is done by giving piglets with pre-starter feed as early as

a. 5 days old
b. 12 days old
c. 9 days old
d. 11 days old

Answer: A

Explanation: Start feeding the piglets on the fifth day. Creep feeding initiates and promotes gut and digestive enzyme development, which enables the piglet to digest nutrients from food sources other than that of milk.

26. The removal of unwanted pigs is called culling. Some pointers in culling gilt and sow are, except

a. Reproductive failures
b. Poor maternal instinct
c. Produces 12 piglets per farrowing
d. Low fecundity

Answer: C

Explanation: The most common reasons for culling a gilt or sow are diseases and low reproductive performance. Having a litter size of 12 per farrowing is considered an average performance.

27. It is a breeding technology where the semen is deposited to the sow's genitalia in time conducive to fertilization through artificial means.

a. Embryo transfer
b. Artificial insemination
c. Superovulation
d. A & C

Answer: B

Explanation: Artificial insemination is the process of introducing the collected boar semen into the sow's cervix or uterine cavity with the use of a catheter and squeeze bottle during estrus.

28. This practice is done to piglets at birth to prevent them from inflicting wounds on the udder of their dam and litter mates.

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a. Needle teeth cutting
b. Wolf teeth cutting
c. Black teeth cutting
d. All of the above

Answer: A

Explanation: Right after birth, clipping of the eight sharp needle teeth is done to prevent injuries to the sow's udder and laceration to the piglets as well when they fight for teat

29. The system of identifying pigs commonly practiced in any pig farm.

a. Ear notching
b. Ear tagging
c. Tattooing
d. None

Answer: A

Explanation: Ear notching is the most commonly used method of pig identification, which is done 24 hours after birth. One can design his own code in the farm.

30. It is a fraction of semen containing viable sperm cells ideal for insemination

a. Gel fraction
b. Pre-sperm fraction
c. Sperm-rich fraction
d. Clear fluid fraction

Answer: C

Explanation: It is grayish and milky. This fraction is collected as it contains viable sperm cells ideal for insemination. But in practice, it is often very difficult to differentiate between the clear fractions and the sperm rich fraction.

31. The scientific name of European wild boar believed to be the principal descendant of our domesticated pig today

a. *Sus scrofa*
b. *Sus indicus*
c. *Sus philippinensis*
d. *Sus domesticus*

Answer: A

Explanation: Domestic pig originates from Eurasian wild boar (*Sus scrofa*).

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32. A breed of pig characterized with a deep red or rusty color or cherry red color developed in the USA and known as "Red power" since males belonging to this breed are noted for their aggressiveness.

a. Pietrain
b. Duroc
c. Hampshire
d. Large White

Answer: B

Explanation: This breed exhibits considerable variations in color ranging from light to golden to very dark red but cherry red is the most common color and popularly known as "red power" because of its aggressiveness.

33. The theoretical production cycle of a sow is ____ days

a. 148
b. 151
c. 155
d. 163

Answer: B

Explanation:
Gestation = 114 days
Lactation = 30 days
Dry period = 7 days
TOTAL = 151 days

34. If a sow can wean a total of 9 piglets per litter, how many can it wean in a year assuming it has a litter index of 2.2?

a. 19.5
b. 19.8
c. 20.2
d. 20.5

Answer: B

Explanation:
Weaned piglets in a year = Piglets weaned per litter x Litter index
= 9 x 2.2
= 19.8

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35. If a 200 sow-level farm targets a 2.2 Litter Index with a 10 Piglets Born Alive / Litter, with a 6% pre-weaning mortality and 3% post-weaning mortality, how many growers is the farm owner expecting in a year?

a. 4,012
b. 4,021
c. 4,136
d. 4,316

Answer: A

Explanation:
Compute for the mortality: 10 PBA x 6 % = 0.6 pre weaning mortality
10 – 0.6 = 9.4 piglets left

9.4 x 3% = 0.282 post weaning mortality
9.4 – 0.282 = 9.12 piglets left

Number of growers expected in a year = (Sow level x LI) x weaned piglets
= (200 x 2.2) x 9.12
= 440 x 9.12
= 4,012

36. Mang Vico bought 10 weaners with an average weight of 12 kg to be reared as finishers. The pigs took 150 days to reach the preferred market weight of 90 kg. How much is the average daily gain of the pigs?

a. 480 g
b. 520 g
c. 600 g
d. 720 g

Answer: B

Explanation:
ADG = (Total gain in weight / Total fattening days) x 1000
= [(90-12) / 150] x 1000
= (78 / 150) x 1000
= 0.52 x 1000
= 520 g

37. This system of breeding is done to purify the bloodline of the animals, however, it is not recommended for continuous use due to its negative effect on production performance of the animal.

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a. Crossbreeding
b. Inbreeding
c. Purebreeding
d. Upgrading

Answer: B

Explanation: This is done to purify the bloodline of the animals since the similarity of the animals within the group is increased. However, it is not recommended for continuous use. It usually decreases the vigor of the animal because it brings together the recessive genes with undesirable effects in the resulting crossbred.

38. The following are effects of inbreeding, EXCEPT

a. Decreased number of pigs born and weaned
b. Hastened sexual maturity
c. Reduced strength and vigor in piglets
d. Reduced ovulation and sperm production

Answer: B

Explanation: Retarded and delayed sexual maturity. Seemingly, the rate of physiologic maturity of both male and female pigs is retarded slightly by inbreeding. It was observed in some lines that the age at which inbred boars begin to mate seem to lag a little as inbreeding becomes more intense.

39. This system of breeding is recommended for the production of slaughter pigs because it takes advantage of heterosis or hybrid vigor

a. Crossbreeding
b. Inbreeding
c. Purebreeding
d. Upgrading

Answer: A

Explanation: It is highly recommended for the production of slaughter hogs for two reasons: (1) to obtain hybrid vigor or heterosis which is defined as the change in performance of the crossbred progeny over the average of the parental breeds and (2) to bring together the desirable characteristics of the parental breeds

40. A hybrid pig which is a result of the cross between Duroc, Hampshire, Landrace and Large white

a. Babcock
b. Dalland

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c. Hypig
d. Genemaster

Answer: A

Explanation:
Dalland – result of a 4-way cross of Large white, Pietrain/Landrace, Daltrain and Large white
Hypig – result of the crosses between Landrace, Yorkshire and own breeding lines
Genemaster – result of the crosses between Large white, and Landrace and Yorkshire

41. Animal geneticists and nutritionists claim that a defect in the animal is likely to be hereditary if:

a. There had been inbreeding
b. The frequency of its occurrence is more with certain families than others
c. It occurred irrespective of season and various rations have been fed
d. All of the above

Answer: D

42. Which among the following traits or characteristics have high heritability

a. Average Daily Gain
b. Feed Conversion Ratio
c. Litter size at birth and at weaning
d. Carcass quality

Answer: D

Explanation: ADG and FCR are growth performance of medium heritability; fertility such like litter size at birth and at weaning is of low heritability; carcass quality (% lean) is of high heritability

43. In this method, selection is done on the basis of the individual's record of inheritance

a. Independent culling level
b. Pedigree
c. Selection index
d. Tandem

Answer: B

Explanation:

A pedigree to be of value should include parents of the individual being considered for selection. The boar and sow should be of good ancestry which could be seen in the

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pedigree record of the animal. Such ancestry and breeding give more assurance of the production of high quality pigs that are uniform and true to type.

Independent culling method – establishes minimum standards for each character, and selecting simultaneously but independently for each character. The performance in one trait is considered entirely independent of the performance in other traits, or the performance in one trait does not affect the performance in another. The chief weakness of this method is that an individual may be culled because of being faulty in one character only.

Selection index – this method combines all important traits into one overall value or index. Theoretically, a selection index provides a more desirable way in which to select for several traits than either tandem or independent culling method.

Tandem – there is selection for only one trait at a time until the desired improvement in that particular trait is reached, following which selection is made for another trait. Tandem selection is recommended only in those rare herds where one character only is primarily in need of improvement; for example, where a certain herd of pigs needs improving in litter size.

44. This method combines all important traits into one overall value or index

    a. Independent culling level
    b. Pedigree
    c. Selection index
    d. Tandem

    Answer: C

    Explanation:
    Selection index – this method combines all important traits into one overall value or index. Theoretically, a selection index provides a more desirable way in which to select for several traits than either tandem or independent culling method. Selection indices are designed to accomplish the following: (a) give emphasis to the different traits in keeping with their relative importance; (b) balance the strong points against the weak points of each animal; (c) obtain an over-all total score for each animal, following which animals can be ranked from best to poorest; (d) assure a constant and objective degree of emphasis on each trait being considered, without shifting of ideals from year to year; and (e) provide a convenient way in which to correct for environmental effects such as feeding differences.

45. The following are traits or standards to be examined when selecting boars EXCEPT:

    a. Behavior
    b. Backfat

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c. Mothering ability
d. Reproductive soundness

Answer: C

Explanation:
Mothering ability is a trait examined in gilts during selection. In selecting gilts, you must consider cannibalism which is based on the performance of its mother during its reproductive life. However, it is common that cannibalism can happen during the first parity.
In selecting boars, we consider: (a) Behavioral traits like docility, temperament, sex characteristics, maturity and aggressiveness. These traits are associated with reproductive potential; (b) Backfat – carcass merit is probably best evaluated by taking measurements such as backfat thickness, loin area, or estimating the percent muscle in the animal. Of these measurements, backfat is the single most important and best measure of leanness. These traits have high heritability values. (d) Reproductive soundness – characteristics associated with soundness include: spacing, number and presentation of teats; genetic abnormalities such as hernia and cryptorchidism, and mating ability.

46. Culling is the removal of unwanted animals in the herd, and it should be decided upon based on:

a. Damaged udder section
b. Health status
c. Reproductive failures
d. All of the above

Answer: D

Explanation:

Damaged udder section. The udder is a vital source of nourishment for the piglets. It is highly susceptible to infection especially during lactation because of the presence of milk which is a very good medium for the multiplication of infectious microorganisms. Usually, injured sections become gangrenous and are difficult to treat. If left unchecked, the infection spreads rapidly to other sections of the udder leaving only a few functional teats for the next lactation.

Health status. Animals with health problems should be culled.

Reproductive failures. Some gilts or sows do not come in heat despite proper management. Others simply do not conceive even after two or more matings with proven boars of unquestionable fertility. Reports indicated that more than 50% of all reproductive failures in females are due to anatomical abnormalities of the reproductive tube.

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Occasionally, there are boars without sex desires or who cannot impregnate even the proven sows. This may also be due to endocrine or antomic defects like bilateral or unilateral occlusion or missing segments of the duct system. Although measures to correct reproductive failure in both males and females were successful in a few cases, it is better to replace the affected animals with normal individuals.

47. This refers to the time interval from weaning to service

    a. Dry period
    b. Estrus period
    c. Gestation period
    d. Lactation period

    Answer: A

    Explanation:
    Dry period is the time interval from weaning to service. This is the time when the sow is unproductive. The length of the dry period varies depending on the sow's body condition. On some farms where weaning is at 28-30 days, the sows in good body condition return to heat during the week following weaning. However, if the sow is thin due to large litters or was fed badly during lactation, the dry period could be longer. Normally, sows will come in heat 5 to 10 days after weaning.
    Estrus period – heat period or the time when the sow is receptive to mating
    Gestation period – pregnancy period, 114 days on the average
    Lactation period – the period after farrowing wherein the sow nurses the litter

48. It is the period intervening between two successive heat periods in the absence of pregnancy

    a. Dry period
    b. Estrus period
    c. Estrous cycle
    d. Lactation period

    Answer: C

    Explanation:
    Estrous cycle usually ranges from 18-24 days, with an average of 21 days. The cycle is usually characteristic of the individual, that is, a sow which has a 21-day cycle comes in heat every 21 days unless interrupted by pregnancy.

49. The substance responsible for the manifestation of heat or estrus

    a. Estrogen
    b. Progesterone

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c. Prolactin
d. Prostaglandin

Answer: A

Explanation:
The follicle stimulating hormone (FSH) directly affects the ovaries where several of the globes develop into vesicles or graafian follicles. In turn, the follicle walls produce a hormone, estrogen which is responsible for the manifestation of heat or estrus. When enough estrogen is carried by the blood to the brain, it signals the pituitary to stop the production of FSH because follicles are already there and are mature. The pituitary gland then produces luteinizing hormone (LH) which causes the ovum within the follicular wall to emerge. This process is called ovulation. One ovum is released from each ripe follicle within 4-6 hours.
Progesterone – stimulates the uterine glands of the uterus to produce nutrients to maintain the released ova.
Prostaglandin – if no conception takes place, the uterus produces prostaglandin which causes the regression of the corpus luteum. With this, the "brake" on the pituitary gland is released. The FSH is produced, and the whole cycle begins again.
Prolactin – if the pig conceives, gives birth and nurses the piglets, the suckling stimulates milk let-down causing the release of prolactin from the pituitary gland. It blocks the release of FSH.

50. The most reliable sign of estrus

a. Clear, sticky mucus discharge
b. Frequent urination
c. Red and swollen vulva
d. Standing heat

Answer: D

Explanation:
The sow stands still with arched back in the presence of the boar, when ridden by other sows or farm caretaker

51. A method of heat detection wherein old semen sample is squeezed from a squirt bottle on the snout of the sow

a. Back pressure test
b. Riding-the-back test
c. Semen-on-the-snout test
d. Teaser boar method

Answer: C

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Explanation:
With the boar's smell in the semen, the sow will just stand firm and remain rigid if it is sexually receptive. A similar preparation, called sex odor aerosol (SOA), is also commercially available. This involves spraying the sow's snout with the SOA allowing her to show signs of heat more readily and to exhibit willingness to stand for the boar.

Back pressure test – done by applying pressure on the sow's sides with both hands. If she is receptive, she will just stand firm and rigid, ready to be served.

Riding-the-back test – done by riding on or pressing the back or loin of the sow. If she is receptive, she will just stand firm and rigid, ready to be served.

Teaser boar method – a vasectomized boar is used in this method and is allowed to mount the sow. If the sow stands firm, remains rigid with cocked ears and willingly submits itself to the advances of the boar, it is sexually receptive.

52. Which of the following statements is TRUE

a. Ovulation occurs 36-40 hours after the start of standing heat hence, live sperm must be in the reproductive tract a few hours before the occurrence of ovulation
b. Ovulation occurs 36-40 hours after the start of standing heat hence, live sperm must be in the reproductive tract a few hours after the occurrence of ovulation
c. Both A and B
d. None of the above

Answer: A

Explanation:

Ovulation is the process of releasing the ova from each ripe follicle inside the ovary. It occurs 36 to 40 hours after the start of standing heat when the most number of eggs are being shed. Live sperm must be in the female reproductive tract a few hours before the occurrence of ovulation; otherwise, the litter size will be reduced. Based on the physiology of the female reproductive system and signs and reactions of the animal during the heat period, the best time to inseminate/serve the sow and gilt is 12-24 hours and on or 12 hours after the onset of standing heat, respectively. This will give time to the sperm to be at the female's reproductive tract before ovulation. Sperm is transported from the cervix to the top of the womb by the contractions of the muscles. It takes up to 2 hours for them to arrive at the bottom of the fallopian tubes, the uteru-tubal junction. At this junction, before the sperm cells can fertilize the ova, they have to undergo a second phase of maturation process known as capacitation. This process will take about 6 hours.

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53. The following are advantages of early pregnancy detection EXCEPT:

    a. Identification of sows for culling or rebreed, non-pregnant, feed-wasting sows
    b. Gives early warning of breeding troubles, due to infertile boars and cystic ovaries of sows
    c. Enables the producer to make effective use of his breeding facilities and to plan more adequately for farrowing, nursing and finishing
    d. Selling of pregnant sows

    Answer: D

54. Piglets reared per sow per year is affected by litter index. Which among the following factors influence litter index

    a. Heat occurrence
    b. Abortion
    c. Age of weaning
    d. Number of stillborn

    Answer: D

    Explanation: Number of stillborn is a factor influencing weaned piglets which affects piglets reared per sow per year.

55. A sow with 151 days production cycle will have a theoretical litter index of

    a. 2.35
    b. 2.42
    c. 2.47
    d. 2.24

    Answer: B

    Explanation: Theoretical litter index = 365 days / length of one production cycle
    = 365 days/year / 151 days/cycle
    = 2.42

56. Gilts maybe classified as a sow when

    a. It is served or bred
    b. It farrows
    c. It turns 8 months old
    d. All of the above

    Answer: D

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Explanation:
Gilts maybe classified as a sow using the following criteria: (1) when it is served or bred; (2) when it farrows; (3) when it turns 8 months old. In the Philippines, the commonly used definition of a sow is once an animal farrows and once the animal is served.

57. This is expressed as the average daily increase in the liveweight during the fattening period

    a. Average Daily Gain
    b. Feed Conversion Ratio
    c. Total Gain in Weight
    d. Feed Consumption

    Answer: A

58. If a pig has a feed conversion ratio of 2.75, what does it mean?

    a. The pig needs 2.75 kg of feed to convert into 2.75 kg gain in weight
    b. The pig needs 2.75 % more of feeds to convert into 1 kg gain in weight
    c. The pig needs 2.75 kg of feed to convert into 1 kg gain in weight
    d. None of the above

    Answer: C

    Explanation: Feed Conversion Ratio (FCR) measures the efficiency of the utilization of the major raw material in the production process. In brief, the FCR tells us how many kilograms of feeds is needed for one kilogram gain in weight.

59. Suppose the normal weight gain of a finisher within 110 finishing days is 67 kg. The ADG is 609 g and the liveweight price per kilogram is P70.00. The value of a 10 gram difference on ADG is P77.00. What does the 10 g difference represent?

    a. 10 g additional daily gain would represent P77.00 increase on the market value per finisher
    b. 10 g additional daily gain would represent P77.00 decrease on the market value per finisher
    c. 10 g decrease in daily gain would represent P70.00 increase on the market value per finisher
    d. 10 g additional daily gain would represent P70.00 increase on the market value per finisher

    Answer: A

60. Which among the following indications is TRUE

    a. An average feeding level attended by poor ADG and FCR oftentimes points to poor quality feeds and/or management aspects

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b. A low feed consumption per day is very often caused by low palatability and poor management practices
c. In case the feed consumption per day is too high compared to the ADG obtained,
d. All of the above

Answer: D

Explanation:
An average feeding level attended by poor ADG and FCR oftentimes points to poor quality feeds and/or management aspects (health, housing, ventilation, etc.)

A low feeding level generally leads to poor ADG and FCR. Relatively too much feed is used for maintenance instead of growth. A low feed consumption per day is very often caused by low palatability and poor management practices.

In case the feed consumption per day is too high compared to the ADG obtained, disorders in the digestive system, a poor digestibility of the feeds or a low nutritional value are likely to be the cause. Also, management aspects are possible causes (spillage, low temperature). A high feeding level may also cause a relatively high fat disposition, which usually leads to an average ADG but poor FCR.

61. This refers to the efficiency with which the capacity of the fattening units are used

a. Rounds per year
b. Litter index
c. Occupancy rate
d. Mortality rate

Answer: A

Explanation:
The number of rounds per year refers to the efficiency with which the capacity of the fattening units is used. The capacity of the units can be expressed either as the number of pig places or as the average number of pigs (stock) on hand. The number of pig places is a fixed number. On the other hand, the average stock on hand may vary considerably depending on: (1) occupancy rate of the units/pens; (2) mortality rate; (3) inter batch periods; and (4) stocking system.

62. This is the most essential but most often neglected nutrient

a. Carbohydrates
b. Protein
c. Vitamin
d. Water

Answer: D

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Explanation: Water is the most essential but most often neglected nutrient. Many pig farmers tend to underestimate the value of water. It is not enough to make water available but the quality should also be considered. Pigs need plenty of clean, fresh water at all times. Performance of all types of pigs is reduced when they are lacking in water. The rate of gain and the amount of feed consumed are all affected by water restriction.

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Unit VII
RUMINANT PRODUCTION

Instructions: Read each question carefully. Choose the best answer that satisfies the question or statement.

1. What is the age of sexual maturity or puberty of bull and heifer calves?

a. 2-4 months
b. 6-8 months
c. 10-12 months
d. 14-16 months

Answer: B

2. What is the estrus period of indigenous and Zebu grades of beef cattle?

a. 2-4 hours
b. 6-8 hours
c. 10-12 hours
d. 14-16 hours

Answer: C

Explanation: Estrus normally lasts from 14 to 18 hours in exotic breeds. Indigenous and zebu grades have shorter estrus period which usually lasts from 10 to 12 hours.

3. What is the recommended minimum interval between calving and first insemination after calving?

a. 30 days
b. 60 days
c. 90 days
d. 120 days

Answer: B

Explanation: a minimum interval of 60 days between calving and first insemination after calving (postpartum breeding) is recommended.

4. These are feed which is relatively high in fiber and low in total digestible nutrients.

a. Carbohydrates
b. Concentrates
c. Protein
d. Roughage

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Answer: D

Explanation: Roughages is defined as feed which is relatively high in fiber and low in total digestible nutrients.

5. Signal Grass is a vigorous trailing perennial which branches at the lower nodes. What is the scientific name of Signal Grass?

   a. *Andropogon gayanus*
   b. *Brachiaria decumbens*
   c. *Brachiaria humidicola*
   d. *Panicum maximum*

   Answer: B

6. The humped cattle include the Zebus from India, Pakistan, Southeast Asia and some islands in the Pacific. What is the scientific name of the humped cattle?

   a. *Bos indicus*
   b. *Bos taurus*
   c. *Bubalus bubalis*
   d. *Capra hircus*

   Answer: A

   Explanation: The humped cattle (*Bos indicus*) include the Zebus from india, Pakistan, Southeast Asia and some islands in the Pacific.

7. It is the recommended breed of beef cattle in the Philippines because of its adaptability under local conditions and resistance against pests and diseases.

   a. Angus
   b. Brahman
   c. Charolais
   d. Guzerat

   Answer: B

   Explanation: In the Philippines, Brahman and its derivatives are recommended because of its adaptability under local conditions and resistance against pests and diseases.

8. This technology involves super ovulatory treatments to induce the donor cows to develop several follicles and multiple ovulations followed by insemination and collection of embryos.

   a. Artificial insemination
   b. Estrus synchronization

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c. Embryo transfer
d. Natural mating

Answer: C

Explanation: Embryo transfer technology involves super ovulatory treatments to induce the donor cows to develop several follicles and multiple ovulations followed by insemination and collection of embryos.

9. For beef cattle shed or barn, what is the proper space requirement of yearling?

a. 1-2 sq.m.
b. 2-3 sq.m.
c. 3-4 sq.m.
d. 4-5 sq.m.

Answer: B

10. The extensive system of beef cattle/buffalo production is also popularly known as

a. Backyard system
b. Cow-calf
c. Feedlot fattening
d. Ranching

Answer: D

11. To measure carrying capacity of pasture, usually one animal unit is equivalent to how many kilograms?

a. 250 kg
b. 275 kg
c. 300 kg
d. 325 kg

Answer: C

12. In beef/buffalo production, one of the improved grasses used as forage in cattle feeding is

a. Calopogonium
b. Centrosema
c. Napier
d. Siratro

Answer: C

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13. The commercial cow-calf operations of the country are located in this province

    a. Bukidnon
    b. Masbate
    c. Pangasinan
    d. All of the above

    Answer: D

14. Usually, the weaning weight of the Brahman and its crosses is

    a. 125 kg
    b. 150 kg
    c. 175 kg
    d. 200 kg

    Answer: B

15. This method is the most common and legal means of identifying beef cattle ownership

    a. Hide branding
    b. Tattooing
    c. Ear notching
    d. Horn branding

    Answer: A

16. Implantation and placentation characterized by the formation of endometrial cups is specific for the species.

    a. Mare
    b. Cow
    c. Cara cow
    d. Sow

    Answer: A

    Explanation: Endometrial cups form from placental cells that invade into the uterine lining (endometrium) of the mare beginning at approximately 35 days of pregnancy.

17. Which among the following herbivores is not a ruminant?

    a. Cattle
    b. Pig
    c. Horse
    d. Goat

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Answer: C

Explanation: Horses do not have multi-compartmented stomachs as cattle do.

18. The young of this species is called kid.

    a. Cattle
    b. Pig
    c. Sheep
    d. Goat

    Answer: D

19. Induced ovulator is an animal that does not ovulate unless there is copulation. Which among the following herbivores is not a ruminant?

    a. Rabbit
    b. Pig
    c. Sheep
    d. Cattle

    Answer: A

    Explanation: Rabbits are monogastric non-ruminant herbivores.

20. The American Brahman breed is an example of

    a. *Bos indicus* (Zebu) cattle
    b. *Bos taurus* humpless cattle
    c. *Bos taurus* humped cattle
    d. *Bos* humpless cattle

    Answer: A

21. The cattle that is European in origin

    a. *Bos taurus*
    b. *Bos indicus*
    c. *Bos bubalis*
    d. *Bos aries*

    Answer: A

22. The true stomach of the ruminant animal

    a. Abomasum
    b. Omasum

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c. Reticulum
d. Rumen

Answer: A

Explanation: The abomasum is the "true stomach" of a ruminant. It is the compartment that is most similar to the stomach of a monogastric animal.

23. All of these animals are monogastric except

a. *Bos indicus*
b. *Sus scrofa*
c. *Meleagris gallopavo*
d. *Equus caballo*

Answer: A

24. Gall bladder is absent in this animal

a. Cattle
b. Horse
c. Dog
d. Cat

Answer: B

Explanation: Horses do not have a gall bladder because horses are designed to eat constantly.

25. Humped cattle usually found in tropical countries

a. *Bos taurus*
b. *Bos indicus*
c. *Bos bubalis*
d. *Bos aries*

Answer: B

Explanation: Zebu cattle (*Bos indicus*) are well adapted to withstanding high temperatures, and are farmed throughout the tropical countries

26. All of these animals are monogastric except

a. Bos indicus
b. Sus scrofa
c. Meleagris gallopavo
d. Equus caballo

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Answer: A

27. These teeth are absent in horse and cattle

a. Canine teeth
b. Molar teeth
c. Milk teeth
d. Incisor

Answer: A

Explanation: Cows cannot bite a person because they don't have any upper front teeth.

28. The scientific name of goats is

a. Bubalus bubalis
b. Capra hircus
c. Equus caballus
d. Ovis aries

Answer: B

29. Communities of plants used as feeds for ruminants by grazing or cutting

a. Fodder
b. Forage
c. Grassland
d. Pasture

Answer: D

Explanation: Pasture grasses and legumes, both native and cultivated, are the most important single source of feed for ruminants such as cattle, horses, sheep, and goats. During the growing season they furnish most of the feed for these animals at a cost lower than for feeds that need to be harvested, processed, and transported. Hundreds of different grasses, legumes, bushes, and trees are acceptable as feeds for grazing animals. The nutritive value of the cultivated varieties has been studied, but information is incomplete for many of those that occur naturally.

30. It refers to the whole plant that is cut and dried to about 15% moisture

a. Hay
b. Silage
c. Straw
d. Green forage

Answer: A

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Explanation: Hay is produced by drying grasses or legumes when they approach the stage of maximum plant growth and before the seed develops. This stage has been shown to give maximum yields of digestible protein and carbohydrates per unit of land area. The moisture content is typically reduced below 18 percent in order to prevent molding, heating, and spoilage during storage.

31. Scientific name of kikuyu grass

a. *Pennisetum clandestinum*
b. *Panicum maximum*
c. *Paspalum atratum*
d. *Penisetum purpureum*

Answer: A

32. *Haemonus contortus* is a common roundworm that causes severe anemia in ruminants. This endoparasite is found in:

a. Cecum
b. Rumen
c. Small intestine
d. Abomasum

Answer: D

Explanation: *Haemonchus contortus*, known as the barber's pole worm, is relatively common in ruminant abomasums. *Haemonchus*, *Ostertagia*, and *Trichostrongylus* spp. The common stomach worms of cattle are *Haemonchus placei* (barber's pole worm, large stomach worm, wire worm), *Ostertagia ostertagi* (medium or brown stomach worm), and *Trichostrongylus axei* (small stomach worm).

33. The distinguishing features of Philippine Carabao are the two diagonal stripes on its brisket. It is known as

a. Liners
b. Chevron
c. Chevon
d. Stripes

Answer: B

Explanation: Chevron - 2 diagonal stripes in the ventral side of the neck; one near the brisket and other near the jaw.

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34. In a cow-calf operation, the standard weaning weight of Brahman and its crosses is

a. 200 kg
b. 125 kg
c. 150 kg
d. 175 kg

Answer: C

35. In beef cattle breeding, the recommended bull to cow ratio under normal situation is:

a. 1:50
b. 1:20
c. 1:30
d. 1:40

Answer: B

Explanation: In natural mating, 1:20 is the recommended ratio under normal situation considering the bull is in good condition.

36. Culling is an effective tool in removing the undesirable animals. The recommended rebreeding time before a cow must be culled is:

a. 2 times
b. 3 times
c. 4 times
d. 5 times

Answer: B

Explanation: Three times is the standard rebreeding guidelines of cows before it is culled because most cattle after calving needs room for improvement specifically the nutritional condition of the cattle.

37. Under continuous breeding program in beef cattle reproduction, bull culling should be done using frequency:

a. Yearly
b. Every 2 years
c. Every 4 years
d. Every 5 years

Answer: C

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38. Stocking rate is measured in terms of animal unit. What is the A.U .equivalent of heifers?

a. 0.5 A.U.
b. 0.25 A.U.
c. 0.75 A.U.
d. 1.0 A.U.

Answer: A

39. To consider as good silage, the proper pH of the ensiling materials should be within this range

a. 4.2-4.5
b. 3.2-3-5
c. 5.2-5.5
d. 3.0-4.0

Answer: A

Explanation: The ideal pH of ensiling materials should be 4.2-4.5 since it will show the final product is well-preserved and well-processed. Usually, wet silage with a high pH will not keep well and in most cases, is accompanied by high ammonia levels.

40. A good silage should contain how many percent lactic acid?

a. 5 - 9%
b. 10 -15%
c. 1 - 5%
d. 15 – 20%

Answer: A

41. The extensive system of beef cattle /buffalo production is also popularly known as

a. Ranching
b. Cow-calf
c. Feedlot fattening
d. Backyard system

Answer: A

42. To measure carrying capacity of pasture, usually one animal unit u=is equivalent to how many kilograms?

a. 300 kg
b. 250 kg
c. 275 kg
d. 325 kg

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Answer: A

43. In beef/buffalo production, one of the improved grasses used as forage in cattle feeding is

a. Napier
b. Centrosema
c. Sirato
d. Calopogonium

Answer: A

44. In buffalo production, what breed is a good example of swamp type?

a. Draft type
b. Philippine Carabao
c. Bulgarian
d. Nili- ravi

Answer: B

45. Paragrass is an improved forage, what is its scientific name?

a. *Brachiara mutica*
b. *Brachiara decumbens*
c. *Brachiara ruzixiensis*
d. *Panicum maximum*

Answer: A

46. Siratro is an improved legume, what is its scientific name?

a. *Macroptilium antropurpureum*
b. *Centrosema pubescens*
c. *Arachis pintoi*
d. *Desmanthus virgatus*

Answer: A

47. Scientific name of ipil-ipil

a. *Calliandra calothyrsus*
b. *Leucaena leucocephala*
c. *Desmodium cinerea*
d. *Desmanthus virgatus*

Answer: B

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